Role : Junior Frontend Developer

Skills : HTML, CSS, Java Script, Git, GitHub, npm, React, Tailwind, Vitest

Learning Roadmap :

Frontend Basics

```
 |
 v

HTML (https://www.youtube.com/watch?
v=kUMe1FH4CHE&list=PLWKjhJtqVAbnSe1qUNMG7AbPmjIG54u88)

 |
 v

CSS  (https://www.youtube.com/watch?
v=kUMe1FH4CHE&list=PLWKjhJtqVAbnSe1qUNMG7AbPmjIG54u88)

 |
 v

JavaScript (https://www.youtube.com/playlist?
list=PLZlA0Gpn_vH9k5ju1yq9qCDqvtuTVgTr6)

 |
 v

Git  (https://www.coursera.org/learn/introduction-git-github)

(Learn about Version Control Systems &
start using Git for your future projects.)
 |
 v

GitHub (Create your GitHub Profile and publish the
        projects you have developed so far to GitHub.)
 |
 v

npm   (Learn to use npm package manager. This should
       be a quick one. Just get the basics and move on.)

-- Direct Links:
    ◦ Account Sign-up: https://www.npmjs.com/signup
    ◦ Two-Factor Auth Config: https://docs.npmjs.com/configuring-two-factor-
authentication
-- Guides:
    ◦ Snyk Modern npm Guide: https://github.com/snyk-snippets/modern-npm-package
(Reference project for production-ready packages).


 |
 v

React  (https://www.youtube.com/watch?v=TtPXvEcE11E)

(Create some project with React. Make sure to
install some external npm packages and include
API calls to fetch some external data.)

 |
 v
```

Tailwind (https://www.youtube.com/watch?v=6biMWgD6_JY)

(There are several CSS frameworks available. You
can pick any. I would recommend learning Tailwind.)

 |
 v

Vitest (https://www.youtube.com/watch?v=k2wi03hw2ek)

(Learn about different Testing types and how to
test your frontend using something like Vitest.)

Certificates :
free :

Offers free certifications for HTML, Responsive Web Design (CSS).
(https://www.freecodecamp.org/learn/)

Paid :

Hosts the HTML+CSS Specialisation
(https://www.coursera.org/search?query=html)

Meta Front-End Developer Professional Certificate
(https://www.coursera.org/professional-certificates/meta-front-end-developer)

Role : Junior Backend Developer

Skills : Python/Go/Java Script, redis, PostgreSQL, Git, GitHub, REST APIs, JSON
APIs, Authentication, JWT,
         Integration Testing, Unit Testing, Fuctional Testing

Learning Roadmap :

Backend
 |
 v
Programming Language (https://www.youtube.com/@NetNinja/search?query=node.js)

- If you are a frontend developer jumping
  into backend development, I would
  recommend picking JavaScript and
  learning Node.js. Alternatively, Go and
  Python are also good options.

Go - (https://www.youtube.com/@TechWorldwithNana/search?query=go)
Python - (https://www.youtube.com/playlist?list=PL-osiE80TeTt2d9bfVyTiXJA-
UTHn6WwU)

 |
 v
Version Control (Git)-(https://www.youtube.com/@TheCodingTrain/search?
query=git%20and%20github)
 |
 v
Repository Hosting Servicec - Create your GitHub profile and start using
                              GitHub for your future projects.
                              See if any of your past projects can be

published on GitHub as OpenSource.

(GitHub)
|
v
Relational Databases - Learn about the relational databases,
normalization, DB migrations, SQL, DB
Indexes and Query performance analysis.

(PostgreSQL) - (https://www.youtube.com/@freecodecamp/search?query=relational%20databases)
|
v
APIs - Learn how to build RESTful APIs and add
JWT authentication to secure your APIs.
Develop some API and add JWT auth.

(REST, JSON APIs, Authentication, JWT) - (https://www.youtube.com/@ProgrammingKnowledge/search?query=apis)
|
v
Caching - Learn about different caching strategies,
including HTTP caching as well.
Use Redis in some of your projects.

(Redis) - (https://www.youtube.com/@Redisinc/search?query=redis)
|
v
Advanced Database Concepts - Learn about additional database topics
including performance analysis, ORMs,
limitations, common issues to keep in
mind with ORMs and in general.
Get some experience with some ORM.

(ORMs, ACID, Transactions, N + 1 problem)
(https://www.youtube.com/@ByteByteGo/search?query=database)
|
v
Testing - Learn about differeng testing techniques and
learn learn how to write automated tests.

(Integration Testing, Unit Testing, Fuctional Testing)

(https://www.youtube.com/@TECHSCHOOLGURU/search?query=testing)

Certificates :

free:

freeCodeCamp Back End Development and APIs - (https://www.freecodecamp.org/learn/back-end-development-and-apis/)

paid :

Python for Everybody (University of Michigan) - (https://www.coursera.org/specializations/python)

Role : AI Agent Developer

Skills : REST API, Python, Git, Transformer Models, LLM Fundamentals, Tokenization,

Context Windows, Prompt Engineering, Chain-of-Thought, ReAct, Agent Design,
Tool Integration, Function Calling, LangChain, LlamaIndex, CrewAI, AutoGen,
Vector Databases, Pinecone, Weaviate, RAG, Retrieval-Augmented Generation,
Agent Memory, Model Context Protocol, MCP, OpenAI API, Anthropic API, Gemini API,
Evaluation, LangSmith, Ragas, Debugging, Observability, Helicone, LangFuse,
Deployment, Prompt Injection, AI Security, Tool Sandboxing, PII Redaction, Bias Mitigation,MLOps

Learning path :

Learn the Pre-requisites
↓
Basic Backend Development (https://www.youtube.com/user/programmingwithmosh)
↓
Git and Terminal Usage (https://www.youtube.com/@coreyms/search?query=git%20)
↓
REST API Knowledge (https://www.youtube.com/@ProgrammingKnowledge/search?query=apis)
↓

LLM Fundamentals (https://www.youtube.com/playlist?list=PLJq-63ZRPdBu38EjXRXzyPat3sYMHbIWU)
↓
Transformer Models and LLMs (https://www.youtube.com/watch?v=Ub3GoFaUcds)
↓
Model Mechanisms (https://www.youtube.com/playlist?list=PLv8Cp2NvcY8BQtdyf29kkoAXpcjaGXHHn)
↓
Tokenization (https://www.youtube.com/watch?v=42Q99S1geQk)
↓
Context Windows (Stanford CME295 Playlist (Lecture 3).)
↓
Token Based Pricing (https://www.youtube.com/watch?v=ZI2PuU6l_ZM)
↓

Generation Controls (Stanford CME295 Playlist (Lecture 3).)
↓
Temperature (https://www.youtube.com/watch?v=42Q99S1geQk)
↓
Frequency Penalty (https://www.youtube.com/watch?v=42Q99S1geQk)
↓
Presence Penalty (https://www.youtube.com/watch?v=42Q99S1geQk)
↓
Top-p (https://www.youtube.com/watch?v=42Q99S1geQk)
↓
Stopping Criteria (https://www.youtube.com/playlist?list=PLoROMvodv4rObv1FMizXqumgVVdzX4_05)
↓
Max Length (https://www.youtube.com/playlist?list=PLoROMvodv4rObv1FMizXqumgVVdzX4_05)
↓

Open Weight Models (https://www.youtube.com/playlist?list=PLv8Cp2NvcY8BQtdyf29kkoAXpcjaGXHHn)
↓
Closed Weight Models (https://www.youtube.com/playlist?list=PLoROMvodv4rObv1FMizXqumgVVdzX4_05)

↓
Model Families and Licences (https://www.youtube.com/watch?v=Ru2jEY4pd7k)
↓
Streamed vs Unstreamed Responses (CampusX LangChain Playlist (below).)
↓
Reasoning vs Standard Models (https://www.youtube.com/playlist?list=PLoROMvodv4rObv1FMizXqumgVVdzX4_05)
↓
Fine-tuning vs Prompt Engineering (https://www.youtube.com/@krishnaik06)
↓
Embeddings and Vector Search (https://www.youtube.com/playlist?list=PLaFUiA52BD9EU--p3iHX5kh5XGapkqP04)
↓
Understand the Basics of RAG (https://www.youtube.com/watch?v=ea2W8IogX80)
↓
Understand the Basics
↓
Pricing of Common Models (Generative AI+LLM Full Course (above).)
↓
_____

AI Agents 101 (https://www.youtube.com/playlist?list=PLKnIA16_RmvYsvB8qkUQuJmJNuiCUJFPL)
↓
What are AI Agents?
↓
What are Tools?
↓
Agent Loop
↓
Perception / User Input
↓
Reason and Plan
↓
Acting / Tool Invocation (https://www.youtube.com/playlist?list=PLKnIA16_RmvYsvB8qkUQuJmJNuiCUJFPL)
↓
Observation & Reflection (https://www.youtube.com/playlist?list=PLKnIA16_RmvYsvB8qkUQuJmJNuiCUJFPL)
↓
_____

Example Usecases (https://www.youtube.com/playlist?list=PLKnIA16_RmvYsvB8qkUQuJmJNuiCUJFPL)
↓
Personal Assistant (https://www.youtube.com/playlist?list=PLKnIA16_RmvYsvB8qkUQuJmJNuiCUJFPL)
↓
Code Generation (Krish Naik Channel (GenAI projects).)
↓
Data Analysis ((https://www.youtube.com/@krishnaik06))
↓
Web Scraping / Crawling (CampusX Agentic AI (tool examples).)
↓
NPC / Game AI (TechWithTim search results.)
↓
_____

Prompt Engineering (https://www.youtube.com/watch?v=DvhFcIRRXyI)
↓
What is Prompt Engineering
↓
Writing Good Prompts
↓
Be specific in what you want
↓
Provide additional context

↓
Use relevant technical terms
↓
Use Examples in your Prompt
↓
Iterate and Test your Prompts
↓
Specify Length, format etc
↓

Tools / Actions (CampusX LangGraph Playlist.)
↓
Tool Definition
↓
Name and Description
↓
Input / Output Schema
↓
Error Handling
↓
Usage Examples
↓

Examples of Tools
↓
Web Search
↓
Code Execution / REPL
↓
Database Queries
↓
API Requests
↓
Email / Slack / SMS
↓
File System Access
↓

Agent Memory -- (youtube - campusx(https://www.youtube.com/@campusx-official))
↓
What is Agent Memory?
↓
Episodic vs Semantic Memory
↓
Short Term Memory
↓
Within Prompt
↓
Long Term Memory
↓
Vector DB / SQL / Custom
↓

Maintaining Memory (CampusX Playlist.)
↓
RAG and Vector Databases (RAG Course Playlist (above).)
↓
User Profile Storage
↓
Summarization / Compression
↓
Forgetting / Aging Strategies (CampusX Memory videos.)

↓
_____
Agent Architectures (CampusX LangGraph Playlist.)
↓
ReAct (Reason + Act)
↓
Chain of Thought (CoT) (Prompt Engineering Course.)
↓
RAG Agent (RAG Playlist.)
↓
_____
Other Architecture Patterns (CampusX Playlist.)
↓
Planner Executor
↓
DAG Agents
↓
Tree-of-Thought
↓
_____
Model Context Protocol (MCP) - (youtube - campusx(https://www.youtube.com/@campusx-official))
↓
Core Components
↓
MCP Hosts
↓
MCP Client
↓
MCP Servers
↓
Creating MCP Servers
↓
_____
Deployment Modes
↓
Local Desktop
↓
Remote / Cloud
↓
_____
Building Agents (CampusX Agentic AI Playlist)
↓
Manual (from scratch)
↓
Direct LLM API calls
↓
Implementing the agent loop - (youtube campux(https://www.youtube.com/@campusx-official)) -- langchain/ langgraph
↓
Parsing model output
↓
Error & Rate-limit handling
↓
_____
LLM Native Function Calling (CampusX LangGraph (function calling modules).)
↓
OpenAI Functions Calling
↓
Gemini Function Calling (Krish Naik videos.)
↓
OpenAI Assistant API (CampusX.)
↓
Anthropic Tool Use

```
        ↓
_______________________
Building Using Frameworks
        ↓
LangChain                       -- ( youtube -
campusx(https://www.youtube.com/@campusx-official))
        ↓
LlamaIndex                        (Krish Naik / CampusX cross-references)
        ↓
Haystack
        ↓
AutoGen
        ↓
CrewAI
        ↓
Smol Depot
        ↓
_______________________
Evaluation and Testing
        ↓
Metrics to Track
        ↓
Unit Testing for Individual Tools
        ↓
Integration Testing for Flows
        ↓
Human in the Loop Evaluation
        ↓
_______________________
Frameworks
        ↓
LangSmith
        ↓
Ragas
        ↓
DeepEval
        ↓
_______________________
Debugging and Monitoring
        ↓
Structured Logging & Tracing
        ↓
Observability Tools
        ↓
    LangSmith
        ↓
    Helicone
        ↓
    LangFuse
        ↓
    openllmetry
        ↓
_______________________
Security & Ethics -- (https://www.youtube.com/shorts/ld1zCXhnJCY)
        ↓
Prompt Injection / Jailbreaks
        ↓
Tool Sandboxing / Permissioning
        ↓
Data Privacy + PII Redaction
        ↓
Bias & Toxicity Guardrails
        ↓
Safety + Red Team Testing
```

Related Roadmaps :

AI Engineer Roadmap
Data Scientist Roadmap
MLOps Roadmap
AI Red Teaming Roadmap
Prompt Engineering Roadmap

Certificates :

Paid :

IBM Generative AI Engineering - (https://www.coursera.org/professional-certificates/ibm-generative-ai-engineering)

Google AI Essentials - (https://grow.google/ai-essentials/)

Role : AI Engineer

Skills : Python, SQL, Data Wrangling (Data Cleaning, Data Preprocessing), Data Visualization, Pandas, NumPy,
Matplotlib, Seaborn, Scikit-learn, Supervised Learning, Unsupervised Learning, Ensemble Methods,
Feature Engineering, Model Evaluation, Deep Learning, Neural Networks, Convolutional Neural Networks (CNN),
Recurrent Neural Networks (RNN), LSTM, Transformers, Transfer Learning, Large Language Models (LLMs),
Prompt Engineering, Retrieval-Augmented Generation (RAG), Logging, AI Safety & Ethics, Debugging
Embeddings, Vector Search, Vector Databases, AI Agent Design, Tool & Function Calling, API Integration,
LangChain, LlamaIndex, AutoGen, CrewAI, OpenAI API, Function calling, Hugging Face, Monitoring
MLOps, Model Deployment, CI/CD for Machine Learning, Model Lifecycle Management

Learning Roadmap :

Pre-requisites (Choose One)
↓
Frontend (https://www.youtube.com/watch?v=1Eq4El-XpTg)- Frontend modules for AI apps.
↓
OR
↓
Backend (https://www.youtube.com/watch?v=1Eq4El-XpTg)- Backend sections focused on AI integration.
↓
OR
↓
Full-Stack (https://www.youtube.com/watch?v=1Eq4El-XpTg)
↓
_____
Introduction (https://www.youtube.com/watch?

v=pSVk-5WemQ0&list=PLKnIA16_RmvaTbihpo4MtzVm4XOQa0ER0)
↓
What is an AI Engineer?
↓
AI Engineer vs ML Engineer
↓
Large Language Models (LLMs)
↓
Inference
↓
Training
↓
Embeddings
↓
Vector Databases
↓
RAG
↓
Prompt Engineering
↓
AI Agents
↓
AI vs AGI
↓
Common Terminology
↓
_____
Impact on Product Development
↓
Roles and Responsibilities (https://www.youtube.com/watch?v=ILrYwyPd1Dc)
↓
_____
Using Pre-trained Models (CampusX GenAI Playlist)
↓
Pre-trained Models
↓
Benefits
↓
Limitations & Considerations
↓
Popular AI Models
↓
OpenAI Models
↓
Capabilities / Context Length
↓
Cut-off Dates / Knowledge
↓
Anthropic Claude (mcp - campusx(https://www.youtube.com/@campusx-official))
↓
Google Gemini
↓
Azure AI
↓
AWS SageMaker (https://www.youtube.com/playlist?list=PLKnIA16_RmvYsvB8qkUQuJmJNuiCUJFPL)
↓
Hugging Face Models (CampusX Open Source AI videos.)
↓
Mistral AI
↓
Cohere
↓

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_________________________
OpenAI Platform                 (Krish Naik OpenAI Series (above).)
        ↓
OpenAI API
        ↓
Chat Completions API
        ↓
Writing Prompts
        ↓
OpenAI Playground
        ↓
Fine-tuning                      (youtube -
campusx(https://www.youtube.com/@campusx-official))
        ↓
Managing Tokens
        ↓
Maximum Tokens
        ↓
Token Counting
        ↓
Pricing Considerations
        ↓
_________________________
Prompt Engineering
        ↓
_________________________
AI Safety & Ethics    (https://www.youtube.com/watch?
v=4FFspU4riHk&list=PLKnIA16_RmvYsvB8qkUQuJmJNuiCUJFPL&index=16)
        ↓
AI Safety Issues
        ↓
Prompt Injection Attacks (CampusX Agentic AI Playlist.)
        ↓
Bias & Fairness
        ↓
Security & Privacy
        ↓
Adversarial Testing
        ↓
OpenAI Moderation API
        ↓
End-user IDs
        ↓
Robust Prompt Engineering
        ↓
Know Your Customers / Usecases
        ↓
Constraining Inputs & Outputs
        ↓
Safety Best Practices
        ↓
_________________________
Open Source AI (Krish Naik OpenAI Series.)
        ↓
Open vs Closed Models
        ↓
Popular Open Source Models
        ↓
Hugging Face  (campusx)
        ↓
Hugging Face Hub
        ↓
Hugging Face Tasks
        ↓
```

Finding Models
↓
Using Open Source Models
↓
Inference SDK
↓
Transformers.js
↓
Ollama (campusx)
↓
Ollama Models
↓
Ollama SDK
↓
_____
Embeddings & Vector Databases (youtube - campusx)
↓
What are Embeddings?
↓
Embedding Use Cases
↓
Semantic Search
↓
Recommendation Systems
↓
Anomaly Detection
↓
Data Classification
↓
OpenAI Embeddings API
↓
Embedding Models
↓
Pricing
↓
Open Source Embeddings
↓
Sentence Transformers
↓
Hugging Face Models
↓
Vector Databases
↓
Chroma
↓
Pinecone
↓
Weaviate
↓
FAISS
↓
LanceDB
↓
Qdrant
↓
Supabase
↓
MongoDB Atlas (https://www.mongodb.com/products/platform/atlas-database)
↓
Indexing Embeddings
↓
Similarity Search
↓

Vector Search Implementation
↓

RAG & Implementation (youtube - campusx)
↓
RAG Usecases
↓
RAG vs Fine-tuning
↓
Chunking
↓
Embedding
↓
Vector Database
↓
Retrieval
↓
Generation
↓
Implementing RAG
↓
Using SDKs Directly
↓
LangChain
↓
LlamaIndex
↓
OpenAI Assistant API
↓
Replicate
↓

AI Agents
↓
RAG as an Alternative
↓
Agent Usecases
↓
Prompt Engineering
↓
ReAct Prompting
↓
Manual Implementation
↓
OpenAI Functions / Tools
↓
OpenAI Assistant API
↓
Building AI Agents
↓

Multimodal AI (https://www.google.com/url?sa=E&q=https%3A%2F%2Fwww.shaip.com%2Fblog%2Fmultimodal-ai-the-complete-guide-to-training-data%2F)
↓
Multimodal Usecases (https://www.google.com/url?sa=E&q=https%3A%2F%2Fwww.gdsonline.tech%2Fmultimodal-ai%2F)
↓
Image Understanding
↓
Image Generation
↓
Video Understanding
↓

Audio Processing
↓
Text-to-Speech
↓
Speech-to-Text
↓
OpenAI Vision API(https://www.google.com/url?sa=E&q=https%3A%2F%2Fplatform.openai.com%2Fdocs%2Fguides%2Fimages-vision)
↓
DALL·E API (https://docs.aimlapi.com/api-references/image-models/openai/dall-e-3)
↓
Whisper API
↓
Hugging Face Models (https://www.google.com/url?sa=E&q=https%3A%2F%2Fhuggingface.co%2Flearn%2Fcomputer-vision-course%2Funit4%2Fmultimodal-models%2Ftasks-models-part1)
↓
LangChain for Multimodal Apps
↓
LlamaIndex for Multimodal Apps (https://www.google.com/url?sa=E&q=https%3A%2F%2Fdevelopers.llamaindex.ai%2Fpython%2Fexamples%2Fmulti_modal%2Fmulti_modal_videorag_videodb%2F)
↓
Implementing Multimodal AI
↓
_____

Development Tools
↓
AI Code Editors (https://www.google.com/url?sa=E&q=https%3A%2F%2Fsealos.io%2Fblog%2Ftop-ai-code-editors)
↓
Code Completion Tools
↓
_____

Scrimba – AI Engineer Path (https://www.google.com/url?sa=E&q=https%3A%2F%2Fscrimba.com%2Flearn%2Faiengineer)
↓
_____

Related Roadmaps :

AI & Data Scientist Roadmap

Prompt Engineering Roadmap

Data Analyst Roadmap

Certificates :

free:

Oracle Gen AI Professional - (https://mylearn.oracle.com/ou/learning-path/become-an-oci-generative-ai-professional-2025/147863)

paid :

AWS AI Practitioner - (https://aws.amazon.com/certification/certified-ai-practitioner/)

Role : Data Scientist

Skills : Python, Sql, Data Wrangling(Data Cleaning, Data Preprocessing, Data Visualization),Supervised Learning,
Unsupervised Learning, Pandas, Numpy, Seaborn, scikit learn, matplotlib, deep learning, CNNs, RNNs, LSTM,
Transformers, Transfer Learning, MLOps, Model Deployment, CI/CD for Machine Learning, Model Lifecycle
Management, Monitoring

Learning Roadmap :

Mathematics (https://www.youtube.com/playlist?list=PLKnIA16_RmvbYFaaeLY28cWeqV-3vADST)
↓
Linear Algebra
↓
Calculus
↓
Differential Calculus
↓
Mathematical Analysis
↓
Mathematics for Machine Learning
↓
_____
Statistics
↓
Probability
↓
Sampling
↓
Central Limit Theorem (CLT)
↓
Hypothesis Testing
↓
A/B Testing
↓
Experiment Design
↓
Test Sensitivity Analysis
↓
Minimum Detectable Effect (MDE)
↓
CUPED / CUPAC
↓
Stratification
↓
Ratio Metrics
↓
Delta Method
↓
_____
Econometrics
↓
Regression Analysis
↓
Time Series Analysis
↓
Distribution Fitting
↓
ARIMA Models
↓

Forecasting Techniques
↓
_____
Programming
↓
Python Programming (https://www.youtube.com/playlist?list=PLKnIA16_Rmvb1RYR-iTA_hzckhdONtSW4)
↓
_____
Data Structures & Algorithms
↓
Data Structures (https://www.youtube.com/playlist?list=PLdo5W4Nhv31bbKJzrsKfMpo_grxuLl8LU)
↓
Algorithms
↓
Problem Solving
↓
_____
Databases
↓
SQL (https://www.youtube.com/playlist?list=PLKnIA16_RmvYun1_5r9Fb4eQigioPB7yn)
↓
Relational Databases
↓
Query Optimization
↓
_____
Exploratory Data Analysis (EDA) (https://www.youtube.com/playlist?list=PLKnIA16_RmvZAqJzKstVHywcRNMn6pcGD)
↓
Data Understanding
↓
Data Analysis (https://www.youtube.com/playlist?list=PLKnIA16_RmvbR85fgbfVRKOiMokUKVupy)
↓
Data Visualization
↓
Pandas
↓
Seaborn
↓
_____
Machine Learning (https://www.youtube.com/playlist?list=PLKnIA16_Rmvbr7zKYQuBfsVkjoLcJgxHH)
↓
Supervised Learning
↓
Unsupervised Learning
↓
Ensemble Methods
↓
Classical Machine Learning
↓
Pattern Recognition
↓
_____
Deep Learning (https://www.youtube.com/playlist?list=PLKnIA16_RmvYuZauWaPlRTC54KxSNLtNn)
↓
Neural Networks
↓
Fully Connected Networks

↓
Convolutional Neural Networks (CNN)
↓
Recurrent Neural Networks (RNN)
↓
LSTM
↓
Transformers
↓
Transfer Learning
↓
Attention Mechanisms
↓
_____
MLOps (https://www.youtube.com/playlist?list=PLKnIA16_RmvaKHYjy5v0dJh8edeaEWb-b) ----> (https://t.me/campus_x_dsmp2/433)
↓
Model Deployment
↓
CI/CD for Machine Learning
↓
Model Lifecycle Management
↓
Monitoring

Related Roadmaps :

-Data Analytics Roadmap
- ML Engineer Roadmap
- AI Engineer Roadmap

Certificates :

free :

Google Data Analytics Certificate - (www.coursera.org/professional-certificates/google-data-analytics)

Paid :

IBM Data Science Professional Certificate - (https://www.coursera.org/professional-certificates/ibm-data-science?utm_source=chatgpt.com)

Microsoft Azure Data Scientist Associate (DP-100) Exam Prep Professional Certificate - (https://www.coursera.org/professional-certificates/azure-data-scientist?utm_source=chatgpt.com)

Role : AI Red Teaming

Skills : AI Security, Machine Learning Fundamentals, Deep Learning, Large Language Models,
Prompt Engineering, Prompt Injection, Jailbreak Techniques, Safety Filter Bypass, Adversarial Machine Learning,
Adversarial Examples, Data Poisoning, Model Extraction, Model Inversion, Model Weight Stealing,
Model Manipulation, Threat Modeling, Risk Assessment, Vulnerability Assessment, Cybersecurity Fundamentals,

Confidentiality Integrity Availability (CIA), Secure System Design, API Security, Authentication, Authorization,
Infrastructure Security, Code Injection, Insecure Deserialization, Remote Code Execution, Black Box Testing,
White Box Testing, Grey Box Testing, Automated Security Testing, Manual Security Testing, Continuous Testing,
Red Team Simulations, LLM Security Testing, Agentic AI Security, Responsible Disclosure, Defensive AI Strategies,
Adversarial Training, Robust Model Design, Monitoring, Logging, Incident Response, Reporting, Ethical Hacking,
Security Research

Learning Roadmap :

Introduction

AI Red Teaming (https://www.youtube.com/playlist?list=PLlrxD0HtieHhXnVUQM42aKRPrirbUIDdh)
↓
Why Red Team AI Systems
↓
Ethical Considerations
↓
Role of Red Teams

Foundational Knowledge

AI / ML Fundamentals (https://www.youtube.com/playlist?list=PLKnIA16_Rmvbr7zKYQuBfsVkjoLcJgxHH)
Supervised Learning
↓
Unsupervised Learning
↓
Reinforcement Learning
↓
Neural Networks
↓
Generative Models (https://www.youtube.com/watch?v=pSVk-5WemQ0&list=PLKnIA16_RmvaTbihpo4MtzVm4XOQa0ER0)
↓
Large Language Models
↓
Prompt Engineering

Cybersecurity Fundamentals (https://www.youtube.com/@AtlantiumAI/search?query=cyber%20security)

Confidentiality, Integrity, Availability (CIA)
↓
Threat Modeling
↓
Risk Management
↓
Vulnerability Assessment

Prompt Hacking

Prompt Injection (Microsoft AI Red Teaming 101 Playlist –

Episodes 4–6.)
↓
Direct Prompt Injection
↓
Indirect Prompt Injection
↓
Jailbreak Techniques (Microsoft AI Red Teaming 101 Playlist (above) – Jailbreak deep dives.)
↓
Safety Filter Bypasses
↓
Prompt Defense & Countermeasures (Microsoft AI Red Teaming 101 Playlist (above) – Defense episodes.)

Model Vulnerabilities

Unauthorized Access (https://www.youtube.com/playlist?list=PL-D1lWyrp4YOmiVHvJbJrEn19OvkjZ1E_)
↓
Model Extraction
↓
Model Weight Stealing
↓
Data Poisoning (Microsoft AI Red Teaming 101 Playlist (above) – Poisoning episodes.)
↓
Adversarial Examples (Jarno Baselier AI Red Team Academy Playlist (above) – Adversarial modules.)
↓
Model Inversion
↓
Model Manipulation
↓
Adversarial Training
↓
Robust Model Design
↓
Continuous Monitoring
↓
Defense Strategies (Microsoft AI Red Teaming 101 Playlist (above) – Defense strategies.)

System Security

Code Injection
↓
Insecure Deserialization
↓
Remote Code Execution (AtlantiumAI Cybersecurity Search)
↓
API Security
↓
Authentication & Authorization
↓
Infrastructure Security

Testing Methodologies

Black Box Testing (Microsoft AI Red Teaming 101 Playlist (above) – Black box sections.)
↓
White Box Testing (Jarno Baselier AI Red Team Academy Playlist (above).)
↓
Grey Box Testing (Microsoft AI Red Teaming 101 Playlist (above).)
↓
Manual Testing
↓
Automated Testing
↓
Continuous Testing

Tools & Frameworks

Testing Platforms (Jarno Baselier AI Red Team Academy Playlist (above) – Tools modules.)
↓
Monitoring Solutions (Microsoft AI Red Teaming 101 Playlist (above).)
↓
Benchmark Datasets (Jarno Baselier AI Red Team Academy Playlist (above).)
↓
Custom Testing Scripts (Microsoft AI Red Teaming 101 Playlist (above) – Scripts in labs.)
↓
Reporting & Documentation Tools (Microsoft AI Red Teaming 101 Playlist (above).

Real-World Applications (bit.ly/llm-security-monitor)

LLM Security Testing (Microsoft AI Red Teaming 101 Playlist (above) – LLM-specific testing.)
↓
Agentic AI Security (Microsoft AI Red Teaming 101 Playlist (above) – Agentic modules.)
↓
Responsible Disclosure (Microsoft AI Red Teaming 101 Playlist (above) – Disclosure practices.)

Professional Development (www.asrg.io)

Hands-on Labs (Microsoft AI Red Teaming 101 Playlist (above) – Lab walkthroughs.)
↓
CTF Challenges (Jarno Baselier AI Red Team Academy Playlist (above) – CTF-style labs.)
↓
Red Team Simulations (Microsoft AI Red Teaming 101 Playlist (above).)
↓
Certifications (Microsoft AI Red Teaming 101 Playlist (above) – Certification prep.)
↓
Industry Research
↓

Conferences & Communities

_____

Future Directions (learning.acm.org)

Emerging AI Threats

Advanced Attack Techniques

AI Security Research

Industry Standards & Policies

Related Roadmaps :

AI and Data Scientist Roadmap
AI Engineer Roadmap
Data Analyst Roadmap
MLOps Roadmap

Certificates :

free :

Generative AI for Penetration Testing: Red Team (https://www.coursera.org/learn/generative-ai-for-penetration-testing-red-team?utm_source=chatgpt.com)

AI Security for Developers – Principles & Practice (https://hiruva.org/?utm_source=chatgpt.com)

paid :

Certified AI Red Teaming Professional (CAIRTP) (https://www.tonex.com/training-courses/certified-ai-red-teaming-professional-cairtp/?utm_source=chatgpt.com)

Role : Android Developer

Skills : Kotlin, Java, Android Studio, Object-Oriented Programming (OOP), Data Structures and Algorithms, Gradle,
Git, GitHub, GitLab, Bitbucket, Android App Components, Activities, Services, Broadcast Receivers,
Content Providers, Intents, Activity Lifecycle, Android Layouts, ConstraintLayout, RecyclerView,
UI Components, Fragments, Jetpack Compose, Navigation Components, Android Architecture Patterns (MVVM, MVI, MVP, MVC),
Design Patterns, Repository Pattern, Factory Pattern, Builder Pattern, Observer Pattern, LiveData, Flow,
RxJava, RxKotlin, Dependency Injection, Dagger, Hilt, Koin, Kodein, SharedPreferences, DataStore,
Room Database, File Storage, REST APIs, Retrofit, OkHttp, Apollo GraphQL, Coroutines, Multithreading,
WorkManager, Firebase, Authentication, Cloud Messaging, Firestore, Crashlytics, Remote Config, Google Maps,
Google Play Services, AdMob, Linting, Ktlint, Detekt, Unit Testing,

JUnit, UI Testing, Espresso, Debugging,
Timber, LeakCanary, Chucker, Performance Benchmarking, APK Signing, App Distribution, Firebase App Distribution,
Google Play Store Deployment

Learning Roadmap :

Pick a Language
↓
Kotlin - (https://www.youtube.com/@PhilippLackner/search?query=kotlin)
↓
_____
The Fundamentals (https://www.youtube.com/watch?v=blKkRoZPxLc)
↓
Development IDE (Android Studio)
↓
Basics of Kotlin
↓
Object-Oriented Programming (OOP) (*https://www.youtube.com/watch?v=yRpLlJmRo2w&list=PLfqMhTWNBTe3LtFWcvwpqTkUSlB32kJop)
↓
Data Structures & Algorithms (https://www.youtube.com/playlist?list=PLQkwcJG4YTCQTEk4J4btiOJBV0PhKjJVS)
↓
Gradle (Build System)
↓
Create a Basic Hello World App
↓
_____
Version Control - (https://www.youtube.com/@ProgrammingKnowledge/search?query=git%20and%20github)
↓
Git
↓
VCS Hosting
↓
GitHub / Bitbucket / GitLab
↓
_____
App Components (https://www.youtube.com/watch?v=BePWJavW_w4&list=PLTV_nsuD2lf4UCTV6xwvNPvFdmCNKyhc8)
↓
Activity and
↓
Services (https://www.youtube.com/@PhilippLackner/search?query=android%20)
↓
Content Provider (Modern Android App Components Playlist (above).)
↓
Broadcast Receiver
↓
Intents
↓
Implicit Intents → Explicit Intents → Intent Filters
↓
Activity Lifecycle
↓
State Changes
↓
Tasks & Back Stack
↓

_____
Interface & Navigation (https://www.youtube.com/watch?v=6_wK_Ud8--0)
↓
Layouts
↓
FrameLayout
↓
LinearLayout
↓
RelativeLayout
↓
ConstraintLayout
↓
RecyclerView
↓
UI Elements
↓
TextView
↓
EditText
↓
Buttons
↓
ImageView
↓
ListView
↓
Drawer
↓
Tabs
↓
Toast
↓
Dialogs
↓
Bottom Sheet
↓
Animations
↓
Fragments
↓
Jetpack Compose
↓
App Shortcuts
↓
Navigation Components (https://www.youtube.com/playlist?list=PLQkwcJG4YTCRjyfVKB8vcK7zeC1BNmBz4)
↓
_____
Design & Architecture (Android & Kotlin Development Masterclass (above) – Architecture sections.)
↓
Architectural Patterns
↓
MVC
↓
MVP
↓
MVVM
↓
MVI
↓
Design Patterns
↓

Repository Pattern
↓
Builder Pattern
↓
Factory Pattern
↓
Observer Pattern
↓
Reactive & State Management (https://www.youtube.com/playlist?list=PLQkwcJG4YTCSVDhww92llY3CAnc_vUhsm)
↓
LiveData
↓
Flow
↓
RxJava (Android & Kotlin Development Masterclass (above).)
↓
RxKotlin
↓

Dependency Injection
↓
Dagger
↓
Hilt
↓
Koin
↓
Kodein
↓

Storage
↓
SharedPreferences
↓
DataStore
↓
Room Database
↓
File System
↓

Networking
↓
Retrofit
↓
OkHttp
↓
Apollo Android (GraphQL)
↓

Asynchronism & Background Work (Coroutines & Flows Masterclass (above).)
↓
Threads
↓
Coroutines
↓
RxJava / RxKotlin (Android & Kotlin Development Masterclass (above).)
↓
WorkManager
↓

Common Services (Android & Kotlin Development Masterclass (above).)
↓
Firebase
↓
Authentication
↓
Crashlytics
↓
Remote Config
↓
Cloud Messaging
↓
Firestore
↓
Google AdMob
↓
Google Play Services
↓
Google Maps
↓

Linting & Code Quality (Android & Kotlin Development Masterclass (above).)
↓
Ktlint
↓
Detekt
↓

Testing (Android & Kotlin Development Masterclass (above).)
↓
JUnit
↓
Espresso
↓

Debugging & Performance (Android & Kotlin Development Masterclass (above).)
↓
Timber
↓
LeakCanary
↓
Chucker
↓
Jetpack Benchmark
↓

Distribution (Android & Kotlin Development Masterclass (above).)
↓
Signed APK
↓
Firebase App Distribution
↓
Google Play Store
↓
Other Distribution Methods

Related Roadmaps :

React Native Developer

Flutter Developer

iOS Developer

Certificates :

free :

Google Android Basics
(https://developer.android.com/courses)

paid :

Meta Android Developer Professional Certificate
(https://www.coursera.org/professional-certificates/meta-android-developer)

Android App Development with Kotlin
(https://www.coursera.org/specializations/android-app-development)

Role : Angular Developer

Skills : JavaScript, TypeScript, Angular Framework, Angular Architecture, Angular CLI, Component-Based Architecture,
Standalone Components, Component Lifecycle, Data Binding, Property Binding, Event Binding, Two-Way Binding,
Template Syntax, Directives, Pipes, Custom Pipes, Routing, Lazy Loading, Route Guards, Reactive Forms,
Template-Driven Forms, Typed Forms, Custom Validators, Control Value Accessor, Dependency Injection,
Services, HTTP Client, REST APIs, Interceptors, RxJS, Observables, Operators, State Management, NgRx,
NGXS, Signals, Zoneless Angular, Change Detection, Performance Optimization, Server-Side Rendering (SSR),
Static Site Generation (SSG), Security (XSS, CSRF, Sanitization), Accessibility (ARIA), Internationalization (i18n),
Animations, Unit Testing, Integration Testing, Jasmine, Karma, Debugging, Angular DevTools, Build Optimization, Deployment

Learning Roadmap :

JavaScript Roadmap
↓
_____
Introduction to Angular (https://www.youtube.com/playlist?list=PLFbEoEbDAzru5xYmFiCXx5RZVVzurqjyc)
↓
Angular Overview & History
↓
Angular Architecture and security
(https://www.youtube.com/@LearnCodeExpress/search?query=Angular%20Security%20Deep%20Dive)
↓
Set Up a New Angular Project
↓
_____
TypeScript Basics (https://www.youtube.com/playlist?list=PLC3y8-

rFHvwhBRAgFinJR8KHIrCdTkZcZ)
↓
———————————————————
Components (Angular Complete Course 2026 Playlist (above) – Components module.)
↓
Component Anatomy
↓
Selector
↓
Template
↓
Styles
↓
Imports
↓
Providers / viewProviders (Angular Complete Course 2026 Playlist (above) – Dependency injection section.)
↓
Change Detection
↓
View Encapsulation
↓
Standalone Components
↓
Metadata
↓
Component Communication
↓
Parent ↔ Child
↓
ContentChild
↓
ViewChild
↓
Component Lifecycle
↓
Dynamic Components
———————————————————
Modules
↓
Module Architecture
↓
Creating Modules
↓
Feature Modules
↓
Lazy Loading Modules
↓
Dependencies
↓
Creating Components
↓
———————————————————
Templates
↓
Interpolation
↓
Template Statements
↓
Binding Concepts
↓
Data Binding
↓

Property Binding
↓
Attribute Binding
↓
Event Binding
↓
Two-way Binding
↓
Template Syntax (Angular Complete Course 2026 Playlist (above) – Control flow.)
↓
@if → @else → @else if
↓
@for
↓
@switch → @case → @default
↓
@let
↓
@defer
↓
Template Reference Variables
↓
@Input & @Output
↓

Directives
↓
Structural Directives
↓
Attribute Directives
↓
Custom Directives
↓

Pipes
↓
Built-in Pipes
↓
Change Detection
↓
Pipe Precedence
↓
Custom Pipes
↓

Routing (Angular Complete Course 2026 Playlist (above) – Routing deep dive.)
↓
Route Configuration
↓
Lazy Loading Routes
↓
Router Outlet
↓
Router Links
↓
Router Events
↓
Route Guards
↓

Forms (Angular Complete Course 2026 Playlist (above) – Forms module.)

↓
Reactive Forms
↓
Typed Forms
↓
Template-driven Forms
↓
Dynamic Forms
↓
Custom Validators
↓
Control Value Accessor
↓
_____
Services & Remote Data (Angular Complete Course 2026 Playlist (above) – Services and HTTP.)
↓
Dependency Injection
↓
HTTP Client
↓
Client Setup
↓
Making Requests
↓
HTTP Interceptors
↓
_____
RxJS Basics (https://www.youtube.com/playlist?list=PLUioGv_6G9YLGX1unj51yidt7bzDORy_G)
↓
Observable Pattern (https://www.youtube.com/@LearningPartnerDigital/search?query=RxJS%20Operators%20in%20Angular)
↓
Observable Lifecycle (RxJS for Angular Developers Playlist (above).)
↓
RxJS vs Promises
↓
Operators
↓
Filtering
↓
Rate Limiting
↓
Transformation
↓
Combination
↓
RxJS Interoperability
↓
_____
Signals (Angular Complete Course 2026 Playlist (above) – Signals introduction.)
↓
Inputs as Signals
↓
Queries as Signals
↓
Model Inputs
↓
_____
State Management (https://www.youtube.com/playlist?list=PLsjpRo2EZP1LwheWj3_gUkpH96WpBv14Y)
↓

NGXS (https://www.youtube.com/watch?v=pxuDBb90t0c)
↓
NgRx
↓
Elf
↓

Zones (Angular Complete Course 2026 Playlist (above) – Zones deep dive.)
↓
Zoneless Applications
↓

Developer Tools (https://www.youtube.com/@AngularUniversity/search?query=%20Understanding%20the%20Angular%20Forms%20ControlValueAccessor%20Interface)
↓
Angular CLI (Angular Complete Course 2026 Playlist (above).)
↓
Local Setup
↓
Build Environments
↓
Deployment
↓
End-to-End Testing
↓
Schematics
↓
CLI Builders
↓
Ahead-of-Time (AoT) Compilation
↓
Angular DevTools
↓
Language Service
↓

Libraries (Angular Complete Course 2026 Playlist (above) – Libraries section.)
↓
Using Libraries
↓
Creating Libraries
↓

Server-Side Rendering (SSR) (Angular Complete Course 2026 Playlist (above) – SSR module.)
↓
Static Site Generation (SSG)
↓
AnalogJS (Angular Complete Course 2026 Playlist (above) – AnalogJS overview.)
↓

Security (Angular Complete Course 2026 Playlist (above) – Security deep dive.)
↓
Cross-Site Scripting (XSS)
↓
Sanitization
↓
Trusted Values
↓

Trusted Types Enforcement
↓
HTTP Vulnerabilities
↓
CSRF
↓
HttpClient CSRF
↓
XSRF Protection
↓
Cross-Site Script Inclusion
↓
_____

Accessibility (Angular Complete Course 2026 Playlist (above) – Accessibility module.)
↓
ARIA Attributes
↓
UI Components
↓
Containers
↓
Routing Accessibility
↓
Link Identification
↓
_____

Performance (Angular Complete Course 2026 Playlist (above) – Performance optimization.)
↓
Deferrable Views
↓
Image Optimization
↓
Zone Pollution
↓
Slow Computations
↓
Hydration
↓
_____

Testing (https://www.youtube.com/@TramoTech/search?query=angular%20unit%20testing)

↓
Testing Services (Angular Complete Course 2026 Playlist (above).)
↓
Testing Pipes
↓
Testing HTTP Requests
↓
Services with Dependencies
↓
Component Bindings
↓
Testing Directives
↓
Component Templates
↓
Debugging Tests
↓
Code Coverage
↓
_____

Internationalization (i18n) (Angular Complete Course 2026 Playlist (above) – i18n section.)
↓
Localize Package
↓
Locales by ID
↓
Translation Files
↓
Multiple Locales
↓
_____
Animations (Angular Complete Course 2026 Playlist (above) – Animations module.)
↓
Transitions & Triggers
↓
Complex Sequences
↓
Reusable Animations
↓
Route Transitions (Angular Complete Course 2026 Playlist (above) – Animations module.)

Related Roadmaps :

TypeScript Developer

Frontend Developer

Vue Developer

Node.js Developer

Certificates :

free:

Angular CLI & Core Concepts (https://v17.angular.io/docs)

paid :

Google Front-End Developer Professional Certificate (https://www.coursera.org/professional-certificates/google-ux-design)

Meta Front-End Developer Professional Certificate (Angular + React) (https://www.coursera.org/professional-certificates/meta-front-end-developer)

Role : API Design Developer

Skills : API Design, RESTful APIs, JSON APIs, GraphQL, gRPC, HTTP Protocols, HTTP Methods, HTTP Status Codes, Headers,
URL Query & Path Parameters, CORS, TCP/IP Basics, DNS Basics, HTTP Caching, CRUD Operations, API Versioning,

Pagination, Rate Limiting, Throttling, Idempotency, HATEOAS, Error Handling, RFC 7807 Problem Details,
Authentication, Authorization, JWT, OAuth 2.0, Token-Based Authentication, Session-Based Authentication,
RBAC, ABAC, API Key Management, API Documentation, OpenAPI Specification, Swagger, Postman, API Security,
OWASP API Security Top 10, Input Validation, API Performance Optimization, Caching Strategies, Load Balancing,
Profiling, Monitoring, API Testing, Unit Testing, Integration Testing, Contract Testing, Mocking APIs,
Load Testing, Webhooks, WebSockets, Server-Sent Events, Event-Driven Architecture, Message Queues,
RabbitMQ, Kafka, Microservices Architecture, API Gateways, Compliance (GDPR, CCPA, PCI DSS, HIPAA), PII Handling,
Backend Integration, DevOps Fundamentals

Learning Roadmap :

Learn the Basics (https://www.youtube.com/watch?v=WXsD0ZgxjRw)
↓
What are APIs
↓
HTTP Fundamentals (https://www.youtube.com/watch?v=2ePf9rue1Ao)
↓
HTTP Versions
↓
HTTP Methods
↓
HTTP Status Codes
↓
HTTP Headers
↓
URL, Query & Path Parameters (APIs for Beginners Full Course (above).)
↓
Cookies (HTTP Crash Course (above).)
↓
Content Negotiation (APIs for Beginners Full Course (above).)
↓
CORS (HTTP Crash Course (above).)
↓
TCP / IP Basics (HTTP Crash Course (above) – Network layer.)
↓
DNS Basics
↓
HTTP Caching
↓
———————————————————
Different API Styles (https://www.youtube.com/playlist?list=PLD-mYtebG3X8mcXS3IfRWVy6H7-qKsAel)
↓
Simple JSON APIs
↓
RESTful APIs
↓
SOAP APIs
↓
GraphQL APIs
↓
gRPC APIs
↓
———————————————————
Building JSON / RESTful APIs (https://www.youtube.com/watch?v=7iHl71nt49o)
↓

REST Principles (Senior Engineer API Design (above).)
↓
URI Design
↓
Versioning Strategies
↓
Pagination
↓
Rate Limiting
↓
Idempotency
↓
HATEOAS
↓
CRUD Operations
↓
Error Handling
↓
RFC 7807 – Problem Details for APIs (Senior Engineer API Design (above) – Error section.)
↓
_____

API Authentication & Authorization (https://www.youtube.com/playlist?list=PLPT-em3BBa7uL4Vk2L_Dlcql9QQPlyUxz)
↓
Authentication
↓
Basic Authentication
↓
Token-Based Authentication
↓
JWT
↓
OAuth 2.0
↓
Session-Based Authentication
↓
Authorization
↓
Role-Based Access Control (RBAC)
↓
Attribute-Based Access Control (ABAC)
↓
API Keys & Key Management
↓
_____

API Documentation (https://www.youtube.com/playlist?list=PLUDwpEzHYYLs3DYFqm79fIj2QOzPke_fW)
↓
OpenAPI / Swagger
↓
Postman
↓
Readme.com
↓
Stoplight
↓
_____

API Security (Senior Engineer API Design (above) – Security module.)
↓
Common API Vulnerabilities
↓
API Security Best Practices

↓
Threat Prevention
↓
———————————————————
API Performance (Senior Engineer API Design (above) – Performance section.)
↓
Performance Metrics
↓
Caching Strategies
↓
Load Balancing
↓
Rate Limiting & Throttling
↓
Profiling & Monitoring
↓
Performance Testing
↓
API Performance Best Practices
↓
———————————————————
API Integration Patterns (https://www.youtube.com/playlist?list=PLsjpRo2EZP1LwheWj3_gUkpH96WpBv14Y)
↓
Synchronous APIs
↓
Asynchronous APIs
↓
Event-Driven Architecture
↓
API Gateways
↓
Microservices Architecture
↓
Webhooks
↓
Polling
↓
Batch Processing
↓
Message Queues
↓
RabbitMQ
↓
Kafka
↓
———————————————————
API Testing (https://www.youtube.com/playlist?list=PLUDwpEzHYYLs3DYFqm79fIj2QOzPke_fW)
↓
Unit Testing
↓
Integration Testing
↓
Functional Testing
↓
Load Testing
↓
Mocking APIs
↓
Contract Testing
↓
Error Handling & Retries

↓
_____
Real-Time APIs (https://www.youtube.com/playlist?list=PLsjpRo2EZP1LwheWj3_gUkpH96WpBv14Y)
↓
WebSockets
↓
Server-Sent Events (SSE)
↓
_____
Standards & Compliance (Senior Engineer API Design (above) – Compliance section.)
↓
API Lifecycle Management
↓
GDPR
↓
CCPA
↓
PCI DSS
↓
HIPAA
↓
PII Handling

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Related Roadmaps :

Backend Roadmap

DevOps Roadmap

Full Stack Roadmap

Certificates :

free :

GraphQL Official Learn (https://graphql.org/learn/)

REST API Design – Open Learning (https://www.restapitutorial.com/)

paid :

IBM Backend Development Professional Certificate (https://www.coursera.org/professional-certificates/ibm-backend-development)

AWS Certified Developer – Associate (https://aws.amazon.com/certification/certified-developer-associate/)

Role : ASP.NET Core Developer

Skills : C#, .NET, ASP.NET Core, MVC, REST APIs, Razor Pages, Razor Components, Minimal APIs, Middleware, Filters, Attributes, Configuration Management,

Dependency Injection, Service Lifetimes (Scoped, Transient, Singleton), Entity Framework Core, Code-First Migrations, Lazy Loading, Eager Loading, Explicit Loading, Dapper, NHibernate, Repository Pattern, SQL, Database Design, Stored Procedures, Constraints, Triggers, Redis, Distributed Caching, Memory Cache, Logging, Serilog, NLog, API Security, Authentication, Authorization, JWT, REST Clients, GraphQL, gRPC, SignalR, WebSockets, AutoMapper, Background Services, Hangfire, Quartz, Unit Testing, Integration Testing, End-to-End Testing, Mocking, Microservices Architecture, Message Brokers, RabbitMQ, Kafka, Azure Service Bus, API Gateway, Docker, Kubernetes, CI/CD Pipelines, GitHub Actions, Azure Pipelines, Software Architecture, Blazor, .NET MAUI, Performance Optimization

Learning Roadmap :

Learn C# Basics (https://www.youtube.com/watch?v=ZPxrO1F5JZg)
↓
_____
General Development Skills (https://www.youtube.com/playlist?list=PLXHXaMtGeiEMviyPyVrJYpL6FtYrntipa)
↓
Git (GitHub / GitLab / Bitbucket)
↓
HTTP / HTTPS (https://www.youtube.com/watch?v=2ePf9rue1Ao)
↓
Data Structures & Algorithms
↓
Database Fundamentals (https://www.youtube.com/playlist?list=PLdo4fOcmZ0oXCPdC3fTFA3Z79-eVH3K-s)
↓
SQL Basics
↓
Database Design
↓
Stored Procedures
↓
Constraints & Triggers
↓
.NET & .NET CLI (C# Full Course 2025 (above) – .NET CLI modules.)
↓
_____
ASP.NET Core Basics (https://www.youtube.com/watch?v=AhAxLiGC7Pc) or (https://www.youtube.com/playlist?list=PL0FcjwFb9rU9sAsurKO9mj3gKiK51vd9p)
↓
MVC
↓
REST APIs
↓
Razor Pages
↓
Razor Components
↓
Middleware
↓
Filters & Attributes
↓
App Configuration
↓
StyleCop (ASP.NET Core Full Course (above) – Code quality.)
↓
Minimal APIs
↓
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Object Relational Mapping (ORM) (https://www.youtube.com/watch?v=lOCUFuTyvaE)
↓
Entity Framework Core
↓
Code-First & Migrations
↓
Lazy / Eager / Explicit Loading
↓
Change Tracker
↓
Dapper
↓
NHibernate
↓

Dependency Injection (ASP.NET Core Full Course (above) – DI deep dive.)
↓
Service Lifetimes
↓
Scoped
↓
Transient
↓
Singleton
↓
DI Containers
↓
Microsoft.Extensions
↓
AutoFac
↓
Scrutor
↓

Caching
↓
In-Memory Cache
↓
EF Second-Level Cache
↓
Distributed Cache
↓
Redis
↓
Memcached (ASP.NET Core Full Course (above) – Cache comparisons.)
↓

Databases (https://www.youtube.com/watch?v=GDkA28nu9SM)
↓
Relational Databases
↓
SQL Server
↓
PostgreSQL
↓
MySQL / MariaDB
↓
NoSQL Databases (Entity Framework Core Full Tutorial (above) – NoSQL extensions.)
↓
MongoDB

(https://learn.mongodb.com/learning-paths/introduction-to-mongodb)
↓
Cassandra
↓
CouchDB
↓
Search Engines (Entity Framework Core Full Tutorial (above) – Search integration.)
↓
Elasticsearch
↓
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Logging (ASP.NET Core Full Course (above) – Logging setup.)
↓
Serilog
↓
NLog
↓
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API Communication (https://www.youtube.com/playlist?list=PLdo4fOcmZ0oXCPdC3fTFA3Z79-eVH3K-s)
↓
REST APIs
↓
OData
↓
GraphQL
↓
gRPC
↓
Real-Time Communication
↓
SignalR
↓
WebSockets
↓
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Object Mapping
↓
AutoMapper
↓
Mapperly
↓
Manual Mapping
↓
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Background Tasks (https://www.youtube.com/playlist?list=PLsjpRo2EZP1LwheWj3_gUkpH96WpBv14Y)
↓
Hosted Services
↓
Hangfire
↓
Quartz
↓
Coravel
↓
———————————————————
Testing (https://www.youtube.com/playlist?list=PLdo4fOcmZ0oXCPdC3fTFA3Z79-eVH3K-s)
↓
Unit Testing
↓

xUnit / NUnit / MSTest
↓
Integration Testing
↓
WebApplicationFactory
↓
Test Containers
↓
Behavior Testing
↓
SpecFlow
↓
End-to-End Testing
↓
Playwright / Cypress
↓
Mocking & Fake Data
↓
Moq / NSubstitute / Bogus
↓
———————————————————
Microservices (https://www.youtube.com/watch?v=DgVjEo3OGBI)
↓
Message Brokers
↓
RabbitMQ
↓
Kafka
↓
Azure Service Bus
↓
Message Bus
↓
MassTransit
↓
API Gateway
↓
Ocelot
↓
YARP
↓
Containerization
↓
Docker
↓
Kubernetes
↓
———————————————————
CI / CD
↓
GitHub Actions
↓
Azure Pipelines
↓
GitLab CI
↓
CircleCI
↓
———————————————————
Architecture & Advanced Topics (https://www.youtube.com/watch?v=4I07X_EGwTY)
↓
Blazor (https://www.youtube.com/watch?v=RBVIclt4sOo)

↓
.NET MAUI (Architecture & Advanced Topics (above) – MAUI section.)
↓
Distributed Systems
↓
Dapr
↓
Orleans
↓
.NET Aspire

Related Roadmap :

Backend Roadmap

DevOps Roadmap

Microservices Roadmap

Certificates :

free :

Entity Framework Core Fundamentals (https://learn.microsoft.com/en-us/ef/core/)

paid :

Microsoft Certified: Azure Developer Associate (AZ-204) (https://learn.microsoft.com/en-us/credentials/certifications/azure-developer/?practice-assessment-type=certification)

Coursera – ASP.NET Core Specialization (https://www.coursera.org/search?query=asp.net%20core)

Role : AWS Cloud Engineer

Skills : Cloud Computing, AWS Core Services, IAM, Identity and Access Management, VPC, Networking, Subnets,
Route Tables, Security Groups, EC2, Auto Scaling, Load Balancing, AMIs, S3, Object Storage, Lifecycle Policies,
CloudFront, CDN, Route 53, DNS Management, CloudWatch, Monitoring, Logging, RDS, DynamoDB, NoSQL Databases,
ElastiCache, Redis, Containerization, Docker, ECS, EKS, Kubernetes, Serverless Computing, AWS Lambda,
API Gateway, EventBridge, Infrastructure Security, High Availability, Fault Tolerance, Backup and Recovery,
Cost Optimization, Well-Architected Framework, CI/CD Basics, DevOps Fundamentals

Learning Roadmap :

Introduction to Cloud Computing (https://www.youtube.com/watch?v=NhDYbskXRgc) / (https://www.youtube.com/watch?v=UmQnenLf_Cs)
↓

IaaS vs PaaS vs SaaS (https://www.youtube.com/watch?v=_SbDJvHreLM)
↓
Public vs Private vs Hybrid Cloud (https://www.youtube.com/watch?v=rsGqAOD1rRM) / (https://www.youtube.com/watch?v=ogz7b2DE32o)
↓
AWS Global Infrastructure (https://www.youtube.com/watch?v=UuRX2gK0IYw)
↓
Shared Responsibility Model (https://www.youtube.com/watch?v=WopTpel6DUY)
↓
Well-Architected Framework (https://www.youtube.com/playlist?list=PLIpE1hDb3gF1epqzvj28WH8zeLB5Plx-H)
↓
IAM (Users, Roles, Policies) (https://www.youtube.com/playlist?list=PL9nWRykSBSFjJK9mFrIP_BPWaC0hAL9dZ)
↓
VPC (CIDR, Subnets, Route Tables, Security Groups, IGW, NAT) (https://www.youtube.com/watch?v=g2JOHLHh4rI)
↓
EC2 (Instances, Storage, Keypairs, Elastic IP, User Data, Pricing) (https://www.youtube.com/watch?v=4dscVzCaXCU)
↓
S3 (Buckets, Objects, Lifecycle, Storage Classes) (https://www.youtube.com/watch?v=A2N9OIun9dU)
↓
SES (Email Sending, DKIM, Limits, Reputation) (https://www.youtube.com/watch?v=tMDRWYggKNs)
↓
Route 53 (Hosted Zones, Routing Policies, Health Checks) (https://www.youtube.com/watch?v=_Fuws3Qivi8)
↓
CloudWatch (Metrics, Logs, Events) (https://www.youtube.com/playlist?list=PLCUXNZJ_mzoLinjfmcJirvVmRDEKg2bfd)
↓
RDS (DB Instances, Storage Types, Backup & Restore) (https://www.youtube.com/watch?v=rM_c7K0-tC0)
↓
DynamoDB (Tables, Keys, Indexes, Streams, Capacity, Backup) (https://www.youtube.com/watch?v=c3Cn4xYfxJY)
↓
ElastiCache (Redis, Memcached) (https://www.youtube.com/watch?v=c3Cn4xYfxJY)
↓
ECS (Clusters, Tasks, Services, Auto Scaling) (https://www.youtube.com/watch?v=c3Cn4xYfxJY)
↓
EKS (Kubernetes on AWS) (https://www.youtube.com/playlist?list=PLiMWaCMwGJXnKY6XmeifEpjIfkWRo9v2l)
↓
Serverless with Lambda (https://www.youtube.com/playlist?list=PL9nWRykSBSFjodfc8l8M8yN0ieP94QeEL)
↓
API Gateway & Lambda@Edge (https://www.youtube.com/watch?v=c3Cn4xYfxJY)
↓
CloudFront (CDN, Distributions, Invalidations) (https://www.youtube.com/watch?v=c3Cn4xYfxJY)
↓
Auto Scaling & Load Balancing (https://www.youtube.com/watch?v=c3Cn4xYfxJY)
↓
Build & Deploy a Complete AWS-Based Application (https://www.youtube.com/watch?v=zA8guDqfv40)

Related Roadmaps :

DevOps Roadmap
↓
Kubernetes Roadmap
↓
Backend Roadmap

Certificates :

free :

AWS Skill Builder (Free Hands-On Labs)
(https://skillbuilder.aws/)

paid :

AWS Certified Cloud Practitioner
(https://aws.amazon.com/certification/certified-cloud-practitioner/)

AWS Certified Solutions Architect – Associate
(https://aws.amazon.com/certification/certified-solutions-architect-associate/)

Role : Backend Developer

Skills : Backend Development, Internet Fundamentals, HTTP, HTTPS, DNS, Hosting, REST APIs, GraphQL, gRPC, JSON APIs,
OpenAPI Specification, HATEOAS, Authentication, Authorization, JWT, OAuth 2.0, OpenID Connect, SAML,
Session Management, Cookie-Based Authentication, Relational Databases, PostgreSQL, MySQL, SQL Server,
Database Design, ACID Properties, Transactions, ORMs, Indexing, Query Optimization, Caching, Redis, CDN,
Web Security, SSL/TLS, CORS, CSP, OWASP Top 10, API Security, Hashing Algorithms, bcrypt, scrypt, SHA,
Unit Testing, Integration Testing, Functional Testing, CI/CD Pipelines, Software Architecture,
Monolithic Architecture, Microservices, Serverless Architecture, Domain-Driven Design (DDD), CQRS,
Event Sourcing, Containerization, Docker, Kubernetes, Web Servers, Nginx, Apache, Message Brokers,
RabbitMQ, Kafka, WebSockets, Server-Sent Events, NoSQL Databases, MongoDB, Cassandra, DynamoDB,
Search Engines, Elasticsearch, System Scaling, Load Balancing, Sharding, Replication, Observability,
Monitoring, Logging, Metrics, DevOps Fundamentals

Learning Roadmap :

Internet Basics (https://www.youtube.com/playlist?list=PLzdnOPI1iJNfMRZm5DDxco3UdsFegvuB7)
↓
How the Internet Works (https://www.youtube.com/playlist?list=PLzdnOPI1iJNfMRZm5DDxco3UdsFegvuB7)
↓
HTTP / HTTPS (https://www.youtube.com/watch?

v=wW2A5SZ3GkI)
↓
Domains & Hosting (https://www.youtube.com/playlist?list=PLzdnOPI1iJNfMRZm5DDxco3UdsFegvuB7)
↓
DNS (https://www.youtube.com/watch?v=NiQTs9DbtW4)
↓
How Browsers Work (https://www.youtube.com/watch?v=n34UjuPKIYI)
↓
_____
Pick a Backend Language
↓
Python / Java / JavaScript / Go / C# / PHP / Ruby (https://www.youtube.com/watch?v=EN6Dx5q8s8w) / (https://www.youtube.com/results?search_query=python+backend+full+course)

(https://www.youtube.com/watch?v=msT0qX9jE5U) / (https://www.youtube.com/watch?v=YS4e4q9oNsQ)
↓
_____
Version Control (https://www.youtube.com/watch?v=RGOj5yH7evk)
↓
Git (https://www.youtube.com/watch?v=zTjRZNkhiEU)
↓
GitHub / GitLab / Bitbucket (https://www.youtube.com/watch?v=rH3zE7VlIMs)
↓
_____
Relational Databases (https://www.youtube.com/watch?v=HXV3zeQKqGY)
↓
PostgreSQL (https://www.youtube.com/watch?v=qw--VYLpxG4)
↓
MySQL / MariaDB (https://www.youtube.com/watch?v=ER8oKX5myE0)
↓
SQL Server / Oracle (https://www.youtube.com/watch?v=HXV3zeQKqGY)
↓
_____
Authentication & Authorization (https://www.youtube.com/watch?v=mbsmsi7l3r4)
↓
Basic Authentication (https://www.youtube.com/watch?v=7iHl71nt49o)
↓
Token-Based Authentication (https://www.youtube.com/watch?v=mbsmsi7l3r4)
↓
JWT (https://www.youtube.com/watch?v=TUconvzw0jw)
↓
OAuth 2.0 (https://www.youtube.com/watch?v=l9EXopnbtKs)
↓
Cookie-Based Authentication (Covered in full auth series above.)
↓
OpenID (Included in OAuth/Enterprise

auth playlists.)
↓
SAML
↓
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APIs (https://www.youtube.com/watch?v=7iHl71nt49o)
↓
REST APIs (https://www.youtube.com/watch?v=7iHl71nt49o)
↓
JSON APIs (Covered in REST series.)
↓
GraphQL (https://www.youtube.com/watch?v=PeAOEAmR0D0)
↓
gRPC (https://www.youtube.com/watch?v=cgaBhA2qt7E)
↓
SOAP (Included in API design course.)
↓
OpenAPI Specification (Covered in senior API design video.)
↓
HATEOAS (Advanced REST topics in playlists.)
↓
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Caching (https://www.youtube.com/results?search_query=caching+backend+full+course)
↓
Server-Side Caching (Covered in above + CloudFront/NGINX sections.)
↓
Client-Side Caching (Covered in above + CloudFront/NGINX sections.)
↓
CDN
↓
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Web Security (https://www.youtube.com/playlist?list=PLxVm-_TFM9JAJTWIAZAh_03_wFwLDWyat)
↓
SSL / TLS (https://www.youtube.com/watch?v=OOYVPKeBmHo)
↓
HTTPS
↓
CORS
↓
CSP
↓
OWASP Top 10 (https://www.youtube.com/playlist?list=PLIbCFt2m6LvK6uYt0JF_WBngkkzgns96R)
↓
API Security Best Practices
↓
Server Security
↓
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Hashing & Encryption (Covered in authentication and security playlists.)
↓
MD5

↓
SHA
↓
bcrypt
↓
scrypt
↓

Testing
(https://www.youtube.com/results?search_query=backend+testing+unit+integration)
↓
Unit Testing
↓
Integration Testing
↓
Functional Testing
↓

CI / CD (https://www.youtube.com/watch?v=Tq0vZU7Hp_M)
↓
Build Pipelines
↓
Automated Testing
↓
Deployment
↓

Advanced Database Concepts
(https://www.youtube.com/watch?v=HXV3zeQKqGY)
↓
ORMs
↓
Normalization
↓
ACID
↓
Transactions
↓
N+1 Problem
↓
Performance Profiling
↓
Failure Modes
↓

Scaling Databases (Included in advanced DB and scaling playlists.)
↓
Indexes
↓
Replication
↓
Sharding
↓
CAP Theorem
↓
Migrations
↓

Software Design & Architecture (https://www.youtube.com/watch?v=le1jS7VkMD4)
↓
Monolithic Architecture

↓
Microservices
↓
Serverless
↓
SOA
↓
Service Mesh
↓
Twelve-Factor Apps
↓
Design Principles
↓
GoF Design Patterns
↓
Domain-Driven Design
↓
Test-Driven Development
↓
CQRS
↓
Event Sourcing
↓
———————————————————
Containerization & Virtualization
(https://www.youtube.com/watch?v=sBBFo2JMpno)
↓
Docker
↓
Kubernetes
↓
LXC
↓
———————————————————
Web Servers (Covered in deployment and DevOps courses.)
↓
Nginx
↓
Apache
↓
Caddy
↓
IIS
↓
———————————————————
Search Engines
↓
Elasticsearch
(https://www.youtube.com/results?search_query=elasticsearch+full+course)
↓
Solr
↓
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Message Brokers
↓
RabbitMQ
(https://www.youtube.com/results?search_query=kafka+rabbitmq+course)
↓
Kafka
↓
———————————————————
Real-Time Communication (Covered in GraphQL and WebSockets playlists.)

↓
WebSockets
↓
Server-Sent Events
↓
Long Polling
↓
Short Polling
↓
GraphQL Subscriptions
↓
_____
NoSQL Databases
(https://www.youtube.com/results?search_query=nosql+databases+full+course)
↓
Document Databases (MongoDB, CouchDB)
↓
Key-Value Stores (Redis, DynamoDB)
↓
Realtime Databases (Firebase, RethinkDB)
↓
Time-Series Databases (InfluxDB, TimescaleDB)
↓
Column Databases (Cassandra, HBase)
↓
Graph Databases (Neo4j, AWS Neptune)
↓
SQLite
↓
_____
Building for Scale
(https://www.youtube.com/results?search_query=observability+prometheus+grafana)
↓
Vertical & Horizontal Scaling
↓
Migration Strategies
↓
Graceful Degradation
↓
Throttling
↓
Backpressure
↓
Load Shifting
↓
Circuit Breaker
↓
Mitigation Strategies
↓
_____
Observability
(https://www.youtube.com/results?search_query=observability+prometheus+grafana)
↓
Instrumentation
↓
Monitoring
↓
Logging
↓
Metrics
↓
Telemetry
↓
_____

Basic Infrastructure Knowledge
(https://www.youtube.com/watch?v=Tq0vZU7Hp_M)
↓
DevOps Fundamentals

Related Roadmaps:

DevOps Roadmap

Full-Stack Roadmap

GraphQL Roadmap

Certificates :

free :

Postman API Fundamentals Student Expert
(https://academy.postman.com/path/postman-api-fundamentals-student-expert)

paid :

AWS Certified Developer – Associate
(https://aws.amazon.com/certification/certified-developer-associate/)

Google Cloud Professional Backend / Developer
(https://cloud.google.com/learn/certification)

Role : Business Intelligence (BI) Analyst

Skills : Business Intelligence, Data Analysis, KPI Development, Metrics Tracking, Stakeholder Management, Business Acumen,
Descriptive Analytics, Diagnostic Analytics, Predictive Analytics, Prescriptive Analytics, Statistics, Hypothesis Testing,
Regression Analysis, SQL, Advanced SQL Queries, Window Functions, Data Cleaning, Exploratory Data Analysis (EDA),
Data Transformation, Data Visualization, Dashboard Design, Power BI, Tableau, Qlik, Looker, Excel, Python, R, Pandas,
Data Warehousing, Data Modeling, Star Schema, Snowflake Schema, ETL Processes, Airflow, dbt, Cloud Data Warehouses,
AWS, Azure, Google Cloud Platform, Time Series Analysis, Forecasting, A/B Testing, Cohort Analysis, Basic Machine Learning,
Data Storytelling, Presentation Design, Data Governance, Data Quality, GDPR, CCPA, Data Ethics

Learning Roadmap :

Introduction to BI
(https://www.youtube.com/watch?v=a1W6l_sIVXE)
↓
What is BI?
(https://www.youtube.com/watch?v=a1W6l_sIVXE)
↓
Why BI Matters
↓
BI Analyst Roles & Responsibilities
(https://www.youtube.com/watch?v=a1W6l_sIVXE)

↓
BI Analyst vs Other Data Roles
(https://www.youtube.com/watch?v=a1W6l_sIVXE)
↓
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Business Fundamentals
(https://www.youtube.com/watch?v=a1W6l_sIVXE)
↓
Metrics & KPIs
(https://www.youtube.com/watch?v=a1W6l_sIVXE)
↓
Stakeholder Identification (Covered in BI Analyst roles videos above.)
↓
Business Functions
(https://www.youtube.com/watch?v=a1W6l_sIVXE)
↓
Finance → Marketing → Operations → HR
↓
Types of BI (Included in Introduction to BI courses above.)
↓
Operational BI → Tactical BI → Strategic BI
↓
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Types of Data Analysis
(https://www.youtube.com/watch?v=a1W6l_sIVXE)
↓
Descriptive Analysis
↓
Diagnostic Analysis
↓
Predictive Analysis
↓
Prescriptive Analysis
↓
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Statistics Basics
(https://www.youtube.com/playlist?list=PLqzoL9-eJTNBZDG8jaNuhap1C9q6VHyVa)
↓
Variables & Data Types
↓
Central Tendency (Mean, Median, Mode)
↓
Dispersion (Range, Variance, Std, IQR)
↓
Distributions (Skewness, Kurtosis)
↓
Inferential Statistics
↓
Hypothesis Testing → p-value → Confidence Intervals
↓
Correlation vs Causation
↓
Regression Analysis
↓
———————————————————
BI Core Data Concepts
(https://www.youtube.com/watch?v=FwjaHCVNBWA)
↓
What is Data? (Included in core concepts videos.)
↓
Structured / Semi-Structured / Unstructured

↓
Data Sources (Covered in BI tools courses.)
↓
Databases → APIs → Web → Cloud → IoT
↓
Data Formats
↓
CSV → JSON → XML → Excel
↓
———————————————————
SQL (https://www.youtube.com/watch?v=SSKVgrwhzus)
↓
SQL Fundamentals
↓
Advanced Queries
↓
Window Functions
↓
Query Optimization
↓
Databases
↓
PostgreSQL → MySQL → Oracle → SQLite
↓
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Data Cleaning (https://www.youtube.com/watch?v=FwjaHCVNBWA)
↓
Exploratory Data Analysis (Included in statistics and SQL courses.)
↓
Handling Missing Values
↓
Removing Duplicates
↓
Outlier Treatment
↓
Standardization
↓
Tools
↓
Excel → SQL → Pandas → dplyr
↓
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Data Visualization (https://www.youtube.com/watch?v=K3pXnbniUcM)
↓
Visualization Fundamentals
↓
Chart Types
↓
Bar → Line → Scatter → Histogram → Heatmap → Map
↓
Visualization Best Practices
↓
Color Theory → Design Principles → Accessibility
↓
———————————————————
BI Tools
↓
Excel (Included in data analyst bootcamps.)

↓
Power BI
(https://www.youtube.com/watch?v=FwjaHCVNBWA)
↓
Tableau
(https://www.youtube.com/watch?v=K3pXnbniUcM)
↓
Qlik (Covered in comparative BI tool playlists.)
↓
Looker
↓
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Cloud Computing for BI
(https://www.youtube.com/watch?v=a1W6l_sIVXE)
↓
Cloud Basics
↓
Cloud Data Warehouses
↓
AWS → Azure → GCP
↓
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Programming
↓
Python
(https://www.youtube.com/playlist?list=PL-osiE80TeTsWmV9i9c895uyw4uZ7f9c)
↓
R (Covered in statistics playlists.)
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Business Applications
↓
Finance Analytics
(https://www.youtube.com/watch?v=a1W6l_sIVXE)
↓
Marketing Analytics
↓
Operations Analytics
↓
Retail & E-commerce Analytics
↓
Healthcare Analytics
↓
Manufacturing Analytics
↓
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BI Techniques
(https://www.youtube.com/watch?v=LZzq1zSL1bs)
↓
Trend Analysis
↓
Seasonality
↓
Forecasting
↓
Time Series Analysis
↓
A/B Testing
↓
Cohort Analysis
↓
Basic Machine Learning

↓
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Data Architecture
(https://www.youtube.com/watch?v=FwjaHCVNBWA)
↓
Data Warehouse
↓
Data Lake
↓
Data Mart
↓
Data Modeling
↓
Fact vs Dimension Tables
↓
Star Schema → Snowflake Schema
↓
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ETL
(https://www.youtube.com/watch?v=Y_vQyMljDsE)
↓
ETL Fundamentals
↓
Airflow
↓
dbt
↓
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Professional Excellence
(https://www.youtube.com/watch?v=K3pXnbniUcM)
↓
Data Storytelling
↓
Dashboard Design
↓
Executive Reporting
↓
Communication Skills
↓
Stakeholder Management
↓
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Data Governance & Ethics
↓
Data Quality
↓
Data Lineage
↓
Privacy
↓
GDPR → CCPA
↓
Bias & Ethical Data Use
↓
———————————————————
Career Development (https://www.youtube.com/watch?v=FwjaHCVNBWA)
↓
End-to-End BI Project
↓
Portfolio Building

Related Roadmaps :

Data Analyst Roadmap
↓
AI & Data Scientist Roadmap
↓
SQL Roadmap

Certificates :

free :

Microsoft Learn – Power BI Learning Path
(https://learn.microsoft.com/en-us/power-bi/)

paid :

Microsoft Power BI Data Analyst (PL-300)
(https://learn.microsoft.com/en-us/credentials/certifications/data-analyst-associate/?practice-assessment-type=certification)

Google Business Intelligence Professional Certificate
(www.coursera.org/professional-certificates/google-business-intelligence)

Role : Blockchain Developer

Skills : Blockchain Fundamentals, Distributed Ledger Technology, Decentralization, Cryptography, Consensus Algorithms,
Proof of Work, Proof of Stake, Blockchain Architecture, Ethereum, EVM, Smart Contracts, Solidity, Vyper, Rust,
Web3 Development, dApp Development, DeFi, NFTs, DAOs, Token Standards (ERC-20, ERC-721, ERC-1155), Smart Contract Security,
Gas Optimization, Blockchain Testing, Hardhat, Truffle, Foundry, Brownie, OpenZeppelin, Chainlink Oracles, IPFS,
Decentralized Storage, Web3.js, Ethers.js, Wallet Integration, MetaMask, Git, GitHub, Blockchain Scaling,
Layer 2 Solutions, Rollups, Zero Knowledge Proofs, Blockchain Security Auditing

Learning Roadmap :

Basic Blockchain Knowledge (https://www.youtube.com/watch?v=tq-ooflj7R4)
↓
What is Blockchain (https://www.youtube.com/watch?v=SSo_EIwHSd4)
↓
Decentralization → Why it matters (https://www.youtube.com/watch?v=A0ehdko1qIM)
↓
Blockchain Structure
↓
Basic Blockchain Operations
↓
Blockchain Applications & Use Cases
↓
_____
General Blockchain Concepts (https://www.youtube.com/watch?v=A0ehdko1qIM)
↓

Mining & Incentive Models
↓
Trust vs Decentralization
↓
Blockchain Forking
↓
Cryptocurrencies
↓
Crypto Wallets
↓
Cryptography
↓
Consensus Protocols
↓
Blockchain Interoperability
↓
———————————————————
Blockchains
(https://www.youtube.com/watch?v=A0ehdko1qIM)
↓
EVM-Based Blockchains
↓
Ethereum → Polygon → Binance Smart Chain → Avalanche → Fantom
↓
TVM-Based Blockchains
↓
TON → Venom → Everscale
↓
Layer-2 Blockchains
↓
Arbitrum → Optimistic Rollups → zk Rollups
↓
———————————————————
Oracles
(https://www.youtube.com/watch?v=8XzRiZKp_iI)
↓
Hybrid Smart Contracts
↓
Chainlink
↓
Oracle Networks
↓
———————————————————
Smart Contracts
(*https://www.youtube.com/watch?v=-1GB6m39-rM)
↓
Programming Languages
↓
Solidity → Vyper → Rust
↓
Smart Contract Frameworks
↓
Hardhat → Truffle → Foundry → Brownie
↓
Smart Contract Security
↓
Best Practices
↓
Static Analysis & Fuzz Testing
↓
Tools
↓
Slither → MythX → Manticore → Echidna
↓

Testing
↓
Unit Tests → Integration Tests → Code Coverage
↓
Management Platforms
↓
OpenZeppelin → Deployment → Upgrades → Monitoring
↓
ERC Tokens
↓
Crypto Wallets
↓
Decentralized Storage
↓

Version Control
(https://www.youtube.com/watch?v=RGOj5yH7evk)
↓
Git
↓
GitHub → GitLab → Bitbucket
↓
---
dApps (Decentralized Applications)
(https://www.youtube.com/watch?v=A0ehdko1qIM)
↓
Use Cases
↓
DeFi → NFTs → DAOs → Payments → Insurance
↓
Node as a Service
↓
Infura → Alchemy → Moralis → QuickNode
↓
Supporting Languages
↓
JavaScript → Python → Go
↓
Frontend Frameworks
↓
React → Angular → Vue
↓
Client Libraries
↓
ethers.js → web3.js
↓
Client Nodes
↓
Geth → Besu → Nethermind → Substrate
↓
Testing → Deployment → Maintenance → Security
↓

Building for Scale (Included in advanced sections of BootCamp 2025 and Patrick Collins (Layer-2, rollups, ZK).)
↓
State Channels
↓
Payment Channels
↓
Optimistic Rollups
↓
Zero Knowledge Proofs
↓

Plasma
↓
Sidechains
↓
Ethereum 2.0
↓
On-Chain Scaling

Related Roadmaps :

Python Roadmap
↓
Rust Roadmap
↓
Backend Roadmap

Certificates :

Free :

Blockchain Basics – University at Buffalo (Coursera)
(https://www.coursera.org/learn/blockchain-basics)

Ethereum Developer Resources (Official)
(https://ethereum.org/en/developers/)

Paid / Professional :

Ethereum & Solidity: The Complete Developer’s Guide (Udemy)
(https://www.udemy.com/course/ethereum-and-solidity-the-complete-developers-guide/)

Certified Blockchain Developer™ (Blockchain Council)
(https://www.blockchain-council.org/certifications/certified-blockchain-developer/)

Role : CloudFlare Developer

Skills : Cloudflare Workers, Edge Computing, Serverless Architecture, JavaScript, TypeScript, HTTP & Web Protocols,
Request/Response Handling, Fetch API, Cloudflare Runtime APIs, Workers Lifecycle, Service Bindings, Middleware Patterns,
Caching Strategies, Workers KV, R2 Object Storage, D1 SQL Database, Durable Objects, Message Queues, Background Jobs,
Event-Driven Architecture, Cloudflare Pages, Wrangler CLI, Miniflare, CI/CD Pipelines, API Development, Webhooks,
Rate Limiting, Web Security Headers, SSL/TLS, Bot Management, Performance Optimization, Observability, Logging & Monitoring,
AI at the Edge, Vector Databases, Git, GitHub

Learning Roadmap :

Prerequisites
→ JavaScript Fundamentals (https://www.youtube.com/watch?v=PkZNo7MFNFg)

→ TypeScript Fundamentals (https://www.youtube.com/watch?v=BCgEBOJ3y5A)
→ HTTP & Web Protocols (https://www.youtube.com/watch?v=2y1t8q7f4k)
→ Command Line Basics
→ Git & GitHub (https://www.youtube.com/watch?v=RGOj5yH7evk)
→ Node.js & NPM (https://www.youtube.com/watch?v=EN6Dx5q8s8w)
→ Serverless Architecture Basics

Core Cloudflare Workers (https://www.youtube.com/watch?v=H7Qe96fqg1M)
→ Workers Runtime Environment
→ Edge Computing Fundamentals
→ Request / Response Handling
→ Fetch API & Runtime APIs
→ Workers Lifecycle (https://www.youtube.com/watch?v=QTsaAhFvX9o)
→ Service Bindings
→ Middleware Patterns
→ Caching Strategies
→

Frameworks & Tooling (https://www.youtube.com/watch?v=PxWleEgi3Hw)
→ API Routers (https://www.youtube.com/watch?v=wIqURcwxB20)
→ Hono → Itty Router
→ Frontend Frameworks (Any)
→ Wrangler CLI
→ Miniflare (Local Dev)
→ DevTools Integration
→ CI / CD Pipelines
→

Storage Solutions (https://www.youtube.com/watch?v=QTsaAhFvX9o)
→ Workers KV
→ Key-Value Operations
→ Metadata & Bulk Ops
→ Caching Patterns
→
→ R2 Object Storage
→ Large File Handling
→ Asset Management
→ Bucket Lifecycle
→
→ D1 Database (https://www.youtube.com/watch?v=PxWleEgi3Hw)
→ Schema Design
→ Migrations
→ Query Optimization
→ Drizzle / Prisma
→

Durable Objects (https://www.youtube.com/watch?v=C5-741uQPVU)
→ Stateful Applications
→ Coordination
→ Persistence
→ Transactional Operations
→

Queues & Workflows
→ Message Queues
→ Background Jobs
→ Rate Limiting
→ Dead Letter Queues
→ Durable Execution
→ Workflows
→

Advanced Cloudflare Features (https://www.youtube.com/playlist?

list=PLzfTyn6__SjiQCn-CDzF-394hUCuGZH4Y) / (https://www.youtube.com/watch?v=zTNV_ryF0Hk)
→ Workers AI
→ Text Generation
→ Image Processing
→ Speech Recognition
→ AI Model Integration
→
→ Vectorize
→ Embeddings
→ Similarity Search
→ AI-Powered Search
→
→ Stream
→ Video Delivery
→ Live Streaming
→ Video Processing
→
→ Images
→ Calls
→ Email Workers
→ AI Gateway
→ Browser Rendering
→ Logging & Monitoring
→ Tunnels
→
_____
Security & Performance (Covered in core and advanced courses.)
→ Workers Security Model
→ Isolates Architecture
→ Web Security Headers
→ Rate Limiting
→ Cache API
→ HTML Rewriting
→ SSL / TLS
→ Bot Management
→
_____
Integration & Architecture (https://www.youtube.com/watch?v=1-dXh8J08UI)
→ Pages Functions
→ Inter-Worker Communication
→ External API Integration
→ Webhooks
→ Event-Driven Systems

Related Roadmaps :

→ DevOps Roadmap
→ AWS Roadmap
→ Node.js Roadmap
→ TypeScript Roadmap

Certificates :

Free :

Cloudflare Workers Documentation & Tutorials https://developers.cloudflare.com/workers/

Paid :

Microsoft Certified: Azure Developer Associate (AZ-204)
(https://learn.microsoft.com/en-us/credentials/certifications/azure-developer/?practice-assessment-type=certification)

Google Cloud Certified – Professional Cloud Developer
(https://cloud.google.com/learn/certification/cloud-developer/)

Role : CSS Developer

Skills : CSS3, Responsive Web Design, Flexbox, CSS Grid, Media Queries, Container Queries, Box Model, Positioning, Typography,
Color Systems, Layout Design, Animations, Transitions, Keyframes, Accessibility (a11y), Cross-Browser Compatibility,
Performance Optimization, Sass/SCSS, PostCSS, BEM Methodology, CSS Modules, CSS-in-JS, Design Systems, UI Styling,
Mobile-First Design

Learning Roadmap :

Frontend Basics (https://www.youtube.com/watch?v=G3e-cpL7ofc) / (https://www.youtube.com/playlist?list=PL4-IK0AVhVjOJs_UjdQeyEZ_cmEV3uJvx)
→ HTML Fundamentals (https://www.youtube.com/watch?v=kUMe1FH4CHE)
→ CSS Inclusion (Inline → Internal → External)
→ Cascade → Specificity → Inheritance

CSS Fundamentals (https://www.youtube.com/playlist?list=PL4-IK0AVhVjOJs_UjdQeyEZ_cmEV3uJvx)
→ Syntax → Selectors → Units → Functions
→ Colors → Typography → Backgrounds

Core Layout (https://www.youtube.com/playlist?list=PL4-IK0AVhVjOJs_UjdQeyEZ_cmEV3uJvx)
→ Box Model
→ Display & Positioning
→ Flexbox (https://www.youtube.com/watch?v=mF33oPW_oVw)
→ CSS Grid (https://www.youtube.com/playlist?list=PL4-IK0AVhVjPv5tfS82UF_iQgFp4Bl998)

Advanced Styling
→ Images & Filters
→ Tables & Lists
→ Multicolumn Layout

Animations (https://www.youtube.com/results?search_query=css+animations+kevin+powell)
→ Transforms
→ Transitions
→ Keyframe Animations

Responsive Design (https://www.youtube.com/playlist?list=PL4-IK0AVhVjOJs_UjdQeyEZ_cmEV3uJvx)
→ Media Queries

→ Container Queries
→ Responsive Typography

Best Practices (https://www.youtube.com/watch?v=e2nkq3h1P68)
→ Accessibility (a11y) (https://www.youtube.com/watch?v=2oiBKSjOOFE)
→ Performance Optimization

Modern CSS (https://www.youtube.com/watch?v=_a5j7KoflTs)
→ Sass / SCSS
→ PostCSS
→ BEM
→ CSS Modules
→ CSS-in-JS

Production Ready (https://www.youtube.com/playlist?list=PL4-IK0AVhVjOJs_UjdQeyEZ_cmEV3uJvx)
→ Design Systems
→ UI Consistency
→ Cross-Browser Testing

Related Roadmaps :

Frontend Roadmap
HTML Roadmap
JavaScript Roadmap
React Roadmap
Full Stack Roadmap

Certificates :

Free :

* **Google Web.dev – Learn CSS** (https://web.dev/learn/css/)

* **MDN Web Docs – CSS Guide** (https://developer.mozilla.org/en-US/docs/Learn/CSS)

Paid

* **Coursera – HTML, CSS, and Javascript for Web Developers (Johns Hopkins)** (https://www.coursera.org/learn/html-css-javascript-for-web-developers)

* **Udemy – Advanced CSS and Sass** (https://www.udemy.com/course/advanced-css-and-sass/)

Role : Cyber Security

Skills : Computer Networking, Linux, Windows Administration, TCP/IP, DNS, HTTP/HTTPS, OSI Model, Subnetting, Firewalls,
IDS/IPS, SIEM, Threat Intelligence, OSINT, Incident Response,

Vulnerability Management, Penetration Testing,
Web Security (OWASP Top 10), Malware Analysis, Digital Forensics,
Cryptography Basics, IAM, MFA, Zero Trust,
Cloud Security (AWS, Azure, GCP), Network Security, Endpoint Security,
Log Analysis, Bash, PowerShell, Python,
Nmap, Wireshark, Metasploit, Burp Suite, Kali Linux

Learning Roadmap :

Fundamentals

Computer Basics (https://www.youtube.com/watch?v=5xWnmUEi1Qw)
↓
Operating Systems (Windows → Linux → macOS)
(https://www.youtube.com/watch?v=RSk3fL_hqWA)
↓
Networking Fundamentals (OSI, TCP/IP, DNS, Subnetting)
(https://www.youtube.com/watch?v=IPvYjXCsTg8)
↓

Core Security Knowledge

CIA Triad (https://www.youtube.com/watch?v=ug8W0sFiVJo)
↓
Authentication & Authorization
↓
Firewalls, IDS, IPS
↓
Threats, Attacks & OWASP Top 10 (https://www.youtube.com/playlist?list=PLBf0hzazHTGOEuhPQSnq-Ej8jRyXxfYvl)
↓

Hands-on Security

Vulnerability Scanning
↓
Penetration Testing (https://www.youtube.com/playlist?list=PLBf0hzazHTGOEuhPQSnq-Ej8jRyXxfYvl) / (https://www.youtube.com/watch?v=ug8W0sFiVJo)
↓
Privilege Escalation (https://www.youtube.com/@_JohnHammond)
↓
Malware Basics & Reverse Engineering (https://www.youtube.com/@_JohnHammond)
↓

Defensive Security

Incident Response Lifecycle
↓
SIEM & Log Analysis (https://www.youtube.com/results?search_query=siem+log+analysis+full+course)
↓
Threat Hunting
↓
Forensics Basics
↓

Advanced & Cloud Security

Zero Trust Architecture
↓
Cloud Security (AWS / Azure / GCP) (https://www.youtube.com/watch?

v=XsRBDJ5hcdg)
↓
IAM, MFA, Secure Networking
↓
Security Automation (Python, Bash) (https://www.youtube.com/results?search_query=python+cybersecurity+david+bombal)
↓

Professional Growth

CTFs & Labs
↓
Industry Certifications (https://www.youtube.com/playlist?list=PLG49S3iH2I5uQdD8gVvL8u9z4f1l8h2kZ)
↓
Real-World Projects & SOC Experience (https://t.me/cybersecurityIn)

🔗 **Related Roadmaps**

* DevOps Roadmap
* Cloud Roadmap
* Backend Roadmap
* Networking Roadmap

Certificates :

Paid :

⭐ Entry → Mid Level

* **CompTIA Security+**
  https://www.comptia.org/certifications/security

* **Cisco CCNA**

https://www.cisco.com/site/us/en/learn/training-certifications/certifications/enterprise/ccna

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⭐ Mid → Advanced

* **CEH (Certified Ethical Hacker – EC-Council)**
  https://www.eccouncil.org/train-certify/certified-ethical-hacker-ceh/

* **OSCP (Offensive Security Certified Professional)** 🔥
  https://www.offsec.com/courses/pen-200/

⭐ Senior / Leadership

* **CISSP (ISC²)**

https://www.isc2.org/certifications/cissp

* **CISM (ISACA)**

https://www.isaca.org/credentialing/cism

Role : Data Analyst

Skills : Data Analysis, Data Cleaning, Data Wrangling, Exploratory Data Analysis (EDA), SQL, Advanced Excel, Python, Pandas,
NumPy, Data Visualization, Power BI, Tableau, Matplotlib, Seaborn, Statistics, Descriptive Statistics, Inferential Statistics,
Hypothesis Testing, Regression Analysis, Correlation Analysis, A/B Testing, Data Modeling, Dashboard Design,
KPI Reporting, Business Analysis, Data Storytelling, APIs, Web Scraping, Machine Learning Basics, Forecasting,
Big Data Basics, Spark Fundamentals

Learning Roadmap :

What is Data Analytics (https://www.youtube.com/watch?v=a1W6l_sIVXE)
↓
Types of Analytics (Descriptive → Diagnostic → Predictive → Prescriptive) (https://www.youtube.com/watch?v=a1W6l_sIVXE)
↓

Core Tools

Excel (Functions → Charts → Pivot Tables) (https://www.youtube.com/watch?v=8aD5w4J4Y5A)
↓
SQL (Basic Queries → Joins → Window Functions → Optimization) (https://www.youtube.com/watch?v=7mz73uXD9DA)
↓
Python / R Basics (https://www.youtube.com/results?search_query=python+for+data+science+full+course) / (https://www.youtube.com/watch?v=_V8eKsto3Ij)
↓
Pandas / dplyr (https://www.youtube.com/playlist?list=PL-osiE80TeTsWmV9i9c895uyw4uZ7f9c)
↓

Data Handling

Data Collection (CSV, Databases, APIs, Web Scraping)
↓
Data Cleaning (Missing Values, Duplicates, Outliers)
↓
Data Transformation
↓

Analysis & Visualization (https://www.youtube.com/watch?v=LZzq1zSL1bs)

Descriptive Statistics
↓
Statistical Analysis (Hypothesis Testing, Correlation, Regression)
↓
Data Visualization (Matplotlib, Seaborn, Tableau, Power BI)

(https://www.youtube.com/watch?v=FwjaHCVNBWA) / (https://www.youtube.com/watch?v=K3pXnbniUcM)
↓

Advanced Skills (https://www.youtube.com/playlist?list=PLKnIA16_Rmvbr7Zq5h_8Tq6z9qL7yL8fK)

Machine Learning Basics
↓
Model Evaluation
↓
Big Data Concepts (Hadoop, Spark) (https://www.youtube.com/watch?v=some-real-spark-link)
↓
(Optional) Deep Learning Basics (https://www.youtube.com/watch?v=aircAruvnKk)
↓

Professional Growth (https://www.youtube.com/watch?v=8aD5w4J4Y5A)

Projects & Portfolio
↓
Kaggle Competitions (https://www.youtube.com/playlist?list=PLKnIA16_Rmvbr7Zq5h_8Tq6z9qL7yL8fK)
↓
Certifications (*https://www.youtube.com/results?search_query=google+data+analytics+certificate+full+course+youtube)
↓
Industry Networking

🔗 **Related Roadmaps**

* AI & Data Scientist Roadmap
* SQL Roadmap
* Python Roadmap
* Prompt Engineering Roadmap

Certificates :

free :

Tableau Free Training & Academic Program (https://www.tableau.com/learn/training)

paid :

* **Google Data Analytics Professional Certificate**
  [https://www.coursera.org/professional-certificates/google-data-analytics] (https://www.coursera.org/professional-certificates/google-data-analytics)

* **IBM Data Analyst Professional Certificate**
  [https://www.coursera.org/professional-certificates/ibm-data-analyst] (https://www.coursera.org/professional-certificates/ibm-data-analyst)

Role : Data Engineer

Skills : Python, SQL, Data Engineering, ETL Pipelines, Data Warehousing, Data Modeling, OLTP, OLAP, Star Schema,
Snowflake Schema, Big Data Processing, Apache Spark, Apache Hadoop, Apache Airflow, dbt, Kafka, Batch Processing,
Stream Processing, Data Ingestion, Data Transformation, Data Cleaning, Data Lakes, Delta Lake, Cloud Data Platforms,
AWS (S3, EC2, RDS, Glue, Redshift), Azure (Data Factory, Azure SQL, Blob Storage), Google Cloud (BigQuery, Cloud Storage, Dataflow),
Docker, Kubernetes, Terraform, CI/CD, Linux, Git, REST APIs, NoSQL Databases (MongoDB, Cassandra, DynamoDB),
Relational Databases (PostgreSQL, MySQL), Data Governance, Data Quality, Data Lineage, Security & Encryption,
Monitoring & Logging, MLOps Basics

Learning Roadmap :

Python → SQL (https://www.youtube.com/playlist?list=PLf0swTFhTI8oRM0Qv2UGijAkeGZDqs-xF) / (https://www.youtube.com/playlist?list=PL50mYnndduIHs5KReaz7F7aSelC-zG2zP)
↓
Linux → Git → Networking (https://www.youtube.com/playlist?list=PLKV41zXnKc9UxKb2SrUrC_HkoUUK66lTA)
↓
Data Structures & Distributed Systems (https://www.youtube.com/results?search_query=data+structures+distributed+systems+data+engineering)
↓
Databases (OLTP → OLAP) (https://www.youtube.com/watch?v=hf2go3E2m8g)
↓
Data Modeling → Warehousing → Data Lakes (https://www.youtube.com/watch?v=hf2go3E2m8g) / (https://www.youtube.com/playlist?list=PLBJe2dFI4sgtlK_zaqaIBdJFgieYPnQ07)
↓
ETL / ELT → Airflow → dbt (https://www.youtube.com/playlist?list=PL_Ct8Cox2p8VnQcGdiznqyTOSUSrxoxui) / (https://www.youtube.com/watch?v=DzxtCxi4YaA)
↓
Big Data → Spark → Hadoop (https://www.youtube.com/watch?v=v_uodKAywXA) / (https://www.youtube.com/playlist?list=PL6UwySlcwEYJ2hFuGIvr4VEHUAfl-GCNT)
↓
Streaming → Kafka → Realtime Pipelines (https://www.youtube.com/watch?v=wanQj6_-8qQ) / (https://www.youtube.com/watch?v=xwtR_GdlkZA)
↓
Cloud (AWS / Azure / GCP) (https://www.youtube.com/watch?v=vQbReDn3GTs) / (https://www.youtube.com/watch?v=bSx3DbJNQk4)
↓
Docker → Kubernetes → CI/CD
↓
Monitoring → Testing → Optimization
↓
Security → Governance → Privacy
↓
MLOps (Optional) (https://www.youtube.com/watch?v=w71RHxAWxaM) / (https://www.youtube.com/watch?v=OLFM0xWf6jU)

Related Roadmaps :

Data Analyst Roadmap

AI & Data Scientist Roadmap

Python Roadmap

SQL Roadmap

MLOps Roadmap

DevOps Roadmap

Certificates :

free :

Google Cloud Skills Boost – Free Paths
(https://www.skills.google/paths/16)

paid :

AWS Certified Data Engineer – Associate
(https://aws.amazon.com/certification/certified-data-engineer-associate/)

Microsoft Certified: Azure Data Engineer Associate (DP-203)
(https://learn.microsoft.com/en-us/credentials/certifications/azure-data-engineer/?practice-assessment-type=certification)

Role : Design System Designer / Engineer

Skills : Design Systems, Atomic Design, UI Components, Design Tokens, Accessibility (WCAG), Typography Systems, Color Systems,
Grid Systems, Responsive Design, UX Writing, Microcopy, Visual Audits, Component Architecture, Figma, Storybook,
Component Libraries, CSS Architecture, Design-Developer Collaboration, Version Control (Git), Documentation,
Governance Models, Theming, Dark Mode Design, Usability Testing, A/B Testing, Analytics, Frontend Basics (HTML, CSS, JavaScript),
React Components, Semantic Versioning

Learning Roadmap :

UI/UX Fundamentals (https://www.youtube.com/playlist?list=PLERed4ILxkJ1_kDOXCrEGmB58IsLOiIc3)
↓
Design System Basics (https://www.youtube.com/playlist?list=PLXDU_eVOJTx6vqOWJSWH87Zb5-riiG63A) / (https://www.youtube.com/playlist?list=PLM2Dd2NIn2CqxUkHHiDvDbNIbqbv8ehy)
(What, Why, Examples, Atomic Design) / (https://www.youtube.com/watch?v=W-h1FtNYim4)
↓
Design Language (https://www.youtube.com/playlist?list=PLRpNMNIF4d9_T5zfabqXAFbRgSTHvbdPZ) / (https://www.youtube.com/playlist?list=PLXDU_eVOJTx5LSjOmeBYMuvaa4UayfMe4)
(Brand, Guidelines, Tone, Accessibility)
↓
Design Tokens (https://www.youtube.com/playlist?list=PLvnhVb8yYRQ1rvEFiPEyNbVPiJxvuCbgL) / (https://www.youtube.com/playlist?list=PLkmvmF0zhgT8IPxMPKFRQHY0GE-YRcnwC)
(Color, Typography, Spacing, Grid, Dark Mode) / (https://www.youtube.com/watch?v=CJyJN0ZdEGA)
↓
Core Components (https://www.youtube.com/playlist?list=PLXDU_eVOJTx5LSjOmeBYMuvaa4UayfMe4) / (https://www.youtube.com/playlist?list=PLkmvmF0zhgT8IPxMPKFRQHY0GE-YRcnwC)

(Buttons, Inputs, Modals, Cards, Navigation)
↓
Tooling (https://www.youtube.com/playlist?list=PLXDU_eVOJTx5LSjOmeBYMuvaa4UayfMe4) / (https://www.youtube.com/playlist?list=PLXDU_eVOJTx7I9CMOHnTEzuxDwoq0Cjqt)
(Figma → Storybook → Version Control) / (https://www.youtube.com/watch?v=O3afqyCS2-o) / (https://www.youtube.com/watch?v=3hTyU3h1i6E)
↓
Developer Handoff (https://www.youtube.com/playlist?list=PLXDU_eVOJTx7I9CMOHnTEzuxDwoq0Cjqt) / (https://www.youtube.com/watch?v=ALkqhXv0GPk)
(Component APIs, Naming, Documentation)
↓
Governance & Versioning (https://www.youtube.com/watch?v=YYt5iW-d-LI) / (https://www.youtube.com/watch?v=5Zc3W4NpiA8) / (https://www.youtube.com/watch?v=vpYN5J78jaA)
(Contribution Model, Releases, SemVer)
↓
Analytics & Adoption (https://www.youtube.com/watch?v=LZQxPHAaEuI)
(Component Usage, Feedback, Metrics)

Related Roadmap :

UX Design Roadmap

Frontend Roadmap

JavaScript Roadmap

React Roadmap

Certificates :

free:

IBM Design Thinking Practitioner (https://www.ibm.com/training/enterprise-design-thinking)

Paid :

NN/g (Nielsen Norman Group) UX Certification (https://www.nngroup.com/ux-certification/)

Google UX Certificate (https://www.coursera.org/professional-certificates/google-ux-design)

Role : DevOps Engineer

Skills : Linux administration, Shell scripting, Python, Git, GitHub/GitLab, Networking fundamentals, DNS, HTTP/HTTPS, SSL/TLS,
SSH, Docker, Kubernetes, Cloud computing, AWS, Azure, GCP, Infrastructure as Code, Terraform, CI/CD pipelines,
GitHub Actions, Jenkins, Configuration management, Ansible, Monitoring, Logging, Prometheus, Grafana, ELK Stack,
Cloud security, Secrets management, Kubernetes security, GitOps, ArgoCD, System design, Troubleshooting

Learning Roadmap :

Programming Language
→ Operating Systems (Linux) (https://www.youtube.com/playlist?list=PL2kSRH_DmWVZp_cu6MMPWkgYh7GZVFS6i) / (https://www.youtube.com/watch?v=29eDuMjsEF8)
→ Terminal & Shell Scripting (https://www.youtube.com/playlist?list=PLIhvC56v63IKioClkSNDjW7iz-6TFvLwS) / (https://www.youtube.com/playlist?list=PL9YnOxYKGRNMqwVoSBPm5o04hnukyju0M)
→ Version Control (Git) (https://www.youtube.com/playlist?list=PLhP5RsB7fhE0Z6UKi7eUA3_xlDpWygZT1)
→ Networking & Protocols (https://www.youtube.com/playlist?list=PLxCzCOWd7aiGFBD2-2joCpWOLUrDLvVV)
→ Web Servers & Proxies (https://www.youtube.com/watch?v=9jZEfW8h5fQ)
→ Containers (Docker) (https://www.youtube.com/playlist?list=PLy7NrYWoggjwPggqtFsI_zMAwvG0SqYCb)
→ Cloud Providers (https://www.youtube.com/playlist?list=PLEiEAq2VkUUIJ3o1tehvtux0_Ynf42CBN)
→ Serverless Computing (https://www.youtube.com/watch?v=5rG-YgTHMC8)
→ Infrastructure Provisioning (IaC) (https://www.youtube.com/playlist?list=PLGyKZJoyKdOfehoKuFC1PKTGzMtRwnGNd)
→ Configuration Management (https://www.youtube.com/playlist?list=PLdpzxOOAlwvLxd5nmtmORCmhD5jkrNbuE)
→ CI/CD Pipelines (https://www.youtube.com/playlist?list=PL0lvsZ5ieQicxwgSw4aLbuc9BMPGSjkYt)
→ Secrets Management (https://www.youtube.com/watch?v=FCWPU3ZuY7A)
→ Monitoring & Logging (https://www.youtube.com/playlist?list=PLy7NrYWoggjxCF3av5JKwyG7FFF9eLeL4)
→ Application Performance Monitoring (https://www.youtube.com/watch?v=AgRNv9W5nJs)
→ Container Orchestration (Kubernetes) (https://www.youtube.com/playlist?list=PLy7NrYWoggjziYQIDorlXjTvvwweTYoNC)
→ Artifact Management (https://www.youtube.com/watch?v=XVTO59pDnpk) / (https://www.youtube.com/watch?v=CNTLIgGGuoQ)
→ GitOps (https://www.youtube.com/watch?v=X5W_706-jSY)
→ Service Mesh (https://www.youtube.com/playlist?list=PLy7NrYWoggjyvPa2FNiLoxqH73rndE4La)
→ Cloud Design Patterns & Architecture (https://www.youtube.com/playlist?list=PLEiEAq2VkUUIJ3o1tehvtux0_Ynf42CBN)

Related Roadmaps :

* Backend Development Roadmap
* Docker Roadmap
* Kubernetes Roadmap
* Linux Roadmap

Certification :

🔷 Beginner / Foundation

* **AWS Certified Cloud Practitioner** (https://aws.amazon.com/certification/certified-cloud-practitioner/)

* **Microsoft Azure Fundamentals (AZ-900)**

(https://learn.microsoft.com/en-us/certifications/azure-fundamentals/)

🔷 DevOps-Focused

* **AWS Certified DevOps Engineer – Professional**
(https://aws.amazon.com/certification/certified-devops-engineer-professional/)

* **Microsoft DevOps Engineer Expert (AZ-400)**
(https://learn.microsoft.com/en-us/certifications/devops-engineer/)

🔷 Kubernetes & Containers

* **CKA – Certified Kubernetes Administrator**
(https://www.cncf.io/certification/cka/)

* **CKAD – Certified Kubernetes Application Developer**
(https://www.cncf.io/certification/ckad/)

Role : Developer Relations

Skills : Public speaking, Technical writing, Developer advocacy, API documentation, Community management,
Content strategy, Blogging, Video creation, Live streaming, Social media strategy, Git, GitHub,
Open source collaboration, Basic programming, API design understanding, SDK usage, Event management,
Developer onboarding, Metrics and analytics, Feedback collection, Personal branding, Developer experience optimization

Learning Roadmap :

Understanding DevRel & Developer Experience
(https://www.youtube.com/c/DevRel) / (https://www.youtube.com/playlist?list=PLWpUQfjuJDXztsNnwYCNeIrQ1W89jRaAn)
→ Communication Skills (Speaking + Writing)
(https://www.youtube.com/playlist?list=PL8EEC66CC5F02545C)
→ Basic Programming & Tooling (https://www.youtube.com/watch?v=mklEhT_RLos)
→ APIs, SDKs & Documentation (https://www.youtube.com/watch?v=GZvSYJDk-us)
→ Git, GitHub & Open Source
(https://www.youtube.com/playlist?list=PLyAb1KN5Zkf-QN8rI3-cs6QAHsyRsIvVc)
→ Community Engagement
(https://www.youtube.com/playlist?list=PLWpUQfjuJDXztsNnwYCNeIrQ1W89jRaAn)
→ Community Building & Management
(https://www.youtube.com/playlist?list=PLWpUQfjuJDXztsNnwYCNeIrQ1W89jRaAn)
→ Event Management (https://www.youtube.com/watch?v=tyrXMWHxUn4)
→ Content Creation (Blogs, Video, Live) (https://www.youtube.com/watch?v=4AAXhQznrAQ)
→ Developer Onboarding & Support
(https://www.youtube.com/playlist?list=PLWpUQfjuJDXztsNnwYCNeIrQ1W89jRaAn)
→ Metrics, Analytics & Reporting (https://www.youtube.com/watch?v=oxpDgWWHJx8)
→ Thought Leadership & Personal Brand
(https://www.youtube.com/playlist?list=PLWpUQfjuJDXztsNnwYCNeIrQ1W89jRaAn)

→ Continuous Learning & Career Growth
(https://www.youtube.com/playlist?list=PLWpUQfjuJDXztsNnwYCNeIrQ1W89jRaAn

Certificates :

free :

Google Technical Writing
(https://developers.google.com/tech-writing)

CNCF / Linux Foundation – Community & OSS Courses
(https://training.linuxfoundation.org/resources/free-courses/)

paid :

API Documentation & Technical Writing
(https://www.coursera.org/search?
query=API%20Documentation%20%26%20Technical%20Writing)

Developer Marketing / Product Marketing
(https://www.pragmaticinstitute.com/)

Role : DevSecOps Engineer

Skills : Secure coding, Cloud security, IAM, Threat modeling, OWASP Top 10, Network security, Encryption,
Kubernetes security, Container security, CI/CD security, Infrastructure as Code security, Vulnerability scanning,
SIEM, Incident response, Zero trust architecture, Supply chain security, Compliance frameworks,
Risk management, Automation security

Learning Roadmap :

Programming Language
→ Python (https://www.youtube.com/playlist?
list=PLsyeobzWxl7poL9JTVyndKe62ieoN-MZ3)
→ Ruby (https://www.youtube.com/watch?v=5gT7L3u3g8k)
→ Rust (https://www.youtube.com/playlist?
list=PL_c9BZzLwBRIymgB73pHXIgazpB-uszKU)
→ Go (https://www.youtube.com/playlist?
list=PL6oD2syjfW7Abh0gohG3JV0PnDmJNhtgx)
→ JavaScript / Node.js (https://www.youtube.com/playlist?list=PLC3y8-
rFHvwh8shCMHFA5kWxD9PaPwxaY)

Scripting Knowledge
→ Bash (https://www.youtube.com/playlist?
list=PLIhvC56v63IKioClkSNDjW7iz-6TFvLwS)
→ PowerShell (https://www.youtube.com/watch?v=UVUd9_k9C6A)

Editors (https://www.youtube.com/watch?v=3G6kAEvbv2A)
→ Vim (https://www.youtube.com/playlist?
list=PLm323Lc7iSW_wuxqmKx_xxNtJC_hJbQ7R)
→ Nano
→ Emacs

Security Foundations (*https://www.youtube.com/playlist?list=PLEiEAq2VkUUJfPOj5nRounXvf3n17PCft) / (https://www.youtube.com/playlist?list=PL9ooVrP1hQOGPQVeapGsJCktzIO4DtI4)
→ CIA Triad
→ Authentication
→ Authorization
→ OWASP Top 10 (https://www.youtube.com/playlist?list=PLyqga7AXMtPOguwtCCXGZUKvd2CDCmUgQ)
→ Encryption
→ Symmetric Encryption
→ Asymmetric Encryption

Networking Basics
→ Firewalls
→ DNS
→ ACLs
→ HTTP
→ VLANs
→ TLS
→ Network Segmentation

Secure Coding Practices (https://www.youtube.com/playlist?list=PLa2xctTiNSCiUcETvRT7lbYuKTpgKDchD)
→ SQL Injection Prevention
→ Input Validation Patterns
→ XSS Prevention

Identity & Access Management (https://www.youtube.com/playlist?list=PLEiEAq2VkUUJfPOj5nRounXvf3n17PCft)
→ IAM
→ Role Based Access Control
→ Least Privilege

Monitoring & Detection (https://www.youtube.com/playlist?list=PLEiEAq2VkUUJfPOj5nRounXvf3n17PCft)
→ Log Analysis
→ Alert Types
→ SIEM

DevSecOps Tooling Basics (https://www.youtube.com/watch?v=7tcX_ndqD68)
→ Burp Suite
→ Wireshark (Basics)
→ Nmap (Basics)

Cloud Security (https://www.youtube.com/playlist?list=PLEiEAq2VkUUJfPOj5nRounXvf3n17PCft)
→ CSPM
→ Key Management Service (KMS)
→ Cloud IAM

Threat Management & Risk (https://www.youtube.com/watch?v=VoiHRBaV8Xk)
→ Threat Modeling
→ STRIDE (https://www.youtube.com/watch?v=rEnJYNkUde0)
→ PASTA (https://www.youtube.com/watch?v=CgW9oKH-76A)
→ Threat Modeling Workflows
→ Attack Surface Mapping

Secure Architecture (https://www.youtube.com/watch?v=ztwNezcel0E) / (https://www.youtube.com/watch?v=geVQlGpg0xg)
→ Defense in Depth
→ Secure API Design
→ Zero Trust Architecture

Container & Platform Security (https://www.youtube.com/playlist?list=PLeLcvrwLe1859Rje9gHrD1KEp4y5OXApB)
→ Docker Security (https://www.youtube.com/watch?v=OjZz4D0B-cA)
→ Kubernetes Security
→ Image Scanning

Security Automation (https://www.youtube.com/watch?v=7tcX_ndqD68)
→ Vulnerability Scanning
→ Nessus
→ Nmap
→ OpenVAS
→ Qualys
→ Automated Patching
→ SOAR Concepts
→ IDS
→ IPS

Incident Response (https://www.youtube.com/watch?v=4MGVTpXAAmo) / (https://www.youtube.com/watch?v=tKXxror5HCE)
→ Incident Response Lifecycle
→ Forensics
→ Containment
→ Root Cause Analysis

Advanced Cryptography (https://www.youtube.com/playlist?list=PLEiEAq2VkUUJfPOj5nRounXvf3n17PCft)
→ Cryptographic Hashing
→ bcrypt
→ SHA-256
→ Advanced Cryptographic Concepts

Enterprise Security Operations (https://www.youtube.com/playlist?list=PLEiEAq2VkUUJfPOj5nRounXvf3n17PCft)
→ Multi-Region Security Planning
→ Large-Scale Identity Strategy

→ Certificate Lifecycle Management
→ PKI Design & Failover
→ DDoS Mitigation Strategy
→ Build Pipeline Hardening
→ Secure Network Zoning

Supply Chain Security (https://www.youtube.com/watch?v=7tcX_ndqD68)
→ SBOMs
→ Dependency Risk Management

Endpoint Security (https://www.youtube.com/playlist?list=PLEiEAq2VkUUJfPOj5nRounXvf3n17PCft)
→ EDR Strategy
→ SOAR Automation
→ Response Strategy

Governance, Risk & Compliance (https://www.youtube.com/playlist?list=PLEiEAq2VkUUJfPOj5nRounXvf3n17PCft)
→ Audit & Compliance Mapping
→ Risk Quantification
→ SOC 2
→ ISO 27001
→ NIST

Related Roadmaps :

DevOps Roadmap
Cybersecurity Roadmap

Certificates :

free :

OWASP WebGoat / OWASP (https://owasp.org/www-project-webgoat/)

Linux Foundation – Secure Software Development (https://training.linuxfoundation.org/resources/free-courses/)

Paid :

Certified Kubernetes Security Specialist (CKS) (https://www.cncf.io/training/certification/cks/)

AWS Certified Security – Specialty (https://aws.amazon.com/certification/certified-security-specialty/)

Role : Engineering Manager

Skills : Technical leadership, System design, Architecture decision-making, Code

review, CI/CD understanding,
Incident management, Hiring, Team building, Performance management, Mentoring, Coaching,
Conflict resolution, Emotional intelligence, Stakeholder management, Project planning,
Agile methodologies, Risk management, KPI tracking, Business strategy, ROI analysis, Budgeting,
Cost optimization, Vendor management, Communication, Executive reporting, Change management,
Engineering culture building, Crisis management

Learning Roadmap :

What is Engineering Management (https://www.youtube.com/playlist?list=PLR_jPfME-b_XULtZBm6nvQuDmqg4KlHlY)
→ EM vs Tech Lead vs Individual Contributor (https://www.youtube.com/playlist?list=PLdxiGiTC1t46zjoJRmeFvlrauEwsVLNso)

Key Focus Areas (https://www.youtube.com/playlist?list=PLR_jPfME-b_XULtZBm6nvQuDmqg4KlHlY)
→ People
→ Product
→ Process

Technical Leadership Foundations (https://www.youtube.com/@techleadjournal)
→ Software Engineering Background
→ System Design & Architecture
→ Technical Debt Identification & Management
→ Technical Documentation
→ Code Review Best Practices

Technical Leadership – Advanced (https://www.youtube.com/@techleadjournal) / (https://www.youtube.com/playlist?list=PLR_jPfME-b_XULtZBm6nvQuDmqg4KlHlY)
→ Architectural Decision-Making
→ System Monitoring & Performance
→ Scaling Infrastructure
→ Technical Roadmapping
→ Build vs Buy Evaluation
→ Technical Risk Assessment
→ Technical Strategy Definition
→ CI/CD Implementation
→ Development & Release Workflows
→ Technical Standards Setting
→ Security Best Practices
→ Testing Strategies
→ Quality & Process Ownership
→ Incident Management

People Management (https://www.youtube.com/playlist?list=PLR_jPfME-b_XULtZBm6nvQuDmqg4KlHlY) / (https://www.youtube.com/watch?v=9YXOg00EzeA)
→ Hiring & Recruitment
→ Team Structure & Design
→ Performance Evaluations
→ Mentoring & Coaching

→ Career Development Planning

Leadership Skills (https://www.youtube.com/playlist?list=PLR_jPfME-b_XULtZBm6nvQuDmqg4KlHlY)
→ Delegation
→ Conflict Resolution
→ Feedback Delivery
→ Team Motivation
→ Trust & Influence Building
→ Emotional Intelligence

Communication (https://www.youtube.com/playlist?list=PLR_jPfME-b_XULtZBm6nvQuDmqg4KlHlY)
→ One-on-One Meetings
→ Team Meetings
→ Status Reporting
→ Stakeholder Management
→ Cross-functional Collaboration

Project Management – Execution (https://www.youtube.com/playlist?list=PLR_jPfME-b_XULtZBm6nvQuDmqg4KlHlY)
→ Resource Allocation
→ Sprint Planning
→ Agile Methodologies
→ Dependency Management
→ Risk Management
→ Release Management
→ Scope Management
→ Timeline Estimation
→ Milestone Management
→ Project Tracking

Project Management – Measurement (https://www.youtube.com/playlist?list=PLR_jPfME-b_XULtZBm6nvQuDmqg4KlHlY)
→ KPI Definition
→ Velocity Tracking
→ Quality Metrics
→ Team Health Metrics
→ Project Postmortems

Business Acumen (https://www.youtube.com/playlist?list=PLR_jPfME-b_XULtZBm6nvQuDmqg4KlHlY)
→ Product Strategy Alignment
→ Business Case Development
→ ROI Analysis
→ Market Awareness
→ Competitive Analysis

Financial Management (https://www.youtube.com/playlist?list=PLR_jPfME-b_XULtZBm6nvQuDmqg4KlHlY)
→ Budget Planning
→ Resource Forecasting
→ Cost Optimization

→ Vendor Management

Organizational Awareness (https://www.youtube.com/playlist?list=PLR_jPfME-b_XULtZBm6nvQuDmqg4KlHlY)
→ Company Culture Understanding
→ Organizational Structure
→ Change Management
→ Politics Navigation
→ Cross-department Collaboration

Engineering Culture (https://www.youtube.com/playlist?list=PLR_jPfME-b_XULtZBm6nvQuDmqg4KlHlY)
→ Defining & Enforcing Values
→ Team Traditions & Rituals
→ Recognition Programs
→ Social Connections
→ Inclusive Environment Creation

Team Culture & Excellence (https://www.youtube.com/playlist?list=PLR_jPfME-b_XULtZBm6nvQuDmqg4KlHlY)
→ Innovation Fostering
→ Learning Culture Development
→ Knowledge Sharing Practices
→ Technical Excellence Mindset
→ Blameless Postmortems
→ Bias Recognition & Mitigation

Crisis Management (https://www.youtube.com/playlist?list=PLR_jPfME-b_XULtZBm6nvQuDmqg4KlHlY)
→ Risk Mitigation
→ Contingency Planning
→ Disaster Recovery
→ Business Continuity
→ Security Incident Handling
→ Production Issue Management

Incident Response (https://www.youtube.com/playlist?list=PLR_jPfME-b_XULtZBm6nvQuDmqg4KlHlY)
→ Emergency Protocols
→ War Room Management
→ Stakeholder Communication
→ Post-Incident Analysis
→ Service Recovery

Stakeholder Management (https://www.youtube.com/playlist?list=PLR_jPfME-b_XULtZBm6nvQuDmqg4KlHlY)
→ Team Support
→ Internal Stakeholder Management

Executive Communication (https://www.youtube.com/playlist?

list=PLR_jPfME-b_XULtZBm6nvQuDmqg4KlHlY)
→ Board Presentations
→ Executive Summaries
→ Strategic Proposals
→ Budget Requests
→ Vision Alignment

Customer Relations (https://www.youtube.com/playlist?
list=PLR_jPfME-b_XULtZBm6nvQuDmqg4KlHlY)
→ Customer Feedback Integration
→ Technical Customer Support
→ Customer Success Alignment
→ Feature Prioritization
→ Technical Partnerships

Partner Management (https://www.youtube.com/playlist?
list=PLR_jPfME-b_XULtZBm6nvQuDmqg4KlHlY)
→ Vendor Relationships
→ Technology Partnerships
→ Integration Management
→ API Strategy
→ External Collaboration

Knowledge Management (https://www.youtube.com/playlist?
list=PLR_jPfME-b_XULtZBm6nvQuDmqg4KlHlY)
→ Architecture Documentation
→ Process Documentation
→ Decision Records
→ Lessons Learned
→ Best Practices

Knowledge Transfer (https://www.youtube.com/playlist?list=PLR_jPfME-
b_XULtZBm6nvQuDmqg4KlHlY)
→ Mentoring Programs
→ Knowledge Bases
→ Brown Bag Sessions
→ Tech Talks
→ Legacy System Retirement

Change Management – Technical (https://www.youtube.com/playlist?
list=PLR_jPfME-b_XULtZBm6nvQuDmqg4KlHlY)
→ Migration Planning
→ Technology Adoption
→ Tool Transitions
→ Process Changes
→ Legacy System Retirement

Change Management – Organizational (https://www.youtube.com/playlist?
list=PLR_jPfME-b_XULtZBm6nvQuDmqg4KlHlY)
→ Change Strategy
→ Impact Assessment
→ Stakeholder Management

→ Communication Planning
→ Resistance Management

Change Management – Team (https://www.youtube.com/playlist?list=PLR_jPfME-b_XULtZBm6nvQuDmqg4KlHlY)
→ Reorganizations
→ Team Mergers
→ Role Transitions
→ Responsibility Shifts
→ Culture Evolution

Foundational & Related Learning Paths
→ Backend Development (https://www.youtube.com/watch?v=8u5fQvX9v0Y)
→ DevOps Engineering (https://www.youtube.com/playlist?list=PLy7NrYWoggjwPggqtFsI_zMAwvG0SqYCb)
→ System Design (https://www.youtube.com/playlist?list=PLMCXH8-0f0uL0f7m5b5q4q4q4q4q4q4q)
→ Software Architecture (https://www.youtube.com/playlist?list=PLbRMhDVUMngc8aY8vK9t2e1y8zB5f5f5f)
→ Software Design & Architecture (https://www.youtube.com/@techleadjournal)

Related Roadmaps :

Backend Development
DevOps Engineering
System Design
Software Architect
Software Design & Architecture

Certificates :

paid :

Certified Scrum Master (CSM)
(https://www.scrumalliance.org/)

Google Engineering Manager / Technical Leadership Programs
(https://www.coursera.org/professional-certificates/google-project-management)

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role : Flutter Developer (Mobile App Developer)

skills :
Dart, Flutter SDK, Widget Tree, Stateless Widgets, Stateful Widgets, Responsive UI, Material UI, Cupertino UI,
State Management (Provider, Riverpod, BLoC, GetX), REST APIs, GraphQL, JSON Parsing,
Firebase (Auth, Firestore, Storage, Cloud Functions, FCM), SQLite, SharedPreferences, Animations, RxDart, Git,
GitHub, CI/CD, App Deployment, Testing, Clean Architecture, SOLID Principles

learning roadmap :
Dart Basics
(https://www.youtube.com/playlist?list=PLkAHsOVSsZBjBrzxXUQVHGokr6HfdzkqO)
→ Dart Advanced Concepts (Futures, Streams, Async/Await, Isolates)

(https://www.youtube.com/playlist?list=PLkAHsOVSsZBjBrzxXUQVHGokr6HfdzkqO)
→ Flutter Setup (Flutter CLI → IDEs → FVM)
(https://www.youtube.com/playlist?list=PL5jb9EteFAODi35jPznP37hnR2sTHOOTU)
→ Flutter Basics (Widgets → Widget Tree → Layouts)
(https://www.youtube.com/playlist?list=PL5jb9EteFAODi35jPznP37hnR2sTHOOTU)
→ UI Widgets (Material → Cupertino → Responsive Widgets)
(https://www.youtube.com/playlist?list=PL9Pi7QVG7pyC3UQEsrRp_hYfpovAnxIlj)
→ Assets Handling (Images → Fonts → Files)
(https://www.youtube.com/playlist?list=PL5jb9EteFAODi35jPznP37hnR2sTHOOTU)
→ Version Control (Git → GitHub)
(https://www.youtube.com/playlist?list=PL9Pi7QVG7pyC3UQEsrRp_hYfpovAnxIlj)
→ Design Principles (OOP → SOLID → Design Patterns → Dependency Injection)
(https://www.youtube.com/playlist?list=PLkAHsOVSsZBjBrzxXUQVHGokr6HfdzkqO)
→ Package Management (pub.dev → flutter pub)
(https://www.youtube.com/playlist?list=PL5jb9EteFAODi35jPznP37hnR2sTHOOTU)
→ API Integration (REST → JSON → GraphQL → WebSockets)
(https://www.youtube.com/playlist?list=PLU9c2hbou-oTI58_p8ADwmHNXB7jkQEOs)
→ Local Storage (SharedPreferences → SQLite)
(https://www.youtube.com/playlist?list=PL9Pi7QVG7pyC3UQEsrRp_hYfpovAnxIlj)
→ Firebase (Auth → Firestore → Storage → Notifications → Functions)
(https://www.youtube.com/playlist?list=PL1WkZqhlAdC9TgTee50FWiiwVZ6kQg4W7)
→ State Management (Provider → Riverpod → BLoC → GetX → Redux)
(https://www.youtube.com/playlist?list=PLNnAcB93JKV8L5sSNDQ0-wxCDUPZXZM7i)
→ Reactive Programming (RxDart)
(https://www.youtube.com/playlist?list=PLNnAcB93JKV8L5sSNDQ0-wxCDUPZXZM7i)
→ Animations (Implicit → Explicit → Hero → Curves)
(https://www.youtube.com/playlist?list=PL4cUxeGkcC9gP1qg8yj-Jokef29VRCLt1)
→ Testing (Unit → Widget → Integration → TDD/BDD)
(https://www.youtube.com/playlist?list=PLlzmAWV2yTgAW2rVT0sqRmtBXc-pmBnJG)
→ Dev Tools (Flutter Inspector → Performance → Memory)
(https://www.youtube.com/watch?v=J7s5bu49k3U)
→ Flutter Internals (Render Objects → 3 Trees → Immutability)
(https://www.youtube.com/watch?v=xiW3ahr4CRU)
→ CI/CD (GitHub Actions → Codemagic → Fastlane → Firebase App Distribution)
(https://www.youtube.com/playlist?list=PLq83k-ITj6lSEhobRX2E94fNSZRs1Uwtm)
→ Analytics (Firebase Analytics → Google Analytics → Mixpanel)
(https://www.youtube.com/playlist?list=PL1WkZqhlAdC9TgTee50FWiiwVZ6kQg4W7)
→ Deployment (Play Store → App Store → iOS Guidelines)
(https://youtube.com/playlist?list=PL-iyKgrrGFZAzrNCRzbm-UyLb43adNNv1)
→ Continuous Learning & Open Source Contributions
(https://www.youtube.com/c/Flutter)

related roadmaps :
React Native Developer Roadmap
Android Developer Roadmap
iOS Developer Roadmap
Full Stack Mobile Developer Roadmap

certificates :

Free :

Flutter & Dart – Coursera (Free with audit option)
(https://www.coursera.org/search?query=flutter)

Paid :

Flutter & Dart - The Complete Guide
(https://www.udemy.com/course/learn-flutter-dart-to-build-ios-android-apps/?srsltid=AfmBOor6s9aXPabMifMD7sJIw3IuF_HOmFW4EQXHNQd0hPQXgGPZFkoT&utm_source=chatgpt.com)
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role : Frontend Developer

skills :
HTML5, Semantic HTML, Accessibility (a11y), SEO Basics, CSS3, Responsive Design, Flexbox, Grid, Tailwind CSS, Sass,
PostCSS, JavaScript (ES6+), DOM Manipulation, Fetch API, Git, GitHub, npm, TypeScript, React, Vue, Angular,
Next.js, Nuxt.js, Vite, Webpack, ESLint, Prettier, Vitest, Jest, Cypress, Playwright, Web Security, Authentication,
GraphQL, REST APIs, Performance Optimization, PWAs, Browser APIs

learning roadmap :
How Internet Works (https://www.youtube.com/c/Computerphile)
→ HTTP → DNS → Hosting → Browsers (https://www.youtube.com/playlist?list=PLxCzCOWd7aiGFBD2-2joCpWOLUrDLvVV)
→ HTML Basics → Semantic HTML → Forms & Validation → Accessibility → SEO (https://www.youtube.com/playlist?list=PL0eyrZgxdwhwNC5ppZo_dYGVjerQY3xYU)
→ CSS Basics → Layouts (Flexbox → Grid) → Responsive Design (https://www.youtube.com/playlist?list=PL0eyrZgxdwhwNC5ppZo_dYGVjerQY3xYU)
→ JavaScript Basics → DOM Manipulation → Fetch API / Ajax (https://www.youtube.com/playlist?list=PLGjplNEQ1it_oTvuLRNqXfz_v_0pq6unW)
→ Version Control (Git) (https://www.youtube.com/playlist?list=PLhP5RsB7fhE0Z6UKi7eUA3_xlDpWygZT1)
→ VCS Hosting (GitHub → GitLab → Bitbucket) (https://www.youtube.com/playlist?list=PLhP5RsB7fhE0Z6UKi7eUA3_xlDpWygZT1)
→ Package Managers (npm → yarn → pnpm) (https://www.youtube.com/playlist?list=PLGjplNEQ1it_oTvuLRNqXfz_v_0pq6unW)
→ JavaScript Framework (React / Vue / Angular / Svelte) (https://www.youtube.com/playlist?list=PL4cUxeGkcC9gZD-Tvwfod2gaISzfRiP9d)
→ CSS Writing (Tailwind CSS) (https://www.youtube.com/playlist?list=PL5jb9EteFAODi35jPznP37hnR2sTHOOTU)
→ CSS Architecture (BEM) (https://www.youtube.com/playlist?list=PL0eyrZgxdwhwNC5ppZo_dYGVjerQY3xYU)
→ CSS Preprocessors (Sass → PostCSS) (https://www.youtube.com/playlist?list=PL4cUxeGkcC9gE9g4f5a0f0a0f0a0f0a0f0a)
→ Build Tools (Vite → Webpack → Rollup → Parcel → esbuild → SWC) (https://www.youtube.com/playlist?list=PL4cUxeGkcC9gZD-Tvwfod2gaISzfRiP9d)
→ Code Quality (ESLint → Prettier) (https://www.youtube.com/playlist?list=PL4cUxeGkcC9gZD-Tvwfod2gaISzfRiP9d)
→ Testing (Vitest → Jest → Playwright → Cypress) (https://www.youtube.com/playlist?list=PL4cUxeGkcC9gE9g4f5a0f0a0f0a0f0a0f0a)
→ Web Security (HTTPS → CORS → CSP → OWASP Risks) (https://www.youtube.com/playlist?list=PLxCzCOWd7aiGFBD2-2joCpWOLUrDLvVV)
→ Authentication (JWT → OAuth → SSO → Sessions) (https://www.youtube.com/playlist?list=PL4cUxeGkcC9gZD-Tvwfod2gaISzfRiP9d)
→ TypeScript (https://www.youtube.com/playlist?list=PLEjh-YiSBCDJcayJNll1NjCOKMcIO4L0T)
→ Web Components (Custom Elements → Shadow DOM) (https://www.youtube.com/playlist?list=PL4cUxeGkcC9gE9g4f5a0f0a0f0a0f0a0f0a)
→ SSR (Next.js / Nuxt.js / SvelteKit) (https://www.youtube.com/playlist?list=PL4cUxeGkcC9gZD-Tvwfod2gaISzfRiP9d)
→ GraphQL (Apollo → Relay) (https://www.youtube.com/playlist?list=PL4cUxeGkcC9gZD-Tvwfod2gaISzfRiP9d)
→ Static Site Generators (Next.js → Astro → Eleventy) (https://www.youtube.com/playlist?list=PL4cUxeGkcC9gZD-Tvwfod2gaISzfRiP9d)
→ Performance Optimization (PRPL → RAIL → Lighthouse → DevTools) (https://www.youtube.com/playlist?list=PL4cUxeGkcC9gZD-Tvwfod2gaISzfRiP9d)
→ Browser APIs (Storage → Service Workers → WebSockets → Notifications) (https://www.youtube.com/playlist?list=PL4cUxeGkcC9gZD-Tvwfod2gaISzfRiP9d)
→ PWAs (https://www.youtube.com/playlist?list=PL4cUxeGkcC9gZD-Tvwfod2gaISzfRiP9d)
→ Cross-Platform Apps (React Native / Flutter / Electron / Tauri) (https://www.youtube.com/playlist?list=PL4cUxeGkcC9gZD-Tvwfod2gaISzfRiP9d)

related roadmaps :
TypeScript Roadmap
Node.js Roadmap
Full Stack Developer Roadmap
React Developer Roadmap

certificates :

Free

Responsive Web Design – freeCodeCamp
(https://www.freecodecamp.org/learn/2022/responsive-web-design/)

JavaScript Algorithms and Data Structures – freeCodeCamp
(https://www.freecodecamp.org/learn/javascript-algorithms-and-data-structures/)

Paid

Front-End Web Developer Professional Certificate – Meta (Coursera)
(https://www.coursera.org/professional-certificates/meta-front-end-developer)

The Complete JavaScript + React Course – Udemy
(https://www.udemy.com/course/the-complete-javascript-course/)

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role : Full Stack Developer

skills :
HTML5, CSS3, JavaScript (ES6+), Responsive Design, Git, GitHub, npm, Tailwind CSS, React, Node.js, Express.js,
REST APIs, PostgreSQL, SQL, JWT Authentication, Redis, CLI Development, Linux Basics,
AWS (EC2, S3, VPC, Route53, SES), CI/CD, GitHub Actions, Monitoring, Deployment, Infrastructure as Code, Ansible,
Terraform

learning roadmap :
HTML → CSS → JavaScript (https://www.youtube.com/watch?v=G3e-cpL7ofc) &
(https://www.youtube.com/playlist?list=PLGjplNEQ1it_oTvuLRNqXfz_v_0pq6unW)
→ Build Static Web Pages (Portfolio / Landing Page)
(https://youtube.com/playlist?list=PLzwXZ53h1qj5hONbzUzYQm3_spnB-dm4e)
→ JavaScript Interactivity
(https://www.youtube.com/playlist?list=PLGjplNEQ1it_oTvuLRNqXfz_v_0pq6unW)
→ npm → External Packages
(https://www.youtube.com/playlist?list=PL4cUxeGkcC9jsz4LDYc6kv3ymONOKxwBU)
→ Git → GitHub → Collaborative Work
(https://www.youtube.com/playlist?list=PLhP5RsB7fhE0Z6UKi7eUA3_xlDpWygZT1)
→ Tailwind CSS
(https://www.youtube.com/playlist?list=PL4cUxeGkcC9gpXORlEHjc5bgnIi5HEGhw)
→ React
(https://www.youtube.com/playlist?list=PLYXp821z9yw687QYidfKjkUfFfdd7K57I)
→ Build Frontend Apps (Todo App / Weather App)
(https://www.youtube.com/playlist?list=PLxSn8YMF02XX_qVqzvd8zVSZB4tCoDeDj)
→ Backend Basics (Node.js)
(https://www.youtube.com/playlist?list=PL4cUxeGkcC9jsz4LDYc6kv3ymONOKxwBU)
→ Build CLI Applications
(https://www.youtube.com/playlist?list=PL4cUxeGkcC9jsz4LDYc6kv3ymONOKxwBU)
→ Databases (PostgreSQL)
(https://www.youtube.com/playlist?list=PLL_LQvNX4xKxq57Vs1PYCfH6FI4socDWS)
→ RESTful APIs
(https://www.youtube.com/playlist?list=PL4cUxeGkcC9jsz4LDYc6kv3ymONOKxwBU)

→ Build CRUD Applications
→ Authentication (JWT)
→ Caching (Redis)
(https://www.youtube.com/watch?v=0Df3LfS4HSY)
→ Build Complete Full Stack App (Frontend + Backend)
(https://www.youtube.com/playlist?list=PL4cUxeGkcC9gpXORlEHjc5bgnIi5HEGhw)
→ Linux Basics
(https://www.youtube.com/playlist?list=PLtK75qxsQaMLZSo7KL-PmiRarU7hrpnwK)
→ AWS Basics (EC2 → S3 → VPC → Route53 → SES)
(https://www.youtube.com/playlist?list=PL0X6fGhFFNTcU-_MCPe9dkH6sqmgfhy_M)
→ Deploy Full Stack Application
→ Monitoring (Monit)
→ CI/CD (GitHub Actions)
(https://www.youtube.com/playlist?list=PLArH6NjfKsUhvGHrpag7SuPumMzQRhUKY)
→ Automation (Ansible)
(https://www.youtube.com/playlist?list=PL2_OBreMn7FqZkvMYt6ATmgC0KAGGJNAN)
→ Infrastructure as Code (Terraform)
(https://www.youtube.com/playlist?list=PLGyKZJoyKdOfehoKuFC1PKTGzMtRwnGNd)

related roadmaps :
Frontend Developer Roadmap
Backend Developer Roadmap
DevOps Roadmap
AWS Cloud Roadmap

certificates :

Free

Full Stack Development – freeCodeCamp
(https://www.freecodecamp.org/learn)

AWS Cloud Practitioner Essentials (Free Tier Learning) – AWS Skill Builder
(https://skillbuilder.aws)

Paid

1. Full Stack Web Development with JavaScript – Coursera (IBM)
(https://www.coursera.org/professional-certificates/ibm-full-stack-cloud-developer)

2. Full Stack Developer Career Path – Frontend Masters
(https://frontendmasters.com/learn/fullstack/)

3. The Complete Web Developer Bootcamp – Udemy
(https://www.udemy.com/course/the-complete-web-development-bootcamp/)

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role : Game Developer

skills :
Game Mathematics, Linear Algebra, Vectors, Matrices, Geometry, Quaternions, Game Physics, Collision Detection,
Game Engines (Unity, Unreal, Godot), C++, C#, GDScript, Graphics Programming, Computer Graphics, Shaders, OpenGL,
DirectX, Vulkan, Game AI, Behavior Trees, State Machines, Pathfinding, Multiplayer Basics, Performance Optimization,
Rendering Pipelines, Ray Tracing, Physically Based Rendering (PBR)

learning roadmap :
Game Math Basics (Vectors → Matrices → Linear Transformations)
(https://www.youtube.com/playlist?list=PLZHQObOWTQDPD3MizzM2xVFitgF8hE_ab)

→ Geometry (Affine Space → Affine Transformations)
(https://www.youtube.com/playlist?list=PLP8hWD_Wx5cb7nK9q9a0QcaS4cI1sX4oY)
→ Orientation (Euler Angles → Quaternions)
→ Curves (Bezier → Catmull-Rom → Splines)
→ Projection (Perspective → Orthographic)
→ Game Physics Basics (Force → Velocity → Acceleration → Friction)
(https://www.youtube.com/playlist?list=PLbuK0gG93AsENAa67XysaOr5K0cczxye)
→ Rigid Body Dynamics (Center of Mass → Inertia → Joints)
(https://www.youtube.com/playlist?list=PLbuK0gG93AsENAa67XysaOr5K0cczxye)
→ Collision Detection (Broad Phase → Narrow Phase → SAT → GJK → EPA)
(https://www.youtube.com/playlist?list=PLbuK0gG93AsENAa67XysaOr5K0cczxye)
→ Spatial Partitioning (AABB → OBB → BVH → Sweep & Prune)
(https://www.youtube.com/playlist?list=PLbuK0gG93AsENAa67XysaOr5K0cczxye)
→ Pick a Game Engine (Unity / Unreal / Godot)
(https://www.youtube.com/playlist?list=PLpj8TZGNIBNxe-Sba4mohPSXOxoyicLYs)
→ Programming Language (C++ / C# / GDScript)
(https://www.youtube.com/playlist?list=PLlrATfBNZ98dC-V-N3m0Go4deliWHPFwT)
→ Game Engine Internals
(https://www.youtube.com/playlist?list=PLlrATfBNZ98dC-V-N3m0Go4deliWHPFwT)
→ Computer Graphics Fundamentals (Rasterization → Graphics Pipeline)
(https://www.youtube.com/watch?v=tt4H_u25mX0)
→ Shaders (Vertex → Fragment → Compute)
→ Lighting & Shadows (Shadow Maps → Light Types → Culling)
→ Graphics APIs (OpenGL → DirectX → Vulkan → Metal)
→ Game AI Basics (FSM → Behavior Trees → Decision Trees)
(https://www.youtube.com/watch?v=aR6wt5BlE-E)
→ Pathfinding & Movement
→ Board Game AI (Minimax → Alpha-Beta → MCTS)
(https://www.youtube.com/watch?v=1budEg62yYI)
→ Advanced Game AI (Reinforcement Learning → Neural Networks)
→ Advanced Rendering (PBR → Real-Time Ray Tracing)
→ Multiplayer Concepts (Client-Server → Sync → Lag Compensation)
→ Optimization & Profiling
→ Build & Publish Games

related roadmaps :
Backend Developer Roadmap
API Design Roadmap
Graphics Programmer Roadmap
AI / ML Roadmap

certificates :

Paid

Unreal Engine C++ Developer Courses – Epic Games (via Coursera / Udemy partners)
(https://www.coursera.org/search?query=Unreal%20Engine%20C%2B%2B%20Developer%20Courses%20%E2%80%93%20Epic%20Games%20)

Game Development Specialization – Coursera (Michigan State University)
(https://www.coursera.org/specializations/game-design-and-development)

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role : iOS Developer

skills :
Swift, Objective-C Basics, iOS Architecture, UIKit, SwiftUI, Auto Layout, Xcode, Git, GitHub, MVC, MVVM, VIPER, TCA, Combine, RxSwift, REST APIs, GraphQL, URLSession, Alamofire, Core Data, UserDefaults, Keychain, GCD, Async/Await, Core Animation, Accessibility, XCTest, CI/CD, App Store Deployment, ASO

learning roadmap :
Pick Language (Swift → Objective-C Interoperability)
(https://www.youtube.com/watch?v=CwA1VWP0Ldw)
→ Swift Basics → Advanced Swift
(https://www.youtube.com/watch?v=CwA1VWP0Ldw)
→ iOS Architecture (Core OS → Core Services → Cocoa Touch)
(https://www.youtube.com/results?search_query=stanford+cs193p+2025)
→ Core Concepts (OOP → FP → Memory Management → ViewController Lifecycle)
(https://www.youtube.com/watch?v=CwA1VWP0Ldw)
→ Version Control (Git → GitHub)
→ Xcode Setup → Project Structure → Debugging
(https://www.youtube.com/watch?v=CwA1VWP0Ldw)
→ UIKit Basics (Views → ViewControllers → Storyboards → Xibs)
(https://www.youtube.com/playlist?list=PL8seg1JPkqgF7hGmB0gUU5DKA7wrrMfZg)
→ Navigation (Stacks → Segues → Modals)
→ SwiftUI Basics → State Management → Navigation
(https://www.youtube.com/watch?v=b1oC7sLIgpI)
→ UI Design (Apple HIG)
(https://www.youtube.com/c/AppleDeveloper)
→ Core Animation (Animations → Lottie)
→ Architecture Patterns (MVC → MVVM → VIPER → TCA)
→ Reactive Programming (Combine → RxSwift)
(https://www.youtube.com/playlist?list=PLWHegwAgjOkoIMgZ7QF_SHUtEB_rWXtH0)
→ Callbacks → Closures → Async/Await → Concurrency
(https://www.youtube.com/watch?v=CwA1VWP0Ldw)
→ Data Persistence (UserDefaults → Core Data → Keychain → SQLite)
(https://www.youtube.com/watch?v=rTXkGCg58w0)
→ JSON / XML Parsing
→ Networking (HTTP → REST → GraphQL → URLSession → Alamofire)
→ Dependency Managers (SPM → CocoaPods → Carthage)
→ Frameworks & Libraries (Static → Dynamic → XCFramework)
→ Accessibility (VoiceOver → Dynamic Type)
→ Apple Frameworks (MapKit → Core ML → ARKit → HealthKit)
(https://www.youtube.com/c/AppleDeveloper)
→ Code Quality (SwiftLint → SwiftFormat)
→ Debugging & Profiling (Instruments)
→ Testing (Unit → UI → XCTest → XCUITest)
→ CI/CD (GitHub Actions → Jenkins → Fastlane)
→ App Distribution (TestFlight → App Store)
→ ASO
→ Continuous Learning (WWDC → New iOS SDKs)
(https://www.youtube.com/c/AppleDeveloper)

related roadmaps :
Android Developer Roadmap
Flutter Developer Roadmap
React Native Roadmap
Mobile App Developer Roadmap

certificates :

Free (Best)

1. Stanford CS193p – Developing Apps for iOS (Free, Highly Respected)
   https://cs193p.sites.stanford.edu

2. Swift Playgrounds – Learn Swift (Apple)

https://developer.apple.com/swift-playgrounds/

3. Apple iOS Development Tutorials (Official)
   https://developer.apple.com/tutorials/

Paid (Best – Working & Industry-Accepted)

iOS & Swift – The Complete iOS App Development Bootcamp (Udemy)
(https://www.udemy.com/course/ios-13-app-development-bootcamp/)

Kodeco (Ray Wenderlich) iOS Developer Path
(https://www.kodeco.com/ios)

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role : Java Developer

skills :
Java Syntax, OOP, Collections Framework, Generics, Exception Handling, Multithreading, Concurrency,
Java Memory Model, Streams API, Functional Programming, Lambda Expressions, JVM Basics, Maven, Gradle, Spring Boot,
REST APIs, Hibernate, JPA, JDBC, Unit Testing, Logging, Dependency Injection

learning roadmap :
Java Basics (Syntax → Data Types → Variables → Control Statements) -
(https://www.youtube.com/playlist?list=PLEiEAq2VkUUKgly9dVxyTi_gXFMX5MPfR)
→ Strings → Arrays → Methods → Math Operations
(https://www.youtube.com/playlist?list=PLEiEAq2VkUUKgly9dVxyTi_gXFMX5MPfR)
→ OOP Basics (Classes → Objects → Access Modifiers → Static → Nested Classes)
(https://www.youtube.com/playlist?list=PLEiEAq2VkUUKgly9dVxyTi_gXFMX5MPfR)
→ Advanced OOP (Inheritance → Polymorphism → Abstraction → Interfaces)
(https://www.youtube.com/playlist?list=PLEiEAq2VkUUJITAa4rvcToEG3pLcL0Onh)
→ Object Lifecycle → Enums → Final → Records → Packages
(https://www.youtube.com/playlist?list=PLEiEAq2VkUUJITAa4rvcToEG3pLcL0Onh)
→ Method Overloading & Overriding
(https://www.youtube.com/playlist?list=PL9gnSGHSqcno1G3XjUbwzXHL8_EttOuKk)
→ Encapsulation → Binding (Static vs Dynamic)
(https://www.youtube.com/playlist?list=PL9gnSGHSqcno1G3XjUbwzXHL8_EttOuKk)
→ Initializer Blocks → Pass by Value / Reference
(https://www.youtube.com/playlist?list=PLEiEAq2VkUUKgly9dVxyTi_gXFMX5MPfR)
→ Advanced Java (Annotations → Modules → Optionals → Exception Handling)
(https://www.youtube.com/playlist?list=PLEiEAq2VkUUJITAa4rvcToEG3pLcL0Onh)
→ Collections Framework (List → Set → Map → Queue → Stack → Deque)
(https://www.youtube.com/playlist?list=PLqq-6Pq4lTTYwNIFaCLE9oS6npVj4RzHy)
→ Generics → Iterators
(https://www.youtube.com/playlist?list=PLA3GkZPtsafZZsLj0Tybu3y0HVl-hp1ea)
→ Concurrency (Threads → Virtual Threads → Java Memory Model → volatile)
(https://www.youtube.com/playlist?list=PLL8woMHwr36EDxjUoCzboZjedsnhLP1j4)
→ Functional Programming (Lambda → Functional Interfaces → Streams API)
(https://www.youtube.com/playlist?list=PLqq-6Pq4lTTa9YGfyhyW2CqdtW9RtY-I3)
→ Core Utilities (Date & Time → Regex → Cryptography → Networking)
(https://www.youtube.com/playlist?list=PLEiEAq2VkUUJITAa4rvcToEG3pLcL0Onh)
→ I/O Operations (File Handling)
(https://www.youtube.com/playlist?list=PLEiEAq2VkUUKgly9dVxyTi_gXFMX5MPfR)
→ Dependency Injection
(https://www.youtube.com/playlist?list=PLqq-6Pq4lTTbx8p2oCgcAQGQyqN8XeA1x)
→ Build Tools (Maven → Gradle → Bazel) (https://www.youtube.com/playlist?list=PL92E89440B7BFD0F6) & (https://www.youtube.com/playlist?list=PLzS3AYzXBoj-BsD10hlMzfqN027nrWdWJ)
→ Web Frameworks (Spring Boot → Quarkus → Play → Javalin)
(https://www.youtube.com/playlist?list=PLUcsbZa0qzu0g) &
(https://www.youtube.com/playlist?list=PL6oD2syjfW7ADAkICQr-SQcEqsenVPfqg)
→ Database Access (JDBC → Hibernate → JPA → Spring Data JPA)
(https://www.youtube.com/playlist?list=PLGRDMO4rOGcP4wZM9s3RPsRbUkg27EScN)

→ Logging (SLF4J → Logback → Log4j2)   (https://www.youtube.com/playlist?list=PLUcsbZa0qzu0gVRFlVfscqjD84TqMssOt)
→ Testing (JUnit → TestNG → Mockito → REST Assured → Cucumber) (https://www.youtube.com/playlist?list=PLqq-6Pq4lTTa4ad5JISViSb2FVG8Vwa4o)
→ Build Java Backend Applications (https://www.youtube.com/playlist?list=PLUcsbZa0qzu0gVRFlVfscqjD84TqMssOt)
→ Explore Backend Roadmap for Advanced Topics (https://www.youtube.com/playlist?list=PLL8woMHwr36EDxjUoCzboZjedsnhLP1j4)

related roadmaps :
Backend Developer Roadmap
Spring Boot Roadmap
Full Stack Java Roadmap
Microservices Roadmap

certificates :

Free

1. Java Programming – University of Helsinki (MOOC, Highly Respected)
   https://java-programming.mooc.fi

2. Java Tutorials – Oracle (Official Documentation & Tutorials)
   https://docs.oracle.com/javase/tutorial/

3. Introduction to Java – Codecademy (Free Tier)
   https://www.codecademy.com/learn/learn-java

Paid

1. Java Programming and Software Engineering Fundamentals – Coursera (Duke University) (https://www.coursera.org/specializations/java-programming)

2. Spring & Hibernate for Beginners – Udemy
 (https://www.udemy.com/course/spring-hibernate-tutorial/)

3. Oracle Java Certification (OCP Java SE) (https://www.oracle.com/developer/java-developers/)

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role : Machine Learning Engineer

skills :
Python, NumPy, Pandas, Matplotlib, Seaborn, Statistics, Probability, Linear Algebra, Data Cleaning,
Feature Engineering, Scikit-learn, Supervised Learning, Unsupervised Learning, Reinforcement Learning,
Model Evaluation, Hyperparameter Tuning, TensorFlow, Keras, PyTorch, Deep Learning, CNNs, RNNs, LSTMs,
Transformers, NLP, Explainable AI, Model Deployment Basics

learning roadmap :
Programming Fundamentals
→ Python Basics (Syntax → Data Types → Control Flow → OOP) (https://www.youtube.com/watch?v=u1RKh1kQqaE&list=PLKnIA16_Rmvb1RYR-iTA_hzckhdONtSW4)
→ Python Libraries (NumPy → Pandas → Matplotlib → Seaborn)
→ Math Foundations (Linear Algebra → Calculus → Probability → Statistics)
→ Data Collection (APIs → Databases → Files)

→ Data Cleaning & Preprocessing
→ Feature Engineering (Scaling → Selection → Dimensionality Reduction)
(https://www.youtube.com/watch?v=ZftI2fEz0Fw&list=PLKnIA16_Rmvbr7zKYQuBfsVkjoLcJgxHH)
→ Introduction to Machine Learning
→ Supervised Learning (Regression → Classification → Tree-Based Models)
→ Unsupervised Learning (Clustering → PCA → Autoencoders)
→ Model Evaluation (Metrics → Cross Validation)
→ Reinforcement Learning (Q-Learning → DQN → Policy Gradient)
(https://www.youtube.com/@campusx-official/search?query=reinforcement%20)
→ Deep Learning Basics (Neural Networks → Backpropagation)
(https://www.youtube.com/watch?v=2dH_qjc9mFg&list=PLKnIA16_RmvYuZauWaPlRTC54KxSNLtNn)
→ Deep Learning Libraries (TensorFlow → Keras → PyTorch)
→ CNNs (Image Classification → Segmentation)
→ RNNs & LSTMs
→ Attention & Transformers
→ Advanced Topics (NLP → Explainable AI → Regularization)
→ Model Selection & Hyperparameter Tuning
→ End-to-End ML Projects
(https://www.youtube.com/watch?v=1xtrIEwY_zY&list=PLKnIA16_RmvY5eP91BGPa0vXUYmIdtfPQ)
→ Explore MLOps & AI Engineer Roadmaps
(https://www.youtube.com/@campusx-official/search?query=mlops)

related roadmaps :
AI Engineer Roadmap
MLOps Roadmap
Data Scientist Roadmap
AI Agents Roadmap

certificates :

Free (Best)

1. Machine Learning with Python – freeCodeCamp
   https://www.freecodecamp.org/learn/machine-learning-with-python/

2. Google Machine Learning Crash Course
   https://developers.google.com/machine-learning/crash-course

3. Introduction to Machine Learning – Kaggle Learn

https://www.kaggle.com/learn/intro-to-machine-learning

Paid (Best – Working & Industry-Accepted)

1. Machine Learning Specialization – Andrew Ng (Coursera)
   https://www.coursera.org/specializations/machine-learning-introduction

2. IBM Machine Learning Professional Certificate – Coursera
   https://www.coursera.org/professional-certificates/ibm-machine-learning

3. Deep Learning Specialization – DeepLearning.AI (Coursera)

https://www.coursera.org/specializations/deep-learning

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role : MLOps Engineer

skills :
Python, Bash, Go, Git, GitHub, Linux, Cloud Computing (AWS, Azure, GCP), Docker, Kubernetes,
Machine Learning Fundamentals, Data Engineering, Apache Airflow, Spark, Kafka, CI/CD, ML Pipelines,
Model Deployment, Monitoring, Feature Stores, Experiment Tracking, Infrastructure as Code, Terraform,
ML System Design

learning roadmap :
Programming Fundamentals (Python → Bash → Go)
(https://www.youtube.com/playlist?list=PLWKjhJtqVAblStefaz_YOVpDWqcRScc2s) /
(https://www.youtube.com/playlist?list=PLqq-6Pq4lTTYwNIFaCLE9oS6npVj4RzHy)
→ Version Control (Git → GitHub)
(https://www.youtube.com/playlist?list=PLvCYkSSakdPipXSHcwxPokjDVA3ZRUZDf)
→ Linux Basics
(https://www.youtube.com/playlist?list=PL8A83A276F0D85E70)
→ Cloud Computing (AWS / Azure / GCP)
(https://www.youtube.com/playlist?list=PLWKjhJtqVAblStefaz_YOVpDWqcRScc2s) /
(https://t.me/CloudCertifications)
→ Cloud-native ML Services
(https://www.youtube.com/playlist?list=PLIivdWyY5sqIOyeovvRapCjXCZykZMLAe)
→ Containerization (Docker)
(https://www.youtube.com/playlist?list=PLqq-6Pq4lTTYwNIFaCLE9oS6npVj4RzHy)
→ Container Orchestration (Kubernetes)
(https://www.youtube.com/playlist?list=PLqq-6Pq4lTTZSKAFG6aCDVDP86Qx4lNas)
→ Machine Learning Fundamentals
(https://www.youtube.com/playlist?list=PLoROMvodv4rMiGQp3WXShtMGgzqpfVfbU)
→ Data Engineering Basics
(https://www.youtube.com/playlist?list=PL3MmuxUbc_hJed7dXYoJw8DoCuVH7V1Sb)
→ Data Pipelines (Airflow)
→ Data Ingestion Systems (Kafka → Spark → Flink)
→ Data Lakes & Data Warehouses
→ MLOps Principles
(https://www.youtube.com/playlist?list=PL3MmuxUbc_hIUISrluw_A7wDSmfOhErJK)
→ CI/CD for ML
(https://www.youtube.com/playlist?list=PL3MmuxUbc_hIUISrluw_A7wDSmfOhErJK)
→ Experiment Tracking & Model Registry
(https://www.youtube.com/playlist?list=PLqq-6Pq4lTTa4ad5JISViSb2FVG8Vwa4o)
→ Feature Stores & Data Lineage
(https://www.youtube.com/playlist?list=PLqq-6Pq4lTTa4ad5JISViSb2FVG8Vwa4o)
→ Model Training Pipelines
(https://www.youtube.com/playlist?list=PLqq-6Pq4lTTa4ad5JISViSb2FVG8Vwa4o)
→ Model Serving (Batch → Real-time)
(https://www.youtube.com/playlist?list=PLqq-6Pq4lTTa4ad5JISViSb2FVG8Vwa4o)
→ Monitoring & Observability (Drift → Performance → Logs)
→ Infrastructure as Code (Terraform)
(https://www.youtube.com/playlist?list=PLdpzxOOAlwvLUH6ww022l7OZGakJYP9WY)
→ Scaling ML Systems
→ Continuous Learning & Optimization

related roadmaps :
AI & Data Scientist Roadmap
DevOps Roadmap
Backend Developer Roadmap
Python Roadmap
Go Developer Roadmap

certificates :

Free (Best)

1. MLOps Zoomcamp – DataTalks.Club
   https://github.com/DataTalksClub/mlops-zoomcamp

Paid (Best – Working & Industry-Accepted)

1. Machine Learning Engineering for Production (MLOps) – DeepLearning.AI
   https://www.coursera.org/specializations/machine-learning-engineering-for-production-mlops

2. Google Professional Machine Learning Engineer Certification
   https://cloud.google.com/certification/machine-learning-engineer

3. AWS Certified Machine Learning – Specialty
   https://aws.amazon.com/certification/certified-machine-learning-specialty/

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role : Node.js Developer (Backend)

skills :

JavaScript (ES6+), Node.js, Express.js, REST APIs, HTTP/HTTPS, Async Programming, Promises, async/await,
Event Loop, npm, npx, Package Management, Environment Variables, dotenv, JWT Authentication, Passport.js,
API Security, Middleware, MVC Architecture, MongoDB, Mongoose, SQL Basics, Prisma, Sequelize, File System (fs),
Streams, Error Handling, Logging (Winston, Morgan), Testing (Jest, Vitest), API Testing, Git, GitHub, CLI Tools,
PM2, Docker Basics, Debugging, Performance Optimization

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learning roadmap :

JavaScript Fundamentals (https://www.youtube.com/playlist?list=PL4cUxeGkcC9i9Ae2D9Ee1RvylH38dKuET)
→ Node.js Basics (What/Why/Runtime) (https://www.youtube.com/playlist?list=PLC3y8-rFHvwh8shCMHFA5kWxD9PaPwxaY)
→ Modules (CommonJS → ESM) (https://www.youtube.com/playlist?list=PLC3y8-rFHvwh8shCMHFA5kWxD9PaPwxaY)
→ npm & npx (https://www.youtube.com/playlist?list=PLC3y8-rFHvwh8shCMHFA5kWxD9PaPwxaY)
→ Async Programming (Callbacks → Promises → async/await) (https://www.youtube.com/playlist?list=PLC3y8-rFHvwh8shCMHFA5kWxD9PaPwxaY)
→ Event Loop & Event Emitter (https://www.youtube.com/playlist?list=PLC3y8-rFHvwh8shCMHFA5kWxD9PaPwxaY)
→ File System & Path Handling (https://www.youtube.com/playlist?list=PLC3y8-rFHvwh8shCMHFA5kWxD9PaPwxaY)
→ CLI Tools with Node.js (https://www.youtube.com/playlist?list=PLC3y8-rFHvwh8shCMHFA5kWxD9PaPwxaY)
→ Express.js / Fastify (https://www.youtube.com/watch?v=f2EqECiTBL8)
→ REST API Design (https://www.youtube.com/watch?v=f2EqECiTBL8)
→ Authentication (JWT, Passport)

(https://www.youtube.com/watch?v=f2EqECiTBL8)
→ Environment Configuration
→ Databases (MongoDB → SQL)
(https://www.youtube.com/watch?v=f2EqECiTBL8)
→ ORM / ODM (Mongoose, Prisma)
(https://www.youtube.com/watch?v=f2EqECiTBL8)
→ Testing (Jest / Vitest)
→ Logging & Monitoring
(https://www.youtube.com/watch?v=f2EqECiTBL8)
→ Process Management (PM2)
(https://www.youtube.com/playlist?list=PLC3y8-rFHvwh8shCMHFA5kWxD9PaPwxaY)
→ Performance & Debugging
(https://www.youtube.com/playlist?list=PLC3y8-rFHvwh8shCMHFA5kWxD9PaPwxaY)
→ Backend System Design
(https://www.youtube.com/watch?v=f2EqECiTBL8)
→ Production Deployment
(https://www.youtube.com/playlist?list=PLC3y8-rFHvwh8shCMHFA5kWxD9PaPwxaY)
→ Backend Roadmap (Advanced)
(https://t.me/JavaScript_js_learn)

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related roadmaps:

JavaScript Roadmap

Backend Roadmap

DevOps Basics Roadmap

System Design Roadmap

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certificates :

free :

1. **Node.js Certification – OpenJS Foundation (Free Courses)**

   * [https://training.linuxfoundation.org/training/introduction-to-nodejs/]
(https://training.linuxfoundation.org/training/introduction-to-nodejs/)

2. **Backend Development with Node.js – FreeCodeCamp**

   * [https://www.freecodecamp.org/learn/back-end-development-and-apis/]
(https://www.freecodecamp.org/learn/back-end-development-and-apis/)

3. **JavaScript & Node.js Courses – OpenClassrooms**

   * [https://openclassrooms.com/en/search?page=1&query=node.js](https://
openclassrooms.com/en/search?page=1&query=node.js)

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paid :

2. **Meta Back-End Developer Professional Certificate (Coursera)**
   (Meta / Facebook)

   * [https://www.coursera.org/professional-certificates/meta-back-end-
developer](https://www.coursera.org/professional-certificates/meta-back-end-
developer)

3. **IBM Backend Development Professional Certificate**

   * https://www.coursera.org/professional-certificates/ibm-backend-development

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role : PostgreSQL Database Administrator (PostgreSQL DBA)

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skills :

PostgreSQL, Relational Databases, SQL, ACID, MVCC, Transactions, Write-Ahead Logging (WAL), Database Design,
Normalization, Indexing, Query Optimization, EXPLAIN & ANALYZE, Backup & Recovery, pg_dump, pg_restore,
pg_basebackup, Replication (Streaming & Logical), High Availability, Patroni, PgBouncer, Database Security,
Roles & Privileges, Row-Level Security, SSL, Performance Tuning, Vacuum & Autovacuum,
Monitoring (Prometheus, pg_stat_activity), Linux System Administration, Docker, Kubernetes, Ansible,
Capacity Planning, Troubleshooting, PL/pgSQL, Stored Procedures, Triggers

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learning roadmap :

Relational Database Fundamentals (https://www.youtube.com/playlist?list=PLf0swTFhTI8oQNiY2ZTf81zvKB7-y3y4e)
→ PostgreSQL Overview & Comparison (https://www.youtube.com/playlist?list=PLf0swTFhTI8oQNiY2ZTf81zvKB7-y3y4e)
→ Core RDBMS Concepts (Tables, Schemas, Constraints) (https://www.youtube.com/playlist?list=PLf0swTFhTI8oQNiY2ZTf81zvKB7-y3y4e)
→ ACID, MVCC & Transactions (https://www.youtube.com/playlist?list=PLdTaEgcmPg9KiTCPWh-K961tiZrvhgfFu)
→ PostgreSQL Installation (Local → Docker → Cloud) (https://www.youtube.com/playlist?list=PLL_LQvNX4xKxq57Vs1PYCfH6FI4socDWS)
→ psql & Server Management (https://www.youtube.com/playlist?list=PLL_LQvNX4xKxq57Vs1PYCfH6FI4socDWS)
→ SQL Fundamentals (DDL → DML → Joins) (https://www.youtube.com/watch?v=U4qvLA1QwEE)
→ Advanced SQL (CTE, Subqueries, Window Functions) (https://www.youtube.com/watch?v=U4qvLA1QwEE)
→ PostgreSQL Configuration (postgresql.conf) (https://www.youtube.com/playlist?list=PLSU32T6nmU24rB_xwGxwIBuIL-dMG1LAt)
→ WAL, Vacuum & Autovacuum (https://www.youtube.com/playlist?list=PLdTaEgcmPg9KiTCPWh-K961tiZrvhgfFu)
→ Authentication & Security (https://www.youtube.com/playlist?list=PLSU32T6nmU24rB_xwGxwIBuIL-dMG1LAt)
→ Roles, Privileges & RLS (https://www.youtube.com/playlist?list=PLSU32T6nmU24rB_xwGxwIBuIL-dMG1LAt)
→ Backup & Restore Strategies (https://www.youtube.com/playlist?list=PLSU32T6nmU24rB_xwGxwIBuIL-dMG1LAt)
→ Replication (Streaming → Logical) (https://www.youtube.com/playlist?list=PLdTaEgcmPg9KiTCPWh-K961tiZrvhgfFu)
→ Connection Pooling (PgBouncer) (https://www.youtube.com/playlist?list=PL3WN5UzLiADd-sjn-aPVVfBlkXPO9IoG7)
→ Monitoring & Metrics (https://www.youtube.com/playlist?list=PL3WN5UzLiADd-sjn-aPVVfBlkXPO9IoG7)
→ Query Optimization & Indexing

(https://www.youtube.com/playlist?list=PLL_LQvNX4xKxq57Vs1PYCfH6FI4socDWS)
→ High Availability (Patroni)
(https://www.youtube.com/playlist?list=PL3WN5UzLiADd-sjn-aPVVfBlkXPO9IoG7)
→ Kubernetes Deployment
(https://www.youtube.com/playlist?list=PL3WN5UzLiADd-sjn-aPVVfBlkXPO9IoG7)
→ Automation (Shell, Ansible)
(https://t.me/postgres_tech)
→ Advanced Internals
(https://www.youtube.com/playlist?list=PLdTaEgcmPg9KiTCPWh-K961tiZrvhgfFu)
→ Troubleshooting & Performance Tuning
(https://www.youtube.com/playlist?list=PLSU32T6nmU24rB_xwGxwIBuIL-dMG1LAt)
→ PostgreSQL DBA (Production Level)
(https://www.youtube.com/playlist?list=PLSU32T6nmU24rB_xwGxwIBuIL-dMG1LAt)

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related roadmaps:

Backend Roadmap

DevOps Roadmap

Linux Roadmap

Cloud Infrastructure Roadmap

Database Internals Roadmap

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certificates :

free best certificates (3)

1. **PostgreSQL for Everybody – University of Michigan (Free Audit)**

   * [https://www.coursera.org/specializations/postgresql-for-everybody]
(https://www.coursera.org/specializations/postgresql-for-everybody)

2. **PostgreSQL Tutorial – PostgreSQL Official Documentation**

   *
[https://www.postgresql.org/docs/current/tutorial.html](https://www.postgresql.o
rg/docs/current/tutorial.html)

3. **SQL & PostgreSQL – FreeCodeCamp**

   * [https://www.freecodecamp.org/learn/relational-database/](https://
www.freecodecamp.org/learn/relational-database/)

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paid best certificates (mnc-recognized) (3)

3. **AWS Database Specialty (PostgreSQL-heavy workloads)**

   * [https://aws.amazon.com/certification/certified-database-specialty/]
(https://aws.amazon.com/certification/certified-database-specialty/)

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role : Product Manager

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skills :

Product Strategy, Product Lifecycle Management, Market Research, User Research, Competitive Analysis,
Requirement Gathering, PRD Writing, User Stories, Agile Methodologies, Scrum, Kanban, Roadmap Planning,
Feature Prioritization, Stakeholder Management, Go-To-Market Strategy, MVP Development, UX/UI Understanding,
Design Thinking, A/B Testing, Product Analytics, Data-Driven Decision Making, Business Strategy,
Growth Metrics (DAU, MAU, LTV, CAC, Churn, Retention), Risk Management, Leadership, Communication,
Cross-functional Collaboration, Product Launch Strategy, Customer Feedback Analysis, Predictive Analytics,
AI in Product Management, Tools (Jira, Notion, Productboard, Aha, Amplitude, Slack)

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learning roadmap :

Product Management Fundamentals
(https://www.youtube.com/playlist?list=PLlS2e7Ia7AHSP39FK0FgHfOxfFio0Hz_U)
→ Product vs Project Management
(https://www.youtube.com/playlist?list=PLoROMvodv4rMZs66WrOI0mRJ9placS8QP)
→ Product Lifecycle (Idea → Development → Growth → Maturity → Decline)
(https://www.youtube.com/playlist?list=PLekZZ6NGVd_TeGm7XfoXrz3SZBd9z7kBV)
→ Idea Generation & Problem Framing
(https://www.youtube.com/playlist?list=PLlS2e7Ia7AHSP39FK0FgHfOxfFio0Hz_U)
→ Market Research & Competitive Analysis
(https://www.youtube.com/playlist?list=PLoROMvodv4rMZs66WrOI0mRJ9placS8QP)
→ User Research & Personas (https://t.me/practicalpm)
→ Product Positioning & USP
(https://www.youtube.com/playlist?list=PLlS2e7Ia7AHSP39FK0FgHfOxfFio0Hz_U)
→ Product Strategy & Vision Planning
(https://www.youtube.com/playlist?list=PLoROMvodv4rMZs66WrOI0mRJ9placS8QP)
→ Defining Goals & Value Proposition
(https://www.youtube.com/playlist?list=PLlS2e7Ia7AHSP39FK0FgHfOxfFio0Hz_U)
→ Strategic Thinking & Competitive Advantage
(https://www.youtube.com/playlist?list=PLoROMvodv4rMZs66WrOI0mRJ9placS8QP)
→ Product Planning (PRDs → User Stories → Job Stories)
(https://www.youtube.com/playlist?list=PLCzMmE8BpR1-xedOC4Oi4kSH1f7ef-Q8x)
→ Product Roadmapping & Backlog Management
(https://www.youtube.com/playlist?list=PLoROMvodv4rMZs66WrOI0mRJ9placS8QP)
→ UX/UI Fundamentals & Design Thinking
(https://www.youtube.com/playlist?list=PLlS2e7Ia7AHSP39FK0FgHfOxfFio0Hz_U)
→ User Testing & Validation (Usability → A/B Testing)
(https://www.youtube.com/playlist?list=PLoROMvodv4rMZs66WrOI0mRJ9placS8QP)
→ Agile Development (Scrum → Kanban → Sprint Planning)
(https://www.youtube.com/playlist?list=PLN8NhqFJ6PSKHskITUYAeJQ_GLHEaAjh-)
→ MVP Development & Iteration
(https://www.youtube.com/playlist?list=PLlS2e7Ia7AHSP39FK0FgHfOxfFio0Hz_U)
→ Go-To-Market Strategy & Product Launch
(https://www.youtube.com/playlist?list=PLoROMvodv4rMZs66WrOI0mRJ9placS8QP)
→ Product Metrics & Analytics
(https://www.youtube.com/playlist?list=PLCzMmE8BpR1-xedOC4Oi4kSH1f7ef-Q8x)
→ Data-Driven Product Decisions
(https://www.youtube.com/playlist?list=PLoROMvodv4rMZs66WrOI0mRJ9placS8QP)
→ Stakeholder Management & Communication
(https://t.me/practicalpm)
→ Risk Management & Contingency Planning

→ Scaling Products & Growth Strategy
(https://www.youtube.com/playlist?list=PLoROMvodv4rMZs66WrOI0mRJ9placS8QP)
→ Leadership & Team Management
(https://www.youtube.com/playlist?list=PLlS2e7Ia7AHSP39FK0FgHfOxfFio0Hz_U)
→ AI & ML Applications in Product Management
(https://www.youtube.com/playlist?list=PLxS3M9r3X-_lEXQoGmK6UM6l6Ec1kszMA)
→ Senior Product Management / Product Leadership
(https://www.youtube.com/playlist?list=PLoROMvodv4rMZs66WrOI0mRJ9placS8QP)

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related roadmaps:

UX Design Roadmap

Data Analytics Roadmap
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certificates :

free

1. **Digital Product Management – University of Virginia (Free Audit)**

   * [https://www.coursera.org/learn/uva-darden-digital-product-management]
(https://www.coursera.org/learn/uva-darden-digital-product-management)

2. **Product Management Fundamentals – Great Learning**

   * [https://www.mygreatlearning.com/academy/learn-for-free/courses/product-
management](https://www.mygreatlearning.com/academy/learn-for-free/courses/
product-management)

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paid

1. **Google Product Management Professional Certificate**

(https://www.coursera.org/search?
utm_medium=sem&utm_source=gg&utm_campaign=b2c_india_x_coursera_ftcof_courseraplu
s_cx_dr_bau_gg_sem_bd-ph_in_all_m_hyb_23-
09_x&campaignid=20590309416&adgroupid=155702724604&device=c&keyword=coursera%20c
ourses&matchtype=e&network=g&devicemodel=&creativeid=675426312946&assetgroupid=&
targetid=kwd-38640378270&extensionid=&placement=&gad_source=1&gad_campaignid=205
90309416&gclid=Cj0KCQiAnJHMBhDAARIsABr7b84Zg0EBd_xb2YQGEVB7GGOj8-
sx5UosoVE1cm_LTxLnOi55tpaJhuIaAnf0EALw_wcB&query=google%20project%20management%3
A%20professional%20certificate)

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role : Prompt Engineer / LLM Engineer

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skills :

Prompt Engineering, Large Language Models (LLMs), AI Fundamentals, NLP Basics,
Tokenization,
Context Window Management, Prompt Design Patterns, Zero-Shot Prompting, Few-Shot

Prompting,
Chain-of-Thought Prompting, ReAct Prompting, Tree-of-Thoughts, System & Role Prompting,
Structured Output Design (JSON, XML, Markdown), LLM Configuration (Temperature, Top-K, Top-P),
Prompt Optimization, Prompt Injection Defense, RAG (Retrieval Augmented Generation), AI Agents,
Prompt Evaluation & Testing, AI Reliability Techniques, Prompt Versioning, AI Ethics & Bias Mitigation,
OpenAI API, Google Gemini API, Anthropic Claude API, HuggingFace Models, Automation with Python,
LLM Cost & Latency Optimization

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learning roadmap :

AI & LLM Fundamentals (https://www.youtube.com/playlist?list=PLLB0NZP0teC8ddQO6T146KC-mrg4onfR7)
→ What are LLMs & How They Work (https://www.youtube.com/playlist?list=PLLB0NZP0teC8ddQO6T146KC-mrg4onfR7)
→ Prompt Basics & Terminology (https://www.youtube.com/playlist?list=PLh45-N6HwAAJScVqfs1ZLVBd2QhYdwD51)
→ Popular LLM Providers (OpenAI → Google → Anthropic → Meta → xAI) (https://www.youtube.com/playlist?list=PLLB0NZP0teC8ddQO6T146KC-mrg4onfR7)
→ Tokens, Context Windows & Hallucinations (https://www.youtube.com/playlist?list=PLLB0NZP0teC8ddQO6T146KC-mrg4onfR7)
→ Prompt Engineering vs Fine-Tuning (https://www.youtube.com/playlist?list=PLh45-N6HwAAJScVqfs1ZLVBd2QhYdwD51)
→ LLM Configuration (Temperature, Top-K, Top-P, Max Tokens) (https://www.youtube.com/playlist?list=PLLB0NZP0teC8ddQO6T146KC-mrg4onfR7)
→ Output Control & Structured Responses (https://www.youtube.com/playlist?list=PLh45-N6HwAAJScVqfs1ZLVBd2QhYdwD51)
→ Zero-Shot Prompting (https://www.youtube.com/playlist?list=PLh45-N6HwAAJScVqfs1ZLVBd2QhYdwD51)
→ One-Shot & Few-Shot Prompting (https://www.youtube.com/playlist?list=PLh45-N6HwAAJScVqfs1ZLVBd2QhYdwD51)
→ System, Role & Contextual Prompting (https://www.youtube.com/playlist?list=PLh45-N6HwAAJScVqfs1ZLVBd2QhYdwD51)
→ Chain-of-Thought Prompting
→ Self-Consistency Prompting
→ Tree-of-Thoughts Prompting
→ ReAct Prompting
→ Automatic Prompt Engineering
→ Prompt Engineering Best Practices
→ Prompt Injection & Security (https://www.youtube.com/playlist?list=PL-D1lWyrp4YOmiVHvJbJrEn19OvkjZ1E)
→ Improving Reliability (Debiasing, Ensembling, Self-Eval) (https://www.youtube.com/playlist?list=PLZwOUWv9ASxpLe45KE5cZWQLDvRwjVlTk)
→ Prompt Evaluation & Versioning
→ RAG & AI Agents (https://www.youtube.com/playlist?list=PLqKyMPlYCZ6ecWZqJdmD5gyjM5aqcUIRY)
→ AI Red Teaming Basics (https://www.youtube.com/playlist?list=PL-D1lWyrp4YOmiVHvJbJrEn19OvkjZ1E)
→ Production Prompt Engineering (https://www.youtube.com/playlist?list=PLh45-N6HwAAJScVqfs1ZLVBd2QhYdwD51)
→ AI Engineer / LLM Engineer Path (https://www.youtube.com/playlist?list=PLLB0NZP0teC8ddQO6T146KC-mrg4onfR7)

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related roadmaps:

AI Engineer Roadmap

AI Agents Roadmap

MLOps Roadmap

AI Red Teaming Roadmap

Data Scientist Roadmap

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certificates :

free

1. **Prompt Engineering for Developers – DeepLearning.AI**

   * https://www.deeplearning.ai/short-courses/chatgpt-prompt-engineering-for-developers/

2. **Generative AI Fundamentals – Google (Skillshop)**

   * https://www.cloudskillsboost.google/paths/118

3. **Introduction to Generative AI – AWS**

   * https://explore.skillbuilder.aws/learn/course/external/view/elearning/17566/introduction-to-generative-ai

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paid

1. **Generative AI with LLMs – DeepLearning.AI (Andrew Ng)**

   * https://www.coursera.org/learn/generative-ai-with-llms

3. **IBM Generative AI Engineering Professional Certificate**

   * https://www.coursera.org/professional-certificates/ibm-generative-ai-engineering

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role : Python Developer / Software Engineer (Python)

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skills :

Python, Core Python Syntax, Data Structures, Algorithms, Object-Oriented Programming, Exception Handling,
Functional Programming, Regular Expressions, Iterators & Generators, Decorators, Context Managers,
Package Management (pip, PyPI, Conda, Poetry), Virtual Environments, Frameworks (Django, Flask, FastAPI),
REST APIs, Async Programming, Threading, Multiprocessing, GIL, Static Typing

(typing, mypy, Pydantic),
Configuration Management, Code Formatting (Black, Ruff), Unit Testing (pytest, unittest), Documentation (Sphinx),
Debugging, Git, Linux Basics, DevOps Fundamentals, CI/CD, Docker Basics

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learning roadmap :

Python Fundamentals (https://www.youtube.com/playlist?list=PL-osiE80TeTt2d9bfVyTiXJA-UTHn6WwU)
→ Syntax, Variables & Data Types (https://www.youtube.com/playlist?list=PLu0W_9lII9agwh1XjRt242xIpHhPT2llg)
→ Conditionals & Loops (https://www.youtube.com/playlist?list=PLu0W_9lII9agwh1XjRt242xIpHhPT2llg)
→ Functions & Exception Handling (https://www.youtube.com/playlist?list=PL-osiE80TeTt2d9bfVyTiXJA-UTHn6WwU)
→ Built-in Data Structures (List, Tuple, Set, Dict) (https://www.youtube.com/playlist?list=PLu0W_9lII9agwh1XjRt242xIpHhPT2llg)
→ Data Structures & Algorithms (https://www.youtube.com/playlist?list=PLWuFHho1zKhVMGPh4dfGhObiABPuzam5E)
→ Recursion & Sorting Techniques (https://www.youtube.com/playlist?list=PLj8W7XIvO93rJHSYzkk7CgfiLQRUEC2Sq)
→ Modules (Built-in & Custom) (https://www.youtube.com/playlist?list=PL-osiE80TeTt2d9bfVyTiXJA-UTHn6WwU)
→ Advanced Python Features (Lambdas, Decorators, Generators) (https://www.youtube.com/playlist?list=PLqnslRFeH2UqLwzS0AwKDKLrpYBKzLBy2)
→ Regular Expressions & Context Managers (https://www.youtube.com/playlist?list=PL-osiE80TeTt2d9bfVyTiXJA-UTHn6WwU)
→ Object-Oriented Programming (https://www.youtube.com/playlist?list=PL-osiE80TeTsqhIuOqKhwlXsIBIdSeYtc)
→ Package Management (pip → Poetry → Conda) (https://www.youtube.com/playlist?list=PLdpzxOOAlwvKwTyYNJCUwGPvql0TrsPgv)
→ Virtual Environments & Configuration (https://www.youtube.com/playlist?list=PLdpzxOOAlwvKwTyYNJCUwGPvql0TrsPgv)
→ Framework Selection (Flask / Django / FastAPI) (https://www.youtube.com/playlist?list=PLdpzxOOAlwvKwTyYNJCUwGPvql0TrsPgv)
→ REST API Development (https://www.youtube.com/playlist?list=PLL2hlSFBmWwzrhahyQscxASc3ZQFh95xl)
→ Async Programming & Concurrency (https://www.youtube.com/playlist?list=PLyb_C2HpOQSBWGekd7PfhHnb9GnqDgrxS)
→ Threading, Multiprocessing & GIL (https://www.youtube.com/playlist?list=PLGKQkV4guDKEv1DoK4LYdo2ZPLo6cyLbm)
→ Static Typing & Data Validation
→ Code Formatting & Linting (https://www.youtube.com/playlist?list=PLOLrQ9Pn6cazlOQ_2_FkqWzyov8ikiDXN)
→ Documentation with Sphinx (https://www.youtube.com/playlist?list=PLzHdTn7Pdxs4QhD7gJMbJQ59g4pLpmNWJ)
→ Testing (pytest, unittest, tox) (https://www.youtube.com/playlist?list=PLyb_C2HpOQSBWGekd7PfhHnb9GnqDgrxS)
→ Common Python Libraries (https://www.youtube.com/playlist?list=PL-osiE80TeTt2d9bfVyTiXJA-UTHn6WwU)
→ DevOps Integration (https://www.youtube.com/playlist?list=PLdpzxOOAlwvKwTyYNJCUwGPvql0TrsPgv)
→ Backend / AI / DevOps Specialization (https://www.youtube.com/playlist?list=PLdpzxOOAlwvKwTyYNJCUwGPvql0TrsPgv)

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related roadmaps:

Backend Roadmap

DevOps Roadmap

AI & Data Science Roadmap

Cloud Computing Roadmap

System Design Roadmap

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certificates :

free best certificates (3)

1. **Python for Everybody – University of Michigan (Free Audit)**

   * https://www.coursera.org/specializations/python

2. **Scientific Computing with Python – FreeCodeCamp**

   * https://www.freecodecamp.org/learn/scientific-computing-with-python/

3. **Introduction to Python – Google (Crash Course)**

   * https://developers.google.com/edu/python

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paid best certificates (mnc-recognized) (3)

1. **Google IT Automation with Python Professional Certificate**

   * https://www.coursera.org/professional-certificates/google-it-automation

2. **IBM Python for Data Science & AI**

   * https://www.coursera.org/professional-certificates/ibm-python-for-data-science-ai

3. **PCAP – Certified Associate in Python Programming (Python Institute)**

   * https://pythoninstitute.org/pcap

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role : QA Engineer / Software Test Engineer

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skills :

Quality Assurance, Software Testing Fundamentals, QA Mindset, SDLC Models, Agile Testing, Scrum, Kanban,
Test Planning, Test Case Design, Test Scenarios, Test Oracles, Test Prioritization, Manual Testing,
Functional Testing, Non-Functional Testing, Regression Testing, Smoke Testing, Sanity Testing, Exploratory Testing,
Integration Testing, UAT, Automation Testing, Selenium, Playwright, Cypress,

WebdriverIO, API Testing, Postman,
REST Assured, SoapUI, Mobile Testing, Appium, Performance Testing, JMeter, K6, Gatling, Security Testing,
OWASP Top 10, Accessibility Testing, Bug Tracking, Test Management Tools, CI/CD Integration, Git, Jenkins,
Reporting & Metrics, Monitoring & Logs

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learning roadmap :

QA Fundamentals & QA Mindset (https://www.youtube.com/playlist?list=PLUDwpEzHYYLseflPNg0bUKfLmAbO2JnE9)
→ Testing Approaches (Black Box → Gray Box → White Box) (https://www.youtube.com/playlist?list=PLUDwpEzHYYLseflPNg0bUKfLmAbO2JnE9)
→ Project & Test Management Tools (https://www.youtube.com/playlist?list=PLXOcVQr8g7b8Ud4lyMyiOawiBQ6hPMYZu)
→ Test Oracles & Test Prioritization (https://www.youtube.com/playlist?list=PLj5VKaW115t1K8AM6smuZiIENBi_VgfR5)
→ SDLC Models (Waterfall → V-Model → Agile) (https://www.youtube.com/playlist?list=PLsGOlyTzNH6fWJsttBPaoUuH474v7Dlda)
→ Agile Frameworks (Scrum → Kanban → XP → SAFe) (https://www.youtube.com/playlist?list=PLEiEAq2VkUUJKZEL96YBV78Dd5gsHSVZ7)
→ Core Testing Techniques (https://www.youtube.com/playlist?list=PLUDwpEzHYYLseflPNg0bUKfLmAbO2JnE9)
→ Functional Testing (https://www.youtube.com/playlist?list=PLlUdEXI83_Xpuhn-9eWN5bSKmbCmuVy1q)
→ Non-Functional Testing (https://www.youtube.com/playlist?list=PLYieqQNFjoYtU-KenCjRJEIz-tW-tsyT8)
→ Exploratory, Smoke, Sanity & Regression Testing (https://www.youtube.com/playlist?list=PLsGOlyTzNH6cNRE1VEVBNx7qqJBaa3Qmu)
→ Manual Testing (Test Plans → Test Cases → Verification & Validation) (https://www.youtube.com/playlist?list=PLUDwpEzHYYLseflPNg0bUKfLmAbO2JnE9)
→ TDD Basics (https://www.youtube.com/playlist?list=PLngnoZX8cAn-wnIWYBbA45iAGxCxJ87w7)
→ Frontend Automation Fundamentals (https://www.youtube.com/channel/UCdsG6Ebb5UBHOhltsofrsjg)
→ Web Technologies (HTML, CSS, JS, Ajax) (https://www.youtube.com/playlist?list=PLM-9L0IKovKIXbrK68pnz0fybtHNBOxWii)
→ UI Automation Tools (Selenium → Playwright → Cypress) (https://www.youtube.com/playlist?list=PLUDwpEzHYYLtvMVjc-Va3RXLKtdB-ofc2)
→ Backend / API Automation (https://www.youtube.com/playlist?list=PLUDwpEzHYYLuW9XEvFEJk2kqsk6HqscI4)
→ Mobile Automation (https://www.youtube.com/playlist?list=PLhW3qG5bs-L8BQaqLpjt5792eMqirve-p)
→ Performance & Load Testing (https://www.youtube.com/playlist?list=PLhW3qG5bs-L9EF8wFBxBY4RU4WzG9X6gd)
→ Security Testing (OWASP Top 10) (https://www.youtube.com/playlist?list=PLHL1eq0OycZi3C9Q-eSYh51E5C7psWOTM)
→ Accessibility Testing (https://www.youtube.com/playlist?list=PLsvet3tE97XcMTCT6U_tZKkGYIn0qeRuD)
→ Email Testing (https://www.youtube.com/playlist?list=PL0anWyn07uYRZiBCeYWVM8itRcZlN6CSe)
→ Test Reporting & Dashboards (https://www.youtube.com/playlist?list=PLUDwpEzHYYLsA2B806kcYQBdYxeMCk39d)
→ Monitoring & Log Analysis (https://www.youtube.com/playlist?list=PLl00VsGSqbtxjr2XGhEgrm_Fg_V3mTMkV)
→ Version Control (Git & Repos) (https://www.youtube.com/playlist?list=PLUDwpEzHYYLuMNNTu0EdWBeBtXdRhj4Lv)
→ CI/CD for QA (https://www.youtube.com/playlist?list=PLUDwpEzHYYLtrdDyf3z-0fybtHNcqAdj5)
→ Headless Testing (https://www.youtube.com/playlist?list=PLvmaC-XMqeBYL-_0pJu6MpShhmWocXj-G)
→ Advanced QA / SDET Path

(https://www.youtube.com/playlist?list=PLUDwpEzHYYLtEENXAFbVPrLeLNHM6OLk7)

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related roadmaps:

Backend Roadmap

Full Stack Roadmap

DevOps Roadmap

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certificates :

free

1. **Software Testing Fundamentals – Great Learning**

   * https://www.mygreatlearning.com/academy/learn-for-free/courses/software-testing-fundamentals

2. **Quality Assurance Full Course – FreeCodeCamp**

   * https://www.freecodecamp.org/news/software-testing-tutorial/

3. **Agile Testing – Atlassian Agile Coach**

   * https://www.atlassian.com/agile/testing

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paid

1. **ISTQB Certified Tester – Foundation Level (CTFL)**
   (Highly valued by Accenture, TCS, Infosys, Cognizant)

   * https://www.istqb.org/certifications/certified-tester-foundation-level

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role : React Developer / Frontend Engineer

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skills :

JavaScript (ES6+), React.js, Functional Components, JSX, Props & State, Component Lifecycle,
Hooks (useState, useEffect, useContext, useReducer, useMemo, useCallback), Custom Hooks, Conditional Rendering,
Lists & Keys, Event Handling, React Router, State Management (Context API, Zustand, Jotai, MobX), REST APIs,
GraphQL, Axios, React Query, SWR, Redux Toolkit Query, HTML5, CSS3, Tailwind CSS, CSS Modules,
Component Libraries (Material UI, Chakra UI, Shadcn UI), Headless UI (Radix UI, React Aria),

Forms (React Hook Form, Formik), TypeScript, Zod, Testing (Jest, Vitest, React Testing Library, Cypress,
Playwright), Next.js, Performance Optimization, Error Boundaries, Suspense, Git, GitHub, Responsive Design,
Web Accessibility

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learning roadmap :

JavaScript Fundamentals (Beginner Level)
(https://www.youtube.com/playlist?list=PL4cUxeGkcC9i9Ae2D9Ee1RvylH38dKuET)
→ ES6+ Concepts
(https://www.youtube.com/playlist?list=PLDtWoQ-cxqiwamnwULkiPkzx8Iv0ny9eV)
→ CLI Tools (Vite)
(https://www.youtube.com/playlist?list=PLXDouhCU5r6qxq4Ad3bXpzJEQtRvMKuST)
→ React Basics & Setup
(https://www.youtube.com/playlist?list=PLC3y8-rFHvwgg3vaYJgHGnModB54rxOk3)
→ Functional Components & JSX
(https://www.youtube.com/playlist?list=PL0Zuz27SZ-6PrE9srvEn8nbhOOyxnWXfp)
→ Props vs State
(https://www.youtube.com/playlist?list=PLC3y8-rFHvwgg3vaYJgHGnModB54rxOk3)
→ Conditional Rendering & Composition
(https://www.youtube.com/playlist?list=PLC3y8-rFHvwg9D7EOSEBabuutIdKZN5V3)
→ Rendering Concepts (Lists, Keys, Events, Refs)
(https://www.youtube.com/playlist?list=PLC3y8-rFHvwgg3vaYJgHGnModB54rxOk3)
→ Component Lifecycle Concepts
(https://www.youtube.com/playlist?list=PLC3y8-rFHvwgg3vaYJgHGnModB54rxOk3)
→ Hooks (useState → useEffect)
(https://www.youtube.com/playlist?list=PLN3n1USn4xlmyw3ebYuZmGp60mcENitdM)
→ Common Hooks (useMemo, useCallback, useRef, useReducer, useContext)
→ Custom Hooks & Best Practices
→ Routing (React Router → TanStack Router)
(https://www.youtube.com/playlist?list=PLOQjd5dsGSxJilh0lBofeY8Qib98kzmF5)
→ State Management (Context → Zustand → Jotai → MobX)
(https://t.me/javascript_js_learn)
→ Styling in React (Tailwind → CSS Modules → Panda CSS)
(https://www.youtube.com/playlist?list=PLXDouhCU5r6q0PaiOBzMBMOm-c3F2y2oh)
→ UI Component Libraries
(https://www.youtube.com/playlist?list=PLZlA0Gpn_vH-aEDXnaFNLsqiJWFpIWV03)
→ Headless Component Libraries
(https://www.youtube.com/playlist?list=PLZlA0Gpn_vH-aEDXnaFNLsqiJWFpIWV03)
→ API Integration (REST → GraphQL)
(https://www.youtube.com/playlist?list=PL55RiY5tL51rG1x02Yyj93iypVuAqw7tw)
→ Data Fetching Libraries
(https://www.youtube.com/playlist?list=PL4cUxeGkcC9hurlXVUhNOK5bz0HWHgKx5)
→ Forms & Validation
(https://www.youtube.com/playlist?list=PL4cUxeGkcC9hNokByJilIsH3DQ0fW4dqf)
→ TypeScript with React
(https://www.youtube.com/playlist?list=PL4cUxeGkcC9ixPUCt9GKeE3IiTeUjiscl)
→ Testing React Applications
(https://www.youtube.com/playlist?list=PLC3y8-rFHvwirqe1KHFCHJ0RqNuN61SJd)
→ React Frameworks (Next.js → Astro)
(https://www.youtube.com/playlist?list=PL4cUxeGkcC9hcFcSbOgxwvapqcyUTSJrF)
→ Advanced React Topics
(https://www.youtube.com/playlist?list=PL7pEw9n3GkoUZX3e9lBbmQqdCpgAWYooz)
→ Performance Optimization
(https://www.youtube.com/playlist?list=PL4cUxeGkcC9i2v2ZqJgydXIcRq_ZuEHvi)
→ Mobile Apps with React Native
(https://www.youtube.com/playlist?list=PL4cUxeGkcC9hZYDtBw8QBpc4ydtma3z_2)
→ Frontend Engineer / Senior React Developer Path
(https://www.youtube.com/playlist?list=PLWKjhJtqVAbmMuZ3saqRIBimAKIMYkt0E)

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related roadmaps:

JavaScript Roadmap

TypeScript Roadmap

Frontend Roadmap

Full Stack Roadmap

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certificates :

free

1. **React Full Course – FreeCodeCamp**

   * https://www.freecodecamp.org/learn/front-end-development-libraries/

2. **React Official Documentation (New Docs)**

   * https://react.dev/learn

3. **Frontend Development with React – Great Learning**

   * https://www.mygreatlearning.com/academy/learn-for-free/courses/react-js-tutorial

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paid

1. **Google UX + Frontend (React-heavy) Professional Programs**

   * (https://www.coursera.org/professional-certificates/google-ux-design)

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role : React Native Developer / Mobile Application Developer

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skills :

JavaScript (ES6+), React, React Native, JSX, Functional Components, Props & State, Expo, React Native CLI,
Metro Bundler, Mobile UI Development, Core React Native Components, Flexbox Layout, Platform-specific Code,
Android & iOS Basics, Mobile Navigation, Gestures & Animations, Networking (Fetch, WebSockets), Push Notifications,
Mobile Security, Authentication, Secure Storage, AsyncStorage, Expo Secure Store, REST APIs, Debugging & Profiling,
 Performance Optimization, Testing (Jest, React Native Testing Library, Detox), App Publishing (Google Play Store, Apple App Store), Git, GitHub, Mobile App Performance, Accessibility

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learning roadmap :

JavaScript Basics (https://www.youtube.com/playlist?list=PL4cUxeGkcC9i9Ae2D9Ee1RvylH38dKuET)
→ CSS Basics (https://www.youtube.com/playlist?list=PL0Zuz27SZ-6Mx9fd9elt80G1bPcySmWit)
→ React Basics (JSX → Components → State → Props) (https://www.youtube.com/playlist?list=PLC3y8-rFHvwgg3vaYJgHGnModB54rxOk3)
→ React Native Introduction & Use Cases (https://www.youtube.com/playlist?list=PL0Zuz27SZ-6OH3xackTS2KLcnmxx5wW7c)
→ Environment Setup (Expo → React Native CLI → Metro Bundler) (https://www.youtube.com/playlist?list=PL0Zuz27SZ-6OH3xackTS2KLcnmxx5wW7c)
→ Development Workflow (Run on Device → Debugging → Fast Refresh) (https://www.youtube.com/playlist?list=PL0Zuz27SZ-6OH3xackTS2KLcnmxx5wW7c)
→ Core Components (View, Text, Image, Button, Modal, SafeAreaView) (https://www.youtube.com/playlist?list=PL0Zuz27SZ-6OH3xackTS2KLcnmxx5wW7c)
→ Lists & Scrolling (ScrollView → FlatList → SectionList) (https://www.youtube.com/playlist?list=PL0Zuz27SZ-6OH3xackTS2KLcnmxx5wW7c)
→ Platform-Specific Code (Platform Module → File Extensions → react-native-web)
→ Styling (StyleSheet → Flexbox → Accessibility)
→ Networking (Fetch → Connectivity → WebSockets)
→ Push Notifications (https://www.youtube.com/playlist?list=PLF6VVsuwdXUc_cD8avHi_ezCAid95m3l6)
→ User Interactions (Gestures → Navigation → Animations → Deep Linking) (https://www.youtube.com/c/catalinmirondev)
→ Security (Auth → Secure Networking → Storage) (https://www.youtube.com/playlist?list=PL0Zuz27SZ-6OH3xackTS2KLcnmxx5wW7c)
→ Storage (AsyncStorage → Expo Secure Store → SQLite) (https://www.youtube.com/playlist?list=PL0Zuz27SZ-6OH3xackTS2KLcnmxx5wW7c)
→ Testing (Unit → Component → E2E)
→ Performance Optimization & Profiling
→ Native Modules (iOS → Android) (https://www.youtube.com/playlist?list=PLy9JCsy2u97nkJJEO6lhtYm1rPfphER4j)
→ App Publishing (Play Store → App Store) (https://www.youtube.com/playlist?list=PLG02JlJZbKbv0OCtmuKCNHqmi9N2WQMWd)
→ Advanced Mobile Development (https://www.youtube.com/playlist?list=PLTjE4ZUA6UjdXyY5logd7K9kx1tuqBYQg)
→ Senior React Native Developer Path (https://www.youtube.com/playlist?list=PL2HX_yT71umBtWDsrELOCnCq_HU9mWZeS)

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related roadmaps:

React Roadmap

JavaScript Roadmap

Frontend Roadmap

Flutter Roadmap

Full Stack Roadmap

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certificates :

free

1. **React Native Tutorial – Official Documentation**

   * [https://reactnative.dev/docs/getting-started](https://reactnative.dev/

docs/getting-started)

2. **Build Mobile Apps with React Native – FreeCodeCamp**

   *
https://www.freecodecamp.org/news/tag/react-native/

3. **Expo React Native Guide**

   * https://docs.expo.dev/

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paid

1. **Meta React Native Specialization**

   * https://www.coursera.org/specializations/meta-react-native

2. **Google Associate Android Developer (React Native-friendly background)**

   * https://developers.google.com/certification/associate-android-developer

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role : Server-Side Game Developer / Multiplayer Backend Engineer

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skills :

Server-Side Game Development, Networking Fundamentals, TCP/IP, UDP, Socket Programming, Network Protocols,
Latency Optimization, Real-Time Systems, Game Server Architecture, C/C++, C#, Java, Go, JavaScript, Erlang,
Serialization (JSON, Protobuf), Multithreading, Concurrency Control, Synchronization Primitives,
Async & Reactive Programming, Actor Model, RPC (gRPC), REST APIs, Database Design, RDBMS, NoSQL Databases,
Redis, Message Queues (Kafka, RabbitMQ), Distributed Systems, Cloud Computing (AWS, GCP, Azure), Docker,
Kubernetes, Game Security, DDoS Mitigation, Authentication, Encryption (TLS/SSL), Performance Profiling,
Scalability, Load Balancing, Monitoring, CI/CD, Linux Systems, Design Patterns, TDD

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learning roadmap :

Networking Fundamentals (https://www.youtube.com/playlist?list=PLDQaRcbiSnqF5U8ffMgZzS7fq1rHUI3Q8)
→ TCP (Reliability, Congestion Control, Flow Control, Vulnerabilities) (https://www.youtube.com/playlist?list=PLDQaRcbiSnqF5U8ffMgZzS7fq1rHUI3Q8)
→ UDP (Datagrams, Lightweight Communication, Game Networking Use Cases) (https://www.youtube.com/playlist?list=PLDQaRcbiSnqF5U8ffMgZzS7fq1rHUI3Q8)
→ TCP vs UDP (Latency vs Reliability Tradeoffs)
→ IP Fundamentals (Routing, DNS, ARP, IPv4 vs IPv6, TLS) (https://www.youtube.com/playlist?list=PLWkguCWKqN9MdQXjSM5DE17NU7_RQA_MH)
→ Programming Languages (C/C++ → C# → Java → Go → JavaScript)

(https://www.youtube.com/playlist?list=PLBlnK6fEyqRjKA_NuK9mHmlk0dZiuXSEp)
→ Socket Programming (BSD Sockets → Winsock → Byte Manipulation)
(https://www.youtube.com/playlist?list=PLcacUGyBsOIA1YUqvKd7p8uvsG_FCt7-U)
→ Data Serialization (JSON → Protobuf)
(https://www.youtube.com/playlist?list=PLqq-6Pq4lTTYPR2oH7kgElMYZhJd4vOGI)
→ Multithreading & Concurrency
(https://www.youtube.com/playlist?list=PLL8woMHwr36EDxjUoCzboZjedsnhLP1j4)
→ Synchronization (Mutex, Semaphore, Spinlock, Barrier)
→ Async & Reactive Models
(https://www.youtube.com/playlist?list=PLqq-6Pq4lTTYPR2oH7kgElMYZhJd4vOGI)
→ Actor Model & Task-Based Concurrency
(https://www.youtube.com/playlist?list=PLL8woMHwr36EDxjUoCzboZjedsnhLP1j4)
→ Reactor / Proactor Patterns
(https://www.youtube.com/playlist?list=PLMc3Vq8Q6CgbcUlSGQ9wgGZg1wsyxVZuz)
→ Programming Techniques (Design Patterns, TDD, DI)
→ Databases (RDBMS → NoSQL → Key-Value Stores)
(https://www.youtube.com/playlist?list=PLKDJE8BkZ4wxSleTXC9m9HDpPVF9UcSh6)
→ ORM & Data Access Layers
(https://www.youtube.com/playlist?list=PLty0cFLf07jVTv0Y7iwQ0mbLtHquy_OES)
→ RPC & REST APIs
(https://www.youtube.com/playlist?list=PLpSmZmoBaROZNRmR3BHPHY6cqNOLqLkKA)
→ Message Queues (Kafka, RabbitMQ)
(https://www.youtube.com/playlist?list=PLQnljOFTspQVcumYRWE2w9kVxxIXy_AMo)
→ Distributed Game Server Architecture
(https://www.youtube.com/playlist?list=PLMygHu90DtpHqU0wSZ4yUY8yFe2cI56Ok)
→ Cloud Platforms (AWS → GCP → Azure → Serverless)
(https://www.youtube.com/playlist?list=PL1k_gTsSGTwUd_X1URtSe5xzheSXSejmA)
→ Containerization (Docker → Kubernetes)
(https://www.youtube.com/playlist?list=PLy7NrYWoggjwPggqtFsI_zMAwvG0SqYCb)
→ Data Clustering & Analytics (Apache Spark)
(https://www.youtube.com/playlist?list=PLgy71-0-2-F3CTeiwCdbWNx_QVg7d23iv)
→ AI Integration for Games
(https://t.me/gamedevplatform)
→ Scalable Multiplayer Backend Systems
(https://www.youtube.com/playlist?list=PLMygHu90DtpHqU0wSZ4yUY8yFe2cI56Ok)
→ Senior Server-Side Game Developer / Backend Architect
(https://www.youtube.com/playlist?list=PL9n0l8rSshSmgrEXVNQrJDBnsvE2BoCbv)

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related roadmaps:

Game Development Roadmap

API Design Roadmap

Distributed Systems Roadmap

Cloud & DevOps Roadmap

AI Engineer Roadmap

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certificates :

free

2. **Game Networking Tutorials – Gaffer On Games**

   * https://gafferongames.com/

3. **Distributed Systems & Cloud Fundamentals – FreeCodeCamp**

* https://www.freecodecamp.org/news/tag/distributed-systems/

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paid

2. **Google Cloud Professional Cloud Developer**

   * https://cloud.google.com/certification/cloud-developer

3. **Microsoft Azure Developer Associate**

   * https://learn.microsoft.com/en-us/credentials/certifications/azure-developer/

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role : Software Engineer (Computer Science Core)

skills :
Data Structures and Algorithms, Problem Solving, Time and Space Complexity Analysis, Object Oriented Programming,
Design Patterns, System Design, Database Management Systems, Operating Systems, Computer Networks,
Multithreading and Concurrency, REST APIs, GraphQL, gRPC, Security Fundamentals, Cryptography Basics,
Software Architecture, UML Diagrams, Git and Version Control, Debugging, Code Optimization

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learning roadmap :
Pick a Programming Language (https://www.youtube.com/playlist?list=PLu0W_9lII9agwh1XjRt242xIpHhPT2llg)
→ Data Structures (https://www.youtube.com/playlist?list=PLBlnK6fEyqRj9lld8sWIUNwlKfdUoPd1Y)
→ Asymptotic Notation (https://www.youtube.com/playlist?list=PLSVu1-lON6Lwr2u_VtLcAxtVAZge9sttL)
→ Common Algorithms (https://www.youtube.com/playlist?list=PLDN4rrl48XKpZkf03iYFl-O29szjTrs_O)
→ Advanced Trees (Tries, AVL, Red-Black, B-Trees, K-D Trees, Skip Lists) (https://www.youtube.com/playlist?list=PLBlnK6fEyqRhX6r2uhhlubuF5QextdCSM)
→ Design Patterns & Software Architecture (https://www.youtube.com/playlist?list=PLfg9ycqfY2SUMcwK5OFeNxjo4pyLQx6tG)
→ Character Encoding, Bitwise Operations, UML (https://t.me/Codewithrandom)
→ Basic Mathematics (Probability, Combinatorics) (https://www.youtube.com/playlist?list=PLBlnK6fEyqRgMCUAG0XRw78UA8qnv6jEx)
→ Complexity Classes (P, NP, NP-Complete, NP-Hard) (https://www.youtube.com/playlist?list=PLDN4rrl48XKpZkf03iYFl-O29szjTrs_O)
→ System Design Fundamentals (https://www.youtube.com/playlist?list=PLCRMIe5FDPsd0gVs500xeOewfySTsmEjf)
→ Databases (SQL & NoSQL) (https://www.youtube.com/playlist?list=PLu0W_9lII9agS67Uits0UnJyrYiXhDS6q)
→ Networking Concepts (https://www.youtube.com/playlist?list=PLIFyRwBY_4bRLmKfP1KnZA6rZbRHtxmXi)
→ Security Fundamentals (https://www.youtube.com/playlist?list=PLEiEAq2VkUUJfPOj5nRounXvf3n17PCft)
→ How Computers Work

(https://www.youtube.com/playlist?list=PLME-KWdxI8dcaHSzzRsNuOLXtM2Ep_C7a)
→ Processes, Threads & Concurrency
(https://www.youtube.com/playlist?list=PLqq-6Pq4lTTY8JY-DwAfoMiIZWHt2FKVX)
→ Coding Interviews & Competitive Programming Practice
(https://www.youtube.com/playlist?list=PL9gnSGHSqcnr_DxHsP7AW9ftq0AtAyYqJ)

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related roadmaps :

→ API Design Roadmap
→ System Design Roadmap
→ Software Design & Architecture Roadmap

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certificates :

**Free

1. CS50x – Introduction to Computer Science (Harvard)
   https://cs50.harvard.edu/x/

2. Data Structures and Algorithms Specialization (UC San Diego – Free Audit)
   [https://www.coursera.org/specializations/data-structures-algorithms]
(https://www.coursera.org/specializations/data-structures-algorithms)

**Paid

1. Google Professional Software Engineer Preparation (Coursera – Google)
   [https://www.coursera.org/professional-certificates/google-it-automation]
(https://www.coursera.org/professional-certificates/google-it-automation)

2. IBM Full Stack Software Developer Professional Certificate
   https://www.coursera.org/professional-certificates/ibm-full-stack-cloud-developer

3. Microsoft Azure Developer Associate (AZ-204)
   https://learn.microsoft.com/en-us/certifications/azure-developer/

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role : Software Design & Architecture Engineer

skills :
Clean Code Principles, Object Oriented Programming, SOLID Principles, Design Patterns (GoF, PoSA),
Software Architecture, Architectural Styles, Architectural Patterns, Domain Driven Design,
Microservices Architecture, MVC, CQRS, Event Driven Architecture, Serverless Architecture,
Enterprise Application Patterns, UML Diagrams, Code Refactoring, Dependency Management,
Abstraction and Modularity, High Cohesion and Low Coupling, Backend Architecture Knowledge

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learning roadmap :
Clean Code Principles (https://www.youtube.com/playlist?list=PLZlA0Gpn_vH-xGQ-nQ87rXI7QkM6W3E79)

→ Programming Paradigms (Structured → Functional → OOP)
(https://www.youtube.com/playlist?list=PL9D558D49CA734A02)
→ Object Oriented Programming Concepts
(https://www.youtube.com/playlist?list=PL9gnSGHSqcno1G3XjUbwzXHL8_EttOuKk)
→ OOP Design Principles (SOLID, DRY, YAGNI)
(https://www.youtube.com/playlist?list=PLXd94bZTzZ-SniG8i_iDnA8PSQ9Gz9jV)
→ Design Principles
(https://www.youtube.com/playlist?list=PLfg9ycqfY2SUMcwK5OFeNxjo4pyLQx6tG)
→ Design Patterns (GoF → PoSA)
(https://www.youtube.com/playlist?list=PLF206E906175C7E07)
→ Architectural Principles (Boundaries, Coupling, Cohesion)
(https://www.youtube.com/playlist?list=PLfg9ycqfY2SUMcwK5OFeNxjo4pyLQx6tG)
→ Architectural Styles (Monolithic → Layered → Client-Server → Event-Driven)
(https://www.youtube.com/playlist?list=PL_aIDuIpplcnSf5Go8C4YWLB6-V9UFdW7)
→ Architectural Patterns (DDD → MVC → Microservices → CQRS → Event Sourcing → Serverless) (https://www.youtube.com/playlist?list=PLxpwvsFSXIjBUgCe7S1xPMOVv7HlXx26M)
→ Enterprise Application Patterns (https://www.youtube.com/playlist?list=PLfg9ycqfY2SUMcwK5OFeNxjo4pyLQx6tG)
→ Backend Architecture Integration (https://www.youtube.com/playlist?list=PLBg2PQNi6uvmlgScuLqEAsQ-2QaUPm4hQ)
→ System Design Roadmap (https://www.youtube.com/playlist?list=PLLhBy6YSIT0ANaihpjwDBSBju0qUZ82SK)

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related roadmaps :

→ Backend Developer Roadmap
→ System Design Roadmap
→ API Design Roadmap

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certificates :

**Paid

1. Software Architecture by University of Alberta (Coursera)
   [https://www.coursera.org/specializations/software-design-architecture]
(https://www.coursera.org/specializations/software-design-architecture)

3. Microsoft Azure Solutions Architect Expert (AZ-305)
   [https://learn.microsoft.com/en-us/certifications/azure-solutions-architect/]
(https://learn.microsoft.com/en-us/certifications/azure-solutions-architect/)

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role : Software Architect

skills :
Software Architecture Design, High Level System Design, Application Architecture, Solution Architecture,
Enterprise Architecture, Object Oriented Programming, SOLID Principles, Design Patterns, Domain Driven Design,
Microservices Architecture, Distributed Systems, API Design, REST, GraphQL, gRPC, Event Driven Architecture,
CQRS, Cloud Architecture, Serverless Computing, Security Architecture, OWASP, Authentication and Authorization,
Database Design (SQL, NoSQL), Data Warehousing, Big Data Systems, DevOps Fundamentals, CI/CD Pipelines,
Containers and Kubernetes, Networking Fundamentals, Agile and Scaled Agile

Frameworks, Documentation and
Technical Writing, Stakeholder Communication, Decision Making, Mentoring and Coaching

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learning roadmap :
Understand Software Architecture Basics (https://www.youtube.com/playlist?list=PL4JxLacgYgqTgS8qQPC17fM-NWMTr5GW6)
→ Architecture Levels (Application → Solution → Enterprise) (https://www.youtube.com/playlist?list=PLuBBTh-4TzDnxZff6pzO8jDNRcWcCnmNv)
→ Software Architect Responsibilities (https://www.youtube.com/channel/UC-Z7T0lAq_xECevIz8E5R5w/videos)
→ Core Architecture Skills (Decision Making, Simplification, Communication) (https://www.youtube.com/playlist?list=PLfg9ycqfY2SUMcwK5OFeNxjo4pyLQx6tG)
→ Programming Languages (Java / Python / Go / JS / .NET) (https://www.youtube.com/playlist?list=PLBlnK6fEyqRjKA_NuK9mHmlk0dZiuXSEp)
→ OOP & Design Principles (SOLID, DRY, DDD, TDD) (https://www.youtube.com/playlist?list=PLXd94bZTzZ-SniG8i_ihDnA8PSQ9Gz9jV)
→ Architectural Patterns (MVC → Microservices → Event-Driven → CQRS → Serverless) (https://www.youtube.com/playlist?list=PLfg9ycqfY2SUMcwK5OFeNxjo4pyLQx6tG)
→ Distributed Systems & Data Consistency (https://www.youtube.com/playlist?list=PLE3ovFxnzNpZGQut7wYe44UwKcdZXFUA8)
→ API Design & Integrations (REST → GraphQL → gRPC → Messaging Queues) (https://www.youtube.com/playlist?list=PLpSmZmoBaROZNRmR3BHPHY6cqNOLqLkKA)
→ Databases & Big Data Systems (https://www.youtube.com/playlist?list=PLxCzCOWd7aiFAN6I8CuViBuCdJgiOkT2Y)
→ Security Architecture (https://www.youtube.com/playlist?list=PLOspHqNVtKADkWLFt9OcziQF7EatuANSY)
→ Web & Mobile Architecture (https://www.youtube.com/playlist?list=PLBKYrTtNYNa4HFL4HRSGKJiC3PgvVXFjU)
→ DevOps & Operations Knowledge (https://www.youtube.com/playlist?list=PL9ooVrP1hQOE5ZDJJsnEXZ2upwK7aTYiX)
→ Networking Fundamentals (https://www.youtube.com/playlist?list=PLIFyRwBY_4bRLmKfP1KnZA6rZbRHtxmXi)
→ Enterprise Software & Large-Scale Systems (https://www.youtube.com/playlist?list=PLMygHu90DtpHqU0wSZ4yUY8yFe2cI56Ok)
→ Agile, Management & Governance Frameworks (https://www.youtube.com/playlist?list=PLsz8d8r2a994Ar3DSebvhVUwEIAxnG3pF)
→ Mentoring, Consulting & Leadership Skills (https://www.youtube.com/playlist?list=PLZlA0Gpn_vH-xGQ-nQ87rXI7QkM6W3E79)

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related roadmaps :
Software Architect
→ Software Design & Architecture
→ System Design Roadmap
→ Backend Developer Roadmap
→ Full Stack Developer Roadmap
→ DevOps Roadmap

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certificates :

**Free :

2. Google Engineering Practices (Architecture & Design Docs)
   https://google.github.io/eng-practices/

**Paid

1. Software Architecture & Design – University of Alberta (Coursera)
   [https://www.coursera.org/specializations/software-design-architecture]
(https://www.coursera.org/specializations/software-design-architecture)

2. TOGAF® Enterprise Architecture Certification (The Open Group)
   https://www.opengroup.org/togaf

3. Microsoft Azure Solutions Architect Expert (AZ-305)
   [https://learn.microsoft.com/en-us/certifications/azure-solutions-architect/]
(https://learn.microsoft.com/en-us/certifications/azure-solutions-architect/)

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role : Spring Boot Developer

skills :
Java, Spring Framework, Spring Boot, Dependency Injection, Inversion of Control, Spring MVC, REST API Development,
Spring Security, JWT Authentication, OAuth2, Hibernate ORM, JPA, Spring Data JPA, MongoDB Integration, JDBC,
Microservices Architecture, Spring Cloud, Eureka Service Discovery, API Gateway, Circuit Breaker,
Configuration Management, Embedded Servers, Actuators, Transaction Management, Unit Testing, Integration Testing,
MockMvc, CI/CD Basics, Docker Fundamentals

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learning roadmap :
Java Fundamentals (https://www.youtube.com/playlist?list=PLsyeobzWxl7pe_IiTfNyr55kwJPWbgxB5)
→ Core Java Concepts (OOP, Collections, Streams)
(https://www.youtube.com/playlist?list=PLA3GkZPtsafZZsLj0Tybu3y0HVl-hp1ea)
→ Spring Framework Basics (IOC, DI, AOP, Annotations)
(https://www.youtube.com/playlist?list=PLE37064DE302862F8)
→ Spring MVC (Controllers, Servlets, JSP, Architecture)
(https://www.youtube.com/playlist?list=PLGRDMO4rOGcP0B8ZHYbuz8AsvTyvJ-E7)
→ Spring Boot Core (Starters → Auto Configuration → Embedded Server → Actuator)
(https://www.youtube.com/playlist?list=PLrhn-7fWTlehDT5Ah-u3cpEPIOCicJIB1)
→ Spring Security (Authentication → Authorization → JWT → OAuth2)
(https://www.youtube.com/playlist?list=PLR2yPNIFMlL8abIXtXx1x9j3NvRix8ZbA)
→ Hibernate & ORM (Entities → Relationships → Transactions → Lifecycle)
(https://www.youtube.com/playlist?list=PLBlnK6fEyqRgMCUAG0XRw78UA8qnv6jEx)
→ Spring Data (JPA → MongoDB → JDBC)
(https://www.youtube.com/playlist?list=PLGRDMO4rOGcPtMMl1DD2cu3GBPZloHOS)
→ REST API Development
(https://www.youtube.com/playlist?list=PLGRDMO4rOGcPtMMl1DD2cu3GBPZloHOS)
→ Microservices with Spring Cloud
(https://www.youtube.com/playlist?list=PLSVW22jAG8pBnhAdq9S8BpLnZ0_jVBj0c)
→ Service Discovery (Eureka)
(https://www.youtube.com/playlist?list=PLq3uEqRnr_2Ffkep2z3DCnumZVymLtabG)
→ API Gateway & Config Server
(https://www.youtube.com/playlist?list=PLVz2XdJiJQxz3L2Onpxbel6r72IDdWrJh)
→ Resilience (Circuit Breaker, Micrometer)
(https://www.youtube.com/playlist?list=PL4TnYdea-xTLGklgQfWW8ZgpQXm8-FOHt)
→ Testing (JPA Test → MockMvc → @SpringBootTest → @MockBean)
(https://www.youtube.com/playlist?list=PL87padGs1bp82KIX1wjmPIGBHXvI2l1Pn)
→ Containerization & Deployment Basics
(https://www.youtube.com/playlist?list=PL87padGs1bp82KIX1wjmPIGBHXvI2l1Pn)

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related roadmaps :

→ Java Roadmap
→ Backend Developer Roadmap
→ System Design Roadmap
→ DevOps Roadmap
→ Docker Roadmap
→ Kubernetes Roadmap
→ Redis Roadmap
→ Frontend Roadmap
→ Full Stack Roadmap

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certificates :

**Free

1. Spring Boot & Spring Framework – Official Spring Guides
   https://spring.io/guides

2. Java Programming – Oracle University (Free Learning Paths)
   https://learn.oracle.com

**Paid

1. Spring Framework Specialization – VMware (Coursera)
   https://www.coursera.org/specializations/spring-framework

2. Java Backend Development – IBM Professional Certificate
   https://www.coursera.org/professional-certificates/ibm-backend-development

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role : Technical Writer

skills :
Technical Documentation, Technical Writing, Content Structuring, Information Architecture, Markdown,
Git and Version Control, API Documentation, Developer Documentation, Research and Content Analysis, SEO Writing,
Keyword Research, Storytelling, Editing and Proofreading, UX Writing, Documentation Tools, Blogging Platforms,
Content Strategy, Content Optimization, Analytics and Metrics Tracking, Communication Skills,
User Persona Analysis

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learning roadmap :
Introduction to Technical Writing (https://www.youtube.com/playlist?list=PLoynTxuTLXaDtAmzbqdsc7JZMWcIt3nnP)
→ Understanding the Role & Growth of a Technical Writer (https://www.youtube.com/playlist?list=PLoynTxuTLXaBwMkqeenVVxngMdUWkZokK)
→ Core Skills (Technology Knowledge → Language Proficiency → Written Communication) (https://www.youtube.com/playlist?list=PL5210ECBFD8621C83)
→ Tooling (Publishing Tools → Editing Tools → Git → Markdown → SEO Tools) (https://www.youtube.com/playlist?list=PLUsbdJd-lWUkk6aAi-dVqk__QongKOmj8)
→ Writing Best Practices (Storytelling → Structure → CTAs → Style Guides) (https://www.youtube.com/playlist?list=PLS6n3zZvu15swgbbIs5I9iDIeENHZhLLf)
→ Content Research (Keywords → Trends → Communities → Topic Evaluation)

(https://www.youtube.com/playlist?list=PLvJ_dXFSpd2vk6rQ4Rta5MhDIRmakFbp6)
→ Technical Content Types
(https://www.youtube.com/playlist?list=PLUsbdJd-lWUkk6aAi-dVqk__QongKOmj8)
→ Product Documentation & How-To Guides
(https://www.youtube.com/playlist?list=PLPWHBRxq3YRiXkFoK539u9SqE7xdpcXap)
→ Developer Documentation & API References
(https://www.youtube.com/playlist?list=PLpjK416fmKwRMhVX5YYWv_5mubCd7cmgG)
→ Help Content & Troubleshooting
(https://www.youtube.com/playlist?list=PLUsbdJd-lWUkk6aAi-dVqk__QongKOmj8)
→ Developer Support Documentation
(https://www.youtube.com/playlist?list=PLEJZyr6k_ykJ4Sa7B-FMr7zbr4uegNO5)
→ Technical Content Marketing
(https://www.youtube.com/playlist?list=PLlvxvizTHn50Lpf-CEGl9mBqcsla2KnL7)
→ SEO for Technical Content
(https://www.youtube.com/playlist?list=PLvJ_dXFSpd2tThwk2t057HdK9hMLKd5k8)
→ Funnel-Based Content (Top → Mid → Bottom)
(https://www.youtube.com/playlist?list=PLlvxvizTHn50Lpf-CEGl9mBqcsla2KnL7)
→ Content Analysis & Metrics
(https://www.youtube.com/playlist?list=PLpjK416fmKwRMhVX5YYWv_5mubCd7cmgG)
→ Content Optimization
(https://www.youtube.com/playlist?list=PLKfCUN-8tLBwh5JpG3vU-95DaJu-YpraF)
→ Content Distribution & Amplification
(https://www.youtube.com/playlist?list=PLlvxvizTHn50Lpf-CEGl9mBqcsla2KnL7)
→ Continuous Learning & Improvement
(https://www.youtube.com/playlist?list=PL0YHYO9PgjDk5SVg2WYRiccqTPh-rAj9K)

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related roadmaps :

→ Backend Developer Roadmap
→ Software Design & Architecture
→ DevOps Roadmap
→ API Design Roadmap
→ Full Stack Roadmap

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certificates :

**Free

1. Technical Writing Course – Google Developers
   https://developers.google.com/tech-writing

2. Write the Docs – Documentation Resources
   https://www.writethedocs.org/guide/

**Paid

2. UX Writing & Content Design – UX Writers Collective

https://uxwriterscollective.com/course/ux-writing

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role : UX Designer

skills :
User Experience Design, Human-Centered Design, Behavioral Design, User Research, User Personas,

Information Architecture, Interaction Design, Wireframing, Prototyping, Usability Testing, A/B Testing,
UX Strategy, Product Thinking, Design Thinking, Behavior Change Design, Gamification, UX Writing, Visual Hierarchy,
Design Systems, Accessibility, Figma, Adobe XD, Sketch, Stakeholder Communication, Impact Measurement

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learning roadmap :
Human Decision Making & Behavioral Science
(https://www.youtube.com/playlist?list=PLndotT3HR37UehCo0uELDHqS1J1X5VPeF)
→ Behavior Change Frameworks (Fogg Model → Hook Model → Dual Process Theory)
(https://www.youtube.com/playlist?list=PLndotT3HR37UehCo0uELDHqS1J1X5VPeF)
→ Behavior Classification (Existing vs New Behavior)
(https://www.youtube.com/playlist?list=PLndotT3HR37UehCo0uELDHqS1J1X5VPeF)
→ Habit Formation & Behavior Change Strategies
(https://www.youtube.com/playlist?list=PLndotT3HR37UehCo0uELDHqS1J1X5VPeF)
→ Product Understanding (Target User → Target Action → Target Outcome)
(https://www.youtube.com/playlist?list=PLtS8Ubq2bIlUWixdH_1Q2yzZh8QJSBVmT)
→ User Personas & Actor Definition
(https://www.youtube.com/playlist?list=PLtS8Ubq2bIlUWixdH_1Q2yzZh8QJSBVmT)
→ Business Understanding (Business Model Canvas → Lean Canvas → SWOT → Competitor Analysis) (https://www.youtube.com/playlist?list=PLyQoJ1DyyK3LH-0NVlUcgDs2i48gpCUUj)
→ Conceptual Design (User Stories → Backlog → Flow Design)
(https://www.youtube.com/playlist?list=PLgDaZD8y4z0AEOYBxJXL2WGVUa69Vwixx)
→ UX Deliverables (Customer Journey Map → Flowcharts → BPMN / EPC)
(https://www.youtube.com/playlist?list=PLtS8Ubq2bIlUWixdH_1Q2yzZh8QJSBVmT)
→ Prototyping & Wireframing (Figma → Adobe XD → Sketch → Balsamiq)
(https://www.youtube.com/playlist?list=PLEM9N83atjwHnS8vy6WFx1dgtOTZT4U9P)
→ UX Best Practices & UX Patterns
(https://www.youtube.com/playlist?list=PLtS8Ubq2bIlUWixdH_1Q2yzZh8QJSBVmT)
→ Attention, Motivation & Decision-Making Design
(https://www.youtube.com/playlist?list=PLndotT3HR37UehCo0uELDHqS1J1X5VPeF)
→ Usability & Interaction Optimization
(https://www.youtube.com/playlist?list=PLtS8Ubq2bIlUWixdH_1Q2yzZh8QJSBVmT)
→ Measuring Impact (A/B Testing → Multivariate Testing)
(https://www.youtube.com/playlist?list=PL8eNk_zTBST_IbClK_ZV3j3RtlXawUUQr)
→ Iteration, Learning & Continuous Improvement
(https://www.youtube.com/playlist?list=PLtS8Ubq2bIlUWixdH_1Q2yzZh8QJSBVmT)

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related roadmaps :

→ Design System Roadmap
→ Frontend Roadmap
→ Product Manager Roadmap

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certificates :

**Free

1. UX Design Fundamentals – Google UX Resources
   https://design.google/library/

3. Nielsen Norman Group UX Articles & Free Resources
   https://www.nngroup.com/articles/

**Paid

1. Google UX Design Professional Certificate (Coursera)
   [https://www.coursera.org/professional-certificates/google-ux-design]
(https://www.coursera.org/professional-certificates/google-ux-design)

2. Interaction Design Specialization – University of California San Diego
   [https://www.coursera.org/specializations/interaction-design](https://
www.coursera.org/specializations/interaction-design)

3. UX Certification – Nielsen Norman Group
   [https://www.nngroup.com/ux-certification/](https://www.nngroup.com/ux-
certification/)

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role : Vue.js Developer

skills :
JavaScript, ES6+, Vue.js, Single File Components, Component-Based Architecture, Props and Events, v-model, Vue Directives,
Template Syntax, Composition API, Options API, Vue Lifecycle Hooks, Reactive State Management, Computed Properties,
Watchers, Form Handling, Event Handling, Vue Router, Pinia State Management, REST API Integration, Axios, GraphQL Basics,
Nuxt.js, SSR and SSG, Tailwind CSS, UI Frameworks (Vuetify, Element UI), Testing with Vitest and Cypress,
Performance Optimization

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learning roadmap :
JavaScript Fundamentals (https://www.youtube.com/playlist?
list=PLDtWoQ-cxqiwamnwULkiPkzx8Iv0ny9eV)
→ Modern JavaScript (ES6+, Async/Await, Modules)
(https://www.youtube.com/playlist?list=PLDtWoQ-cxqiwamnwULkiPkzx8Iv0ny9eV)
→ Project Bootstrapping (create-vue)
(https://www.youtube.com/playlist?list=PLiaS5OslD9xU2ZE7n5cgk2mbFh9M5wjY4)
→ Vue Components (SFCs → Props → Events → v-model)
(https://www.youtube.com/playlist?list=PLC3y8-rFHvwgeQIfSDtEGVvvSEPDkL_1f)
→ Template Syntax & Directives
(https://www.youtube.com/playlist?list=PLC3y8-rFHvwgeQIfSDtEGVvvSEPDkL_1f)
→ Conditional & List Rendering
(https://www.youtube.com/playlist?list=PLC3y8-rFHvwgeQIfSDtEGVvvSEPDkL_1f)
→ Options API → Composition API
→ Application Configuration
(https://www.youtube.com/playlist?list=PLiaS5OslD9xU2ZE7n5cgk2mbFh9M5wjY4)
→ Lifecycle Hooks
(https://www.youtube.com/playlist?list=PLC3y8-rFHvwgeQIfSDtEGVvvSEPDkL_1f)
→ Forms & Input Handling
(https://www.youtube.com/playlist?list=PLC3y8-rFHvwgeQIfSDtEGVvvSEPDkL_1f)
→ Event Handling & Modifiers
→ Computed Properties & Watchers
→ Advanced Vue Concepts (Slots → Plugins → Custom Directives → Teleport → Animations)
→ Routing with Vue Router
(https://www.youtube.com/playlist?list=PL6tf8fRbavl08OCepj1G3vjV-dNQQJA1E)
→ State Management with Pinia
(https://www.youtube.com/playlist?list=PL6tf8fRbavl218Hl6bHbrd5vJyOVwVswX)
→ API Communication (Axios → Fetch → GraphQL)
(https://www.youtube.com/playlist?list=PLC3y8-rFHvwgeQIfSDtEGVvvSEPDkL_1f)
→ Styling & UI Libraries
→ Testing (Vitest → Vue Testing Library → Cypress → Playwright)
(https://www.youtube.com/playlist?list=PLJNLwTPak6dhC5HGSud_mR1Lchhr8mXeT)

→ SSR & SSG (Nuxt.js → Quasar → VitePress)
(https://www.youtube.com/playlist?list=PL5OhSdfH4uDsunUOrTcN7Fi7Vk_GAgSJM)
→ Mobile Apps with Capacitor
(https://www.youtube.com/playlist?list=PLC3y8-rFHvwgeQIfSDtEGVvvSEPDkL_1f)
→ Performance Optimization & Best Practices
(https://www.youtube.com/playlist?list=PLC3y8-rFHvwgeQIfSDtEGVvvSEPDkL_1f)

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related roadmap :

→ JavaScript Roadmap
→ TypeScript Roadmap
→ Frontend Developer Roadmap
→ React Roadmap
→ Angular Roadmap

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certificates :

**Free

1. Vue.js Official Documentation & Guide
   https://vuejs.org/guide/

2. Frontend Development with Vue – Vue Mastery (Free Courses)
   https://www.vuemastery.com/courses/

3. JavaScript & Frontend Learning – MDN Web Docs

https://developer.mozilla.org/en-US/docs/Learn

**Paid

1. Vue.js Developer Learning Path – Vue Mastery
   https://www.vuemastery.com/

3. Full Stack JavaScript Developer – IBM (Coursera)
   https://www.coursera.org/professional-certificates/ibm-full-stack-cloud-developer

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role : Docker Engineer / Containerization Specialist

skills :
Docker Fundamentals, Containerization, Linux Basics, Namespaces & cgroups, Union Filesystems, Docker CLI, Dockerfile Creation,
Image Building & Optimization, Container Networking, Container Security, Data Persistence (Volumes & Bind Mounts),
Container Registries (Docker Hub, ECR, GCR, ACR), Docker Compose, Runtime Configuration, Third-Party Images,
Docker Deployment, CI/CD Integration, Debugging & Developer Workflows

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learning roadmap :
Introduction to Containers (What → Why → Bare Metal vs VMs vs Containers)
(https://www.youtube.com/playlist?list=PLy7NrYWoggjzfAHlUusx2wuDwfCrmJYcs)
→ Docker & OCI Basics (https://www.youtube.com/playlist?

list=PLy7NrYWoggjzfAHlUusx2wuDwfCrmJYcs)
→ Underlying Technologies (Namespaces → cgroups → Union Filesystems)
(https://www.youtube.com/playlist?list=PLy7NrYWoggjzfAHlUusx2wuDwfCrmJYcs)
→ Linux Fundamentals (Package Managers → Permissions → Shell → Scripting)
(https://www.youtube.com/playlist?list=PLT98CRl2KxKECHltRib03tG8pyKEzwf9t)
→ Prerequisites (Programming → App Architecture → Web Dev Basics)
(https://www.youtube.com/playlist?list=PLu0W_9lII9agS67Uits0UnJyrYiXhDS6q)
→ Docker Installation (Desktop & Engine)
(https://www.youtube.com/playlist?list=PLy7NrYWoggjzfAHlUusx2wuDwfCrmJYcs)
→ Docker Basics
(https://www.youtube.com/playlist?list=PLy7NrYWoggjzfAHlUusx2wuDwfCrmJYcs)
→ Data Persistence (Ephemeral Filesystem → Volumes → Bind Mounts)
→ Using 3rd-Party Container Images
→ Command Line Utilities
→ Runtime Configuration Options
→ Building Images (Dockerfiles → Layer Caching → Size & Security)
→ Container Registries & Tagging Best Practices
→ Running Containers (docker run → docker compose)
→ Docker CLI Mastery (Images → Containers → Volumes → Networks)
→ Container Security (Runtime → Image Hardening)
(https://www.youtube.com/playlist?list=PLBf0hzazHTGNv0-GVWZoveC49pIDHEHbn)
→ Developer Experience (Hot Reload → Debugging → Testing → CI)
(https://www.youtube.com/playlist?list=PLy7NrYWoggjzfAHlUusx2wuDwfCrmJYcs)
→ Deploying Containers (PaaS → Kubernetes → Docker Swarm → Nomad)
(https://www.youtube.com/playlist?list=PLy7NrYWoggjwPggqtFsI_zMAwvG0SqYCb)
→ Practice Real Projects & Deployment
(https://www.youtube.com/playlist?list=PLy7NrYWoggjzfAHlUusx2wuDwfCrmJYcs)

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related roadmaps :

→ Kubernetes Roadmap
→ DevOps Roadmap
→ Backend Roadmap
→ Cloud (AWS / Azure / GCP) Container Services
→ System Design Roadmap

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certificates :

**Free

2. Docker Foundations – Docker Official Learning Resources
   https://docs.docker.com/get-started/

3. Container Fundamentals – IBM (Free Learning on Cognitive Class)
   [https://cognitiveclass.ai/courses/docker-essentials)

**Paid

2. Docker & Kubernetes – The Complete Guide (Udemy – Industry Standard)
   [https://www.udemy.com/course/docker-and-kubernetes-the-complete-guide/]
(https://www.udemy.com/course/docker-and-kubernetes-the-complete-guide/)

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role : Elasticsearch Engineer / Search Platform Engineer

skills :
Elasticsearch, ELK Stack, JSON, REST APIs, Indexing & Searching, Query DSL, ES|QL, EQL, KQL, Data Modeling, Sharding & Replication, Cluster Architecture, CRUD

Operations, Bulk Indexing, Search Optimization, Aggregations, Text Analysis, Custom Analyzers, Relevance Tuning (BM25), Pagination & Sorting, Highlighting, Transform API, ILM & Data Tiers, Snapshots & Restore, Cross-Cluster Replication, Vector Search, Semantic Search, Security (Auth, Roles, API Keys), Kibana, Docker for Elasticsearch

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learning roadmap :
JSON & REST API Basics (https://www.youtube.com/playlist?list=PLy7NrYWoggjwV7qC4kmgbgtFBsqkrsefG)
→ Elasticsearch Introduction (What ES is → Use Cases → ES vs RDBMS → ELK Stack) (https://www.youtube.com/watch?v=a4HBKEda_F8)
→ Environment Setup (Docker → Elastic Cloud → Kibana Console) (https://www.youtube.com/watch?v=a4HBKEda_F8)
→ Core Architecture (Cluster → Nodes → Index → Document → Shards → Replicas → Split Brain) (https://www.youtube.com/watch?v=a4HBKEda_F8)
→ Data Modeling (Mappings → Data Types → Text vs Keyword → Nested & Object Types) (https://www.youtube.com/watch?v=a4HBKEda_F8)
→ Data Ingestion (CRUD → Bulk Indexing → Update/Delete by Query → Migrations)
→ Search Fundamentals (Query DSL → Query vs Filter → Leaf Queries → Bool Queries)
→ Search Result Control (Pagination → Source Filtering → Sorting → Highlighting)
→ Text Analysis (Analyzers → Analyze API → Custom Analyzers)
→ How Search Works (Inverted Index → Doc Values → Fielddata)
→ Aggregations (Metric → Bucket → Pipeline → Nested Aggregations)
→ Transformations (Pivot → Latest → Transform API)
→ Relevance & Tuning (BM25 → Boosting → Function Score → Synonyms → Match Phrase)
→ Production Readiness (Monitoring → CAT APIs → ILM → Rollover → Snapshots → Autoscaling)
→ Advanced Features (Vector Search → Semantic Search → Hybrid Search → Security)
→ Build & Optimize Real Search Applications

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related roadmaps :

→ MongoDB Roadmap
→ Redis Roadmap
→ PostgreSQL Roadmap
→ Data Engineer Roadmap
→ DevOps Roadmap
→ Backend Roadmap

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certificates :

**Free

1. ElasticSearch Engineer Free Training – Elastic Official
   https://www.elastic.co/training/free

2. Elasticsearch Basics – Elastic Documentation & Labs
   https://www.elastic.co/guide/index.html

**Paid

1. Elastic Certified Engineer – Elastic (Official, Highly Valued)
   https://www.elastic.co/training/certification

2. Elasticsearch Engineer Course – Elastic Training (Official)

https://www.elastic.co/training/elasticsearch-engineer

3. Elasticsearch Complete Guide – Udemy (Used in MNCs)
   https://www.udemy.com/course/elasticsearch-complete-guide/

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role : GraphQL Developer / API Engineer

skills :
GraphQL, API Design, Schema Design, Query Optimization, Mutations & Subscriptions, GraphQL Queries, Variables & Fragments,
Directives, Schema Type System, Resolvers, Validation & Execution, GraphQL over HTTP, WebSockets & SSE,
Pagination Strategies, Authorization & Authentication, Caching & Batching,
Apollo Client, Relay, URQL, graphql-js,
GraphQL Java, GraphQL Go, Optimistic UI, Real-time APIs, Backend Integration,
Frontend Data Fetching

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learning roadmap :
GraphQL Introduction (What is GraphQL → Problems it Solves → Thinking in Graphs)
(https://www.youtube.com/playlist?list=PL4cUxeGkcC9gUxtblNUahcsg0WLxmrK_y)
→ GraphQL on Frontend vs Backend
→ Queries (Fields → Arguments → Variables → Aliases → Directives → Fragments)
→ Mutations (Single → Multiple → Optimistic UI)
→ Subscriptions (Event-based → Live Queries → @defer / @stream)
→ Schema Design (Scalars → Enums → Objects → Lists → Interfaces → Unions)
→ Root Fields & Resolvers (Sync → Async → Scalar Coercion → Lists)
→ Validation
→ Execution Flow
→ Serving GraphQL APIs
→ GraphQL over HTTP (Spec → Caching → Batching → Authorization)
→ GraphQL over WebSockets (Real-time → Authorization → Spec)
→ GraphQL over SSE
(https://www.youtube.com/playlist?list=PLT8NQOez-YZ8s7lxUExy3AIr7cl0Dsy2w)
→ Pagination (Cursor-based → Offset-based)
(https://www.youtube.com/playlist?list=PL4cUxeGkcC9gUxtblNUahcsg0WLxmrK_y)
→ Implementations
→ GraphQL Servers (graphql-js → GraphQL Java → GraphQL Go)
(https://www.youtube.com/playlist?list=PLZV0a2jwt22slmUC9iwGGWfRQRIhs1ELa)
→ GraphQL Clients (Apollo Client → Relay → URQL)
(https://www.youtube.com/playlist?list=PL4cUxeGkcC9gUxtblNUahcsg0WLxmrK_y)
→ Production-Ready GraphQL APIs
(https://www.youtube.com/playlist?list=PLZV0a2jwt22slmUC9iwGGWfRQRIhs1ELa)

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related roadmaps :

→ JavaScript Roadmap
→ React Roadmap
→ Frontend Roadmap
→ Backend Roadmap
→ API Design Roadmap

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certificates :

**Free

1. GraphQL Official Learning – GraphQL Foundation
   https://graphql.org/learn/

2. Apollo GraphQL Tutorial (Free Track)
   https://www.apollographql.com/tutorials/

3. Introduction to GraphQL – HowToGraphQL
   https://www.howtographql.com/

**Paid

2. GraphQL Complete Guide – Udemy (Industry Standard)
   https://www.udemy.com/course/graphql-bootcamp/

3. GraphQL Masterclass – Educative (Used in Big Tech Prep)

https://www.educative.io/courses/advanced-graphql

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role : HTML Developer

skills :
HTML5, Semantic HTML, Web Fundamentals, HTML Forms, Accessibility (a11y), SEO Basics, Media Embedding, Tables & Lists,
HTML Attributes, DOM Structure, Browser Rendering Basics, Responsive Markup, Cross-Browser Compatibility, Web Standards,
Clean Markup Practices

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learning roadmap :
Introduction to Web (Markup Languages → How Web Works → HTTP → DNS → Domains → Browsers → Hosting → SEO Basics) (https://www.youtube.com/playlist?list=PL0UJI1nZ56yCDWcDY1kgwH92oXbTXc-Zi)
→ Frontend Basics (HTML → CSS → JavaScript) (https://www.youtube.com/playlist?list=PL0eyrZgxdwhwP0AxnbBiDBCi53LK9uCMZ)
→ First HTML File (Tags → Attributes → Comments → DOCTYPE → html → head → meta → body → Entities) (https://www.youtube.com/playlist?list=PL0eyrZgxdwhwP0AxnbBiDBCi53LK9uCMZ)
→ Basic Tags (h1–h6 → p → br → hr → pre) (https://www.youtube.com/playlist?list=PL0eyrZgxdwhwP0AxnbBiDBCi53LK9uCMZ)
→ Textual Tags (b/strong → i/em → mark → sub → sup → div → span → links → title) (https://www.youtube.com/playlist?list=PL0eyrZgxdwhwP0AxnbBiDBCi53LK9uCMZ)
→ Standard Attributes (id → class → style → data-* attributes)
→ Tables
→ Lists (Ordered → Unordered → Definition → Nested)
→ Embedding Media (Images → figure → audio → video → iframe → CSP)
→ Highlighting Changes (del → s → ins)
→ Quotations & Citations (blockquote → q → cite → dfn → abbr → address)
→ Forms (Inputs → Labels → File Upload → Validation → Limitations)
→ Semantic Markup (header → nav → main → section → article → aside → footer)
→ Styling Basics (Inline CSS → Internal CSS → External CSS)
→ Including JavaScript
→ Accessibility (ARIA → Screen Readers → Semantic HTML)

(https://www.youtube.com/playlist?list=PLEfukpL-3chlbr0sfSKyYmCtb46LcqX0P)
→ SEO Basics (Meta Tags → Headings → Semantic Structure)
(https://www.youtube.com/playlist?list=PL0eyrZgxdwhwP0AxnbBiDBCi53LK9uCMZ)

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related roadmaps :

→ Frontend Roadmap
→ JavaScript Roadmap
→ CSS Roadmap
→ Web Accessibility Roadmap

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certificates :

**Free

1. HTML Introduction – W3Schools
   https://www.w3schools.com/html/

2. HTML Basics – MDN Web Docs

https://developer.mozilla.org/en-US/docs/Learn/HTML

3. Responsive Web Design (HTML + CSS) – freeCodeCamp
   https://www.freecodecamp.org/learn/responsive-web-design/

**Paid

1. HTML, CSS & Web Design Certification – Coursera (Meta)
   https://www.coursera.org/professional-certificates/meta-front-end-developer

2. Frontend Web Developer Nanodegree – Udacity (Used by Google, Amazon hiring paths)
   https://www.udacity.com/course/front-end-web-developer-nanodegree--nd0011

3. HTML & CSS Certification – edX (W3C)
   https://www.edx.org/professional-certificate/w3cx-front-end-web-developer

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role : JavaScript Developer / Frontend Developer

skills :
JavaScript (ES6+), Variables & Scope, Data Types, Type Casting, Arrays & Objects, Maps & Sets, Control Flow, Loops,
Functions, Closures, Hoisting, Equality Comparisons, Error Handling, Asynchronous JavaScript, Promises, async/await,
API Integration (Fetch, XHR), DOM Manipulation, Event Loop, Modules (CommonJS, ESM), Classes, Prototypes, Memory Management,
Browser DevTools, Debugging & Performance Optimization

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learning roadmap :
Introduction to JavaScript (What is JS → History → Versions → How to Run JS)

(https://www.youtube.com/playlist?list=PLGjplNEQ1it_oTvuLRNqXfz_v_0pq6unW)
→ Variables (var → let → const → Hoisting → Naming Rules → Block/Function/Global Scope) (https://www.youtube.com/playlist?list=PLGjplNEQ1it_oTvuLRNqXfz_v_0pq6unW)
→ Data Types (Primitive Types → Object → Prototype → Prototypal Inheritance → typeof → Built-in Objects)
→ Type Casting (Type Conversion vs Coercion → Explicit Casting → Implicit Casting)
→ Data Structures (Arrays → Typed Arrays → Map → WeakMap → Set → WeakSet → JSON) (https://www.youtube.com/playlist?list=PLM68oyaqFM7TNKEsvkrM67WuDJieeITFZ)
→ Equality Comparisons (== → === → Object.is → SameValue → SameValueZero) (https://www.youtube.com/playlist?list=PLGjplNEQ1it_oTvuLRNqXfz_v_0pq6unW)
→ Loops & Iterations (for → while → do…while → for…in → for…of → break/continue)
→ Control Flow (if/else → switch → try/catch/finally → throw → Error Objects)
→ Expressions & Operators (Assignment → Arithmetic → Logical → Bitwise → Unary → Conditional → BigInt → String Operators)
→ Functions (Parameters → Default Params → Rest → Arrow Functions → IIFE → arguments → Recursion → Lexical Scope → Closures → Strict Mode → this Keyword → call/apply/bind)
→ Asynchronous JavaScript (Event Loop → setTimeout → setInterval → Callbacks → Callback Hell → Promises → async/await)
→ Working with APIs (Fetch → XMLHttpRequest)
→ Classes (https://www.youtube.com/playlist?list=PLDtWoQ-cxqiwamnwULkiPkzx8Iv0ny9eV)
→ Iterators & Generators (https://www.youtube.com/playlist?list=PL0zVEGEvSaeG2T5n8FuPGb11JHea7idb9)
→ Modules (CommonJS → ES Modules) (https://www.youtube.com/playlist?list=PLDtWoQ-cxqiwamnwULkiPkzx8Iv0ny9eV)
→ Memory Management (Memory Lifecycle → Garbage Collection) (https://www.youtube.com/playlist?list=PL0k8toYFGjDvGN8J2CIJ-JX-EO8o8G00i)
→ DOM APIs (https://www.youtube.com/playlist?list=PLp57ZqpB-mGF-1hqnfht-jmsI8idoOFkJ)
→ Browser DevTools (Debugging → Memory Leaks → Performance Debugging) (https://www.youtube.com/playlist?list=PLNYkxOF6rcIC74v_mCLUXbjj7Ng7oTAPE)

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related roadmaps :

→ Frontend Roadmap
→ React Roadmap
→ TypeScript Roadmap
→ Node.js Roadmap

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certificates :

**Free

1. JavaScript Algorithms and Data Structures – freeCodeCamp
   https://www.freecodecamp.org/learn/javascript-algorithms-and-data-structures/

2. JavaScript Guide – MDN Web Docs
   https://developer.mozilla.org/en-US/docs/Web/JavaScript/Guide

3. JavaScript Basics – W3Schools
   https://www.w3schools.com/js/

**Paid

1. JavaScript Programming – Meta (Coursera)
   https://www.coursera.org/professional-certificates/meta-front-end-developer

2. Front-End Web Developer Nanodegree – Udacity (used by Google, Amazon hiring tracks)
   https://www.udacity.com/course/front-end-web-developer-nanodegree--nd0011

3. JavaScript Certification – edX (W3C / Microsoft)
   https://www.edx.org/learn/javascript

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role : Kotlin Developer

skills :
Kotlin Basics, Variables & Type Inference, Collections, Control Flow, Functions & Lambdas, Higher-Order Functions,
Extension Functions, Object-Oriented Programming, Generics, Null Safety, Coroutines & Async Programming, File I/O,
Kotlin Standard Library, Gradle & Maven, Documentation (KDoc, Dokka), Kotlin Multiplatform, Android Development,
Server-Side Kotlin (Spring Boot, Ktor), Interoperability with Java, CI/CD Basics

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learning roadmap :
Introduction to Kotlin (Why Kotlin → History → Java Interoperability → Environment Setup) (https://www.youtube.com/playlist?list=PLQkwcJG4YTCRSQikwhtoApYs9ij_Hc5Z9)
→ Language Basics (val vs var → Variables → Type Inference → Type Checks & Casts → Basic Types → Data Types)
→ Code Organization (main Function → Comments)
→ Printing Data (print → println → String Templates)
→ Collections (Lists → Maps → Sets → Read-only vs Mutable → Ranges → Progressions → Sequences → Iterators)
→ Control Flow (if → when → for → while → Label Loops → break/continue → Exceptions)
→ Functions (Parameters → Return → varargs → Standard Functions → Local/Member Functions → Tail-recursive → Lambdas → Anonymous Functions → Higher-order Functions → Function Types → Extension Functions)
→ Collection Operations (Transformations → Filtering → plus/minus → Retrieving Parts → Single Elements → Ordering → Grouping → Aggregates → List/Map/Set-specific Ops)
→ Classes & Objects (Classes → Instances → Constructors → Properties → Methods → Nested/Inner Classes → Object Declarations → Inheritance → Visibility Modifiers → Abstract Classes → Interfaces → Property Delegates → Data/Sealed/Enum/Inline Classes → Generics → Type Aliases)
→ Null Safety (Nullable vs Non-nullable → Safe Calls → Elvis → Not-null Assertion → Safe Casts)
→ Asynchronous Programming (Coroutines → Suspending Functions → Coroutine Builders → Coroutine Behavior → Async Flow → Best Practices) --> (https://www.youtube.com/playlist?list=PLQkwcJG4YTCQcFEPuYGuv54nYai_lwil)
→ I/O (File Reading/Writing → Buffered Streams → Creating Files) (https://www.youtube.com/playlist?list=PLQkwcJG4YTCRSQikwhtoApYs9ij_Hc5Z9)
→ Packages & Ecosystem (Packages → Imports → Kotlin Libraries → Standard Library → Default Imports → Opt-in APIs)
→ Documentation (KDoc → Dokka)
→ Build Tools (Gradle → Maven → Build Tool API → Gradle Plugins)
→ CI/CD Tools (TeamCity)
→ IDEs (IntelliJ IDEA → Android Studio → Kotlin Notebook)
→ Kotlin Platforms (Multiplatform → Compose Multiplatform → Kotlin/JS →

Kotlin/WASM → Kotlin/Native → Java Interop → Scripting)
→ Kotlin Applications (Android → Server-Side → Data Analysis → Competitive Programming → AI Development) (https://www.youtube.com/playlist?list=PLQ_Ai1O7sMV0LRCZJaHpl_9WLQunMqrzG)

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related roadmaps :

→ Android Roadmap
→ Java Roadmap
→ Spring Boot Roadmap
→ Full Stack Roadmap

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certificates :

Free (Best):

1. Kotlin Official Docs & Tutorials – JetBrains
   https://kotlinlang.org/docs/home.html

2. Kotlin for Beginners – JetBrains Academy
   https://www.jetbrains.com/academy/

Paid / Industry-Recognized:

1. Kotlin for Java Developers – Coursera (JetBrains)
   https://www.coursera.org/learn/kotlin-for-java-developers

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role : Kubernetes Engineer

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skills :
Kubernetes Fundamentals, Containers & Docker, Pods, ReplicaSets, Deployments, StatefulSets, Jobs, Kubernetes Networking,
Services & Load Balancing, ConfigMaps, Secrets, Resource Requests & Limits, Namespace Quotas, RBAC, Kubernetes Security,
Monitoring & Logging, Observability, Autoscaling (HPA, VPA, Cluster Autoscaler), Scheduling, Storage & Volumes, Helm,
GitOps, CI/CD with Kubernetes, Canary & Blue-Green Deployments, CRDs, Kubernetes APIs, Multi-Cluster Management

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learning roadmap :
Introduction to Kubernetes (Overview → Why Kubernetes → Core Concepts → Alternatives) (https://www.youtube.com/playlist?list=PLy7NrYWoggjziYQIDorlXjTvvwweTYoNC)
→ Containers
→ Kubernetes Setup (Managed Providers → Local Cluster → First Application Deployment)
→ Running Applications (Pods → ReplicaSets → Deployments → StatefulSets → Jobs)
→ Services & Networking (Service Types → External Access → Load Balancing → Pod-to-Pod Networking)
→ Configuration Management (ConfigMaps → Secrets)

→ Resource Management (Requests & Limits → Namespace Quotas → Resource Optimization) (https://www.youtube.com/playlist?list=PLy7NrYWoggjziYQIDorlXjTvvwweTYoNC)
→ Security (RBAC → Network Policies → Pod & Container Security)
→ Monitoring & Logging (Logs → Metrics → Traces → Resource Health → Observability Tools)
→ Autoscaling (HPA → VPA → Cluster Autoscaler)
→ Scheduling (Taints & Tolerations → Topology Spread Constraints → Pod Priority → Evictions)
→ Storage & Volumes (Persistent Volumes → CSI Drivers → Stateful Applications)
→ Deployment Patterns (CI/CD → GitOps → Helm Charts → Canary Deployments → Blue-Green → Rolling Updates & Rollbacks)
→ Advanced Kubernetes (Custom Controllers → Custom Schedulers → CRDs → Kubernetes APIs & Extensions)
→ Cluster Operations (Control Plane Setup → Worker Node Management → Self-managed vs Managed Clusters → Multi-Cluster Management)
(https://www.youtube.com/playlist?list=PLy7NrYWoggjziYQIDorlXjTvvwweTYoNC)

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related roadmaps :

→ Docker Roadmap
→ DevOps Roadmap
→ Backend Roadmap
→ System Design Roadmap

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certificates :

✅ Free

1. **Introduction to Kubernetes** – The Linux Foundation (edX)
   https://www.edx.org/learn/kubernetes/the-linux-foundation-introduction-to-kubernetes

2. **Kubernetes Official Documentation & Tutorials** – CNCF
   https://kubernetes.io/docs/home/

3. **Kubernetes Basics** – Kubernetes.io Interactive Tutorial
   https://kubernetes.io/docs/tutorials/kubernetes-basics/

💼 Paid

1. **CKA – Certified Kubernetes Administrator** (CNCF / Linux Foundation)
   https://training.linuxfoundation.org/certification/certified-kubernetes-administrator-cka/

2. **CKAD – Certified Kubernetes Application Developer** (CNCF / Linux Foundation)
   https://training.linuxfoundation.org/certification/certified-kubernetes-application-developer-ckad/

3. **CKS – Certified Kubernetes Security Specialist** (CNCF / Linux Foundation)
   https://training.linuxfoundation.org/certification/certified-kubernetes-security-specialist-cks/

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role : Laravel Developer / PHP Backend Developer / Full Stack Developer (Laravel)

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skills :
PHP, Laravel Framework, MVC Architecture, REST API Development, Routing & Controllers, Blade Templating Engine,
Eloquent ORM, MySQL/PostgreSQL, CRUD Operations, Authentication & Authorization, Middleware, Form Validation,
Laravel Sanctum & Passport, Dependency Injection, Database Migrations & Seeders, Queue & Job Processing,
Event-Driven Architecture, Caching, Logging & Debugging, Unit & Feature Testing, Artisan CLI, File Storage Handling,
Deployment & Optimization, Laravel Forge, Performance Optimization, Git Version Control, API Security, Docker (Laravel Sail)

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learning roadmap :
Web Development Fundamentals (HTTP → Browsers → Client-Server Architecture → Databases) (https://www.youtube.com/playlist?list=PL0UJI1nZ56yCDWcDY1kgwH92oXbTXc-Zi)
→ PHP Basics (Syntax → OOP in PHP → Composer → PSR Standards) (https://www.youtube.com/playlist?list=PLQH1-k79HB3_EJxKnKJ-eoAtnnBxHovLb)
→ Introduction to Laravel (What is Laravel → Why Frameworks → Full Stack vs Backend Usage) (https://www.youtube.com/playlist?list=PLb90U00aeJnAgscGwsbok7Bls7SjC44kJ)
→ Laravel Setup (Laravel Herd / Composer → Create Project → Starter Kits: Breeze → Jetstream → Livewire → Inertia) (https://www.youtube.com/playlist?list=PLb90U00aeJnAgscGwsbok7Bls7SjC44kJ)
→ Laravel Project Structure (Request-Response Lifecycle → App → Resources → Routes → Config → Storage → Database → Tests → Public → Vendor)
→ Routing & Controllers (Basic Routes → Route Parameters → Named Routes → Route Groups → Middleware → Model Binding → Resource Controllers → Dependency Injection → Redirect & View Routes)
→ Middleware (Global Middleware → Route Middleware → Request Filtering & Security)
→ Requests & Responses (Form Data Handling → File Uploads → JSON Responses → Redirect Responses → Custom Responses)
→ Views & Blade (Blade Templates → Layouts → Components → Directives → Loops → Conditionals → Blade + Livewire)
→ Databases & Eloquent ORM (Query Builder → CRUD → Relationships → Accessors & Mutators → Query Scopes → Pagination → Migrations → Seeders)
→ Forms & Validation (Manual Validation → Error Handling → Validation Rules)
→ Authentication (Manual Auth → Starter Kit Auth → Laravel Sanctum → Laravel Passport)
→ Authorization (Gates → Policies → Role-based Access Control)
→ Security & Debugging (Encryption → Hashing → Logging → Telescope → Debugbar → Error Handling → Exception Customization)
→ Testing (Unit Testing → Feature Testing → PHPUnit → Pest Testing Framework)
→ Advanced Laravel Features (Queues → Jobs → Events → Listeners → Notifications → Task Scheduling → File Storage → Localization → Facades → Package Development)
→ Performance & Optimization (Caching → Streamed Responses → Code Optimization → Monitoring → Pint Code Style)
→ Deployment & DevOps (Configuration Optimization → Health Checks → Laravel Forge → Laravel Cloud → Docker with Sail → Production Deployment) (https://www.youtube.com/playlist?list=PLb90U00aeJnAgscGwsbok7Bls7SjC44kJ)

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related roadmaps :

→ PHP Roadmap
→ Backend Developer Roadmap
→ Full Stack Developer Roadmap
→ API Development Roadmap
→ DevOps Roadmap

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certificates :

✅ Free

1. **Laravel Official Documentation & Laravel Bootcamp**
   https://laravel.com/docs
   https://bootcamp.laravel.com/

💼 Paid

1. **Laravel Certification Program – Laravel (Official Certification)**
   https://certification.laravel.com/

2. **Meta Backend Developer Professional Certificate (Coursera)**
   [https://www.coursera.org/professional-certificates/meta-back-end-developer]
(https://www.coursera.org/professional-certificates/meta-back-end-developer)

3. **IBM Full Stack Cloud Developer Professional Certificate (Coursera)**
   [https://www.coursera.org/professional-certificates/ibm-full-stack-cloud-
developer](https://www.coursera.org/professional-certificates/ibm-full-stack-
cloud-developer)

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role : Linux System Administrator

skills :
Linux Fundamentals, Command Line Proficiency, File & Directory Management, Vim & Nano Editing, Shell Basics,
Environment Variables, File Permissions, Links & Archiving, Text Processing Tools (grep, awk, sed, cut, sort),
Process Management, User & Group Management, Systemd Service Management, Package Management (apt/yum/snap),
Disks & Filesystems, Networking Fundamentals, Boot & Logs, Shell Scripting, Troubleshooting,
Containers & Container Runtimes (Docker), TCP/IP Networking

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learning roadmap :
Navigation Basics (Basic Commands → Moving Files/Dirs → Creating/Deleting Files/Dirs → Directory Hierarchy) (https://www.youtube.com/playlist?list=PLT98CRl2KxKHaKA9-4_I38sLzK134p4GJ)
→ Editing Files (Vim → Nano) (https://www.youtube.com/playlist?list=PLT98CRl2KxKHaKA9-4_I38sLzK134p4GJ)
→ Shell Fundamentals (Command Path → Environment Variables → Redirects → Super User → Command Help)
→ Working with Files (Permissions → Archiving & Compression → Copying/Renaming → Soft/Hard Links)

→ Text Processing (stdout/stderr/stdin → grep → awk → cut → sort → head → tail → tr → paste → join → split → pipe → tee → nl → wc → expand/unexpand → uniq) (https://www.youtube.com/playlist?list=PLNlxfQ7JYooHd_Ptnzxlc8JWdTczNpUR0)
→ Server Review (Uptime & Load → Authentication Logs → Running Services → Memory & Disk Usage) (https://www.youtube.com/playlist?list=PLT98CRl2KxKHaKA9-4_I38sLzK134p4GJ)
→ Process Management (Foreground/Background Processes → Listing/Searching → Signals → Killing → Priorities → Forking)
→ User & Group Management (Create/Delete/Update Users & Groups → Permission Assignment)
→ Service Management with systemd (Start/Stop/Status → Logs → New Services)
→ Package Management (Repos → Search/Install/Remove/Upgrade → Snap) (https://www.youtube.com/playlist?list=PLJNrjHYcEuBUcech9H4MHC3kHiWQaJVCM)
→ Disks & Filesystems (Inodes → Filesystem Types → Swap → Mounts → Adding Disks → LVM) (https://www.youtube.com/playlist?list=PLT98CRl2KxKHaKA9-4_I38sLzK134p4GJ)
→ Booting Linux (Boot Loaders → Boot Logs)
→ Networking Fundamentals (TCP/IP Stack → Subnetting → IP Routing → DNS → DHCP → arp → Netfilter → SSH/File Transfer) (https://www.youtube.com/playlist?list=PLZorYoZY3amerHPJOhMFQNpAPJ4HE5vED)
→ Shell Programming (Literals → Variables → Loops → Conditionals → Debugging) (https://www.youtube.com/playlist?list=PLIhvC56v63IKioClkSNDjW7iz-6TFvLwS)
→ Troubleshooting (ICMP → ping → traceroute → netstat → Packet Analysis) (https://www.youtube.com/playlist?list=PLZorYoZY3amerHPJOhMFQNpAPJ4HE5vED)
→ Containerization Essentials (ulimits → cgroups → Container Runtimes → Docker) (https://www.youtube.com/playlist?list=PLy7NrYWoggjzfAHlUusx2wuDwfCrmJYcs)

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related roadmaps :

→ DevOps Roadmap
→ Docker Roadmap
→ Kubernetes Roadmap
→ Backend Roadmap
→ System Design Roadmap

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certificates :

**Free

1. **Introduction to Linux – The Linux Foundation (edX)**
   https://www.edx.org/course/introduction-to-linux

**Paid

1. **LFCS – Linux Foundation Certified System Administrator**
   https://training.linuxfoundation.org/certification/linux-foundation-certified-system-administrator-lfcs/

2. **RHCSA – Red Hat Certified System Administrator (Red Hat)**

https://www.redhat.com/en/services/certification/rhcsa

3. **Linux Professional Institute LPIC-1**

[https://www.lpi.org/our-certifications/lpic-1-overview](https://www.lpi.org/

our-certifications/lpic-1-overview)

**role : Next.js Frontend Developer

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skills :
JavaScript (ES6+), React.js, Next.js, SPA, SSR, CSR, SSG, App Router, Pages Router, Routing Patterns, Layouts & Templates,
Middleware, Server Actions, Data Fetching, Caching & Revalidation, Client & Server Components, Node.js Runtime,
Edge Runtime, REST APIs, Tailwind CSS, CSS Modules, Sass, Image & Font Optimization, Metadata & SEO,
Internationalization (i18n), TypeScript, Environment Variables, Testing (Vitest, Jest, Playwright, Cypress),
Performance Optimization, Docker, Production Deployment

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learning roadmap :
JavaScript Basics (https://www.youtube.com/playlist?list=PLu0W_9lII9agS67Uits0UnJyrYiXhDS6q)
→ Why React (https://www.youtube.com/playlist?list=PLC3y8-rFHvwgg3vaYJgHGnModB54rxOk3)
→ Why Frontend Frameworks
→ SPA vs SSR
→ Introduction to React Frameworks (https://www.youtube.com/playlist?list=PLC3y8-rFHvwjOKd6gdf4QtV1uYNiQnruI)
→ Next.js Fundamentals
→ Why Next.js
→ Remix Overview
→ Rendering Strategies (SPA → CSR → SSR → SSG)
→ Getting Started (create-next-app)
→ Routing Fundamentals
→ Next.js Routing Basics
→ Routing Terminology
→ Pages Router → App Router → Why App Router
→ Routing Patterns & Structuring Routes
→ Layouts & Templates
→ Loading UI & Streaming
→ Error States & Handling
→ Intercepting Routes
→ Parallel Routes
→ Static vs Dynamic Routes
→ Redirects
→ API Endpoints
→ Rendering Pages
→ Middleware (Use Cases → Route Matcher → Headers → Cookies)
→ Working with Data (Fetching → Caching → Memoization → React Cache → Revalidation → Sensitive Data)
→ Server Actions
→ Rendering (Client Components → Server Components → Composition)
→ Runtimes (Node.js → Edge)
→ Styling (Global CSS → CSS Modules → Sass → Tailwind → CSS-in-JS)
→ Optimization (Images → Videos → Fonts → Metadata → Scripts → i18n)
→ Advanced Configuration (Bundling → Lazy Loading → Analytics → OpenTelemetry → Static Assets → TypeScript → ESLint → Prettier → Env Variables → MDX → Custom Server)
→ Testing (Vitest → Jest → Playwright → Cypress)
→ Preparing for Production
→ Deployment (Node Server → Docker → Static Export → Adapters)

(https://www.youtube.com/playlist?list=PLC3y8-rFHvwjOKd6gdf4QtV1uYNiQnruI)

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related roadmaps :

→ Frontend Roadmap
→ React Roadmap
→ Next.js Roadmap
→ Full Stack Roadmap

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certificates :

**Free

1. **Next.js Learn – Vercel (Official)**
   https://nextjs.org/learn

2. **React & Next.js Tutorials – freeCodeCamp**
   https://www.freecodecamp.org/news/tag/react/

3. **JavaScript Algorithms & Frontend Concepts – freeCodeCamp**
   https://www.freecodecamp.org/learn/

**Paid

1. **Meta Front-End Developer Professional Certificate (Meta – Coursera)**
   https://www.coursera.org/professional-certificates/meta-front-end-developer

2. **IBM Front-End Developer Professional Certificate**
   https://www.coursera.org/professional-certificates/ibm-frontend-developer

3. **Frontend Masters – React & Next.js Learning Path**
   https://frontendmasters.com/learn/react/

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role : PHP Developer
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skills :
PHP (Core), Web Development Fundamentals, Syntax & Variables, Data Types & Casting, Output & Debugging, Arrays,
Conditionals & Loops, Functions (Named, Anonymous, Arrow, Callback, Variadic), HTTP & Request Handling,
File Handling & Operations, Form Processing, Uploads, State Management (Sessions & Cookies), Security Best Practices,
Database Connectivity (PDO/MySQLi), ORM, Transactions, OOP (Classes, Inheritance, Polymorphism, Traits, Namespaces,
Magic Methods), Composer & Package Management, PHP Frameworks (Laravel, Symfony), Testing (PHPUnit, Pest),
Static Analysis (PHPStan, Psalm), Performance Optimization, Web Servers (Apache/Nginx/PHP-FPM), Debugging Tools (Xdebug)

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learning roadmap :
Introduction to PHP (What PHP Is → History → Versions & Features)
(https://www.youtube.com/playlist?list=PL0eyrZgxdwhwwQQZA79OzYwl5ewA7HQih)
→ Development Environment Setup (Install PHP → WAMP/XAMPP/MAMP/LAMP → Local Server)
→ Fundamentals (Syntax → Variables & Scope → Data Types → Type Casting)
→ Output & Debugging (echo → print → var_dump → print_r)
→ Arrays (Indexed → Associative → Multi-dimensional)
→ Conditionals (if/else → switch → match → Null Coalescing → Null Safe)
→ Loops
→ Constants
→ Functions (Declaration → Parameters → Return Values → Default/Optional → Named Args → Anonymous → Arrow → Callback → Recursion → Variadic)
→ File Handling (include/require → require_once/include_once)
→ File Operations (Reading/Writing → Permissions → JSON/CSV/XML Processing)
→ HTTP & Request Handling (Methods → $_GET/$_POST/$_REQUEST/$_SERVER → Superglobals)
→ Form Processing
→ File Uploads
→ State Management (Cookies → Sessions)
→ Basics of Security (Input Validation → XSS Prevention → SQL Injection → CSRF → Hashing → Auth Mechanisms → Sanitization)
(https://www.youtube.com/playlist?list=PLfdtiltiRHWFsPxAGO-SVPGhCbCwKWF_N)
→ Database Integration (PDO/MySQLi → Connectivity → ORM → Transactions → Connection Pooling → Optimization → Migrations)
(https://www.youtube.com/playlist?list=PL0eyrZgxdwhwwQQZA79OzYwl5ewA7HQih)
→ OOP Fundamentals (Classes/Objects → Constructors/Destructors → Properties → Methods → Access Specifiers → Static Members → Inheritance)
(https://www.youtube.com/playlist?list=PL0eyrZgxdwhypQiZnYXM7z7-OTkcMgGPh)
→ More OOP (Polymorphism → Abstract Classes → Interfaces → Traits → Namespaces → Magic Methods → Dependency Injection → Type Declarations)
→ Frameworks & Tools (Laravel → Symfony → Composer → Packagist → Autoloading)
(https://www.youtube.com/playlist?list=PLb90U00aeJnAgscGwsbok7Bls7SjC44kJ)
→ Testing & Quality Tools (PHPUnit → Pest → PHPCodeSniffer → PHP CS Fixer → Static Analysis with PHPStan/Phan/Psalm)
(https://www.youtube.com/playlist?list=PLQH1-k79HB3-9tf-3bKXjyhQDVJbXYX8f)
→ External Integrations (cURL → Guzzle → PHP-FIG → PSR Standards)
(https://www.youtube.com/playlist?list=PLr3d3QYzkw2xabQRUpcZ_IBk9W50M9pe-)
→ Advanced PHP Topics (Performance Optimization → Caching → Memory Management → Opcode Caching)
→ System Interactions (System Commands → Process Control → Environment Variables) (https://www.youtube.com/playlist?list=PL0eyrZgxdwhwwQQZA79OzYwl5ewA7HQih)
→ Debugging Tools (Xdebug → Zend Debugger)
(https://www.youtube.com/playlist?list=PLdYYj2XLw5BkvJnZWA7LIuKEBdsw1Mi5G)
→ Web Server Config (Apache → Nginx → PHP-FPM)
(https://www.youtube.com/playlist?list=PLr3d3QYzkw2xabQRUpcZ_IBk9W50M9pe-)

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related roadmaps :

→ Backend Development Roadmap
→ DevOps Engineering Roadmap
→ System Design Roadmap
→ Software Architect Roadmap

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certificates :

**Free

1. **PHP Manual & Tutorials – Official PHP Docs**
   https://www.php.net/manual/en/

3. **PHP Tutorials – W3Schools**
   https://www.w3schools.com/php/

**Paid

2. **Zend PHP Certification (Zend / Perforce)**
   https://www.zend.com/en/certification/php-certification

3. **PHP Developer Career Path – Codecademy Pro**
   https://www.codecademy.com/learn/paths/php

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role : Redis Developer / Backend Engineer / Caching & Performance Specialist

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skills :
Redis Fundamentals, In-Memory Data Structures, Key-Value Store Concepts, Caching Strategies, Session Management,
Pub/Sub Messaging, Leaderboards & Counters, Redis CLI, Strings, Lists, Sets, Hashes, Sorted Sets, Bitmaps, HyperLogLog,
Streams, Geospatial Indexes, Transactions, Lua Scripting, Persistence (RDB / AOF / Hybrid), Replication & Clustering,
Redis Sentinel, Security (Authentication & TLS/SSL), Monitoring & Tuning, Memory Management & Policies,
Redis Modules (RedisJSON, RedisSearch, TimeSeries, Bloom), Production Operations, Backup & Disaster Recovery

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learning roadmap :
Introduction to Redis (What Redis Is → In-Memory Store → Key-Value DB → Cache → Message Broker) (https://www.youtube.com/playlist?list=PL4cUxeGkcC9h3V2eqhi8rRdIDJshP-b4P)
→ Core Use Cases (Caching → Real-Time Analytics → Session Management → Pub/Sub → Leaderboards/Counters)
→ Key Features (Persistence Options → Rich Data Structures → Performance & Scalability)
→ Redis vs Other Databases (vs SQL/NoSQL → When to Choose Redis)
→ Getting Started with Redis (Install Locally → Docker Setup → Run Redis → Redis CLI → Basic Commands: SET/GET/DEL/EXPIRE/TTL)
→ Basic Data Structures (Strings → Lists → Sets → Hashes → Sorted Sets)
→ Working with Keys (DEL → expire → TTL)
→ Intermediate Redis Operations (Key Naming → Pattern Retrieval → Expiration → Pipelining → MGET/MSET)
→ Advanced Structures (Bitmaps → HyperLogLog → Streams → Geospatial Indexes)
→ Pub/Sub Messaging (SUBSCRIBE/UNSUBSCRIBE → PUBLISH → Use Cases)
→ Transactions & Atomicity (WATCH → MULTI → EXEC → Optimistic Locking)
→ Lua Scripting (EVAL → EVALSHA → Use Cases)
→ Persistence Strategies (No Persistence → RDB Snapshotting → AOF → Hybrid → Choosing Strategies → Best Practices)
→ Replication, High Availability & Clustering (Replication Basics → Redis Sentinel → Clustering) (https://www.youtube.com/playlist?list=PLbGui_ZYuhijTJDBVBBqvri76uUNfsrCc)

→ Security (Authentication → Network Security → SSL/TLS Encryption)
(https://www.youtube.com/playlist?list=PLS1QulWo1RIYZZxQdap7Sd0ARKFI-XVsd)
→ Monitoring & Optimization (INFO → MONITOR → Slow Log → redis-benchmark → Memory Management → Max Memory Policy → Third-Party Tools like RedisInsight)
→ Redis Modules (RedisJSON → Search → TimeSeries → RedisBloom)
(https://www.youtube.com/playlist?list=PL4cUxeGkcC9h3V2eqhi8rRdIDJshP-b4P)
→ Production Readiness (redis.conf → Config Tuning → Backup/Restore → Upgrading → Minimizing Downtime → Disaster Recovery)
(https://www.youtube.com/playlist?list=PLS1QulWo1RIYZZxQdap7Sd0ARKFI-XVsd)
→ Enterprise Features (Active-Active Geo Distribution → Redis on Flash → Enterprise Security & Compliance) (https://www.youtube.com/playlist?list=PLbGui_ZYuhijTJDBVBBqvri76uUNfsrCc)

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related roadmaps :

→ MongoDB Roadmap
→ PostgreSQL Roadmap
→ SQL Roadmap
→ Backend Developer Roadmap
→ DevOps / System Design Roadmap

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certificates :

**Free

1. **Redis University – Official Courses (Free)**
   https://university.redis.com/

2. **Redis Tutorials – Redis.io Docs**
   https://redis.io/commands

**Paid

1. **Redis Developer Certification – Redis University (Paid Exam Option)**
   https://university.redis.com/certification

**role : MongoDB Developer

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skills :

MongoDB Fundamentals, NoSQL Data Modeling, BSON & JSON, Collections & Documents, CRUD Operations, Query Operators,
Indexing Strategies, Aggregation Framework, Transactions, Performance Optimization, MongoDB Atlas,
Replication & Replica Sets, Sharding & Horizontal Scaling, Backup & Recovery, Security & Authentication,
Encryption (At Rest & In Transit), Monitoring & Tuning, MongoDB Drivers & Connectors, Production Database Management

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learning roadmap :

MongoDB Basics
SQL vs NoSQL (https://www.youtube.com/playlist?list=PL4cUxeGkcC9h77dJ-QJlwGlZlTd4ecZOA)
→ What is MongoDB?
→ When to Use MongoDB
→ MongoDB Atlas (https://www.youtube.com/playlist?list=PL4RCxklHWZ9vmzBP7lybE08CxbIU17PIf)
→ MongoDB Terminology (https://www.youtube.com/playlist?list=PL4cUxeGkcC9h77dJ-QJlwGlZlTd4ecZOA)

Data Model & Data Types
BSON vs JSON
→ Embedded Documents & Arrays
→ MongoDB Data Types (ObjectId, String, Int, Date, Boolean, Decimal128, etc.)

Collections & CRUD Operations
insert()
→ find()
→ update()
→ delete()
→ bulkWrite()
→ validate()
→ Counting Documents

Useful Concepts
Read Concern & Write Concern
→ Cursors
→ Retryable Reads & Writes

Query Operators
Projection Operators
→ Comparison Operators
→ Array Operators
→ Element Operators
→ Logical Operators

Performance Optimization
Indexes (Single, Compound, Text, TTL, Geospatial)
→ Atlas Search Indexes (https://www.youtube.com/playlist?list=PL4RCxklHWZ9vmzBP7lybE08CxbIU17PIf)
→ Query Optimization (https://www.youtube.com/playlist?list=PL4cUxeGkcC9h77dJ-QJlwGlZlTd4ecZOA)

Aggregation Framework
Pipelines (https://www.youtube.com/playlist?list=PLWkguCWKqN9OwcbdYm4nUIXnA2IoXX0LI)
→ Stages & Operators
→ $match, $group, $lookup, $unwind, $project, $sort

Transactions (https://www.youtube.com/playlist?list=PL1BztTYDF-QPwd9Qg-nxQk1UYtorhzXLc)

Developer Tools & Integrations
MongoDB Drivers (https://www.youtube.com/playlist?list=PL4cUxeGkcC9h77dJ-QJlwGlZlTd4ecZOA)
→ Connectors (Kafka, Spark, Elasticsearch)

Backup & Recovery
mongodump
→ mongorestore

Scaling MongoDB
Replica Sets (https://www.youtube.com/playlist?

list=PL1BztTYDF-QPwd9Qg-nxQk1UYtorhzXLc)
→ Sharded Clusters
→ Configuration Tuning
→ Index & Query Optimization (https://www.youtube.com/playlist?
list=PL4cUxeGkcC9h77dJ-QJlwGlZlTd4ecZOA)

MongoDB Security
Role-Based Access Control
→ Authentication (X.509, Kerberos, LDAP)
→ MongoDB Auditing (https://www.youtube.com/playlist?
list=PL1BztTYDF-QPwd9Qg-nxQk1UYtorhzXLc)
→ Encryption (At Rest, Client-Side, Queryable, TLS/SSL)

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related roadmaps :

MongoDB
→ Backend Developer Roadmap
→ Node.js Roadmap
→ Frontend Roadmap (GraphQL Ecosystem)
→ System Design Roadmap
→ DevOps Roadmap

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certificates :

Free (Highly Recommended):

1. **MongoDB University – Official (Free)**
   https://learn.mongodb.com/

2. **MongoDB Docs & Tutorials**
   https://www.mongodb.com/docs/

Paid / Industry-Recognized:

3. **Complete MongoDB Bootcamp – Udemy**
   [https://www.udemy.com/course/mongodb-the-complete-developers-guide/]
(https://www.udemy.com/course/mongodb-the-complete-developers-guide/)

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role :

Shell Script Developer

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skills :

Linux Command Line, Bash Scripting, Shell Programming, File & Directory Management, File Permissions,
Text Processing (grep, awk, sed), Pipes & Redirections, Environment Variables, Process Management, Job Control,
System Monitoring, Networking Commands, Automation Scripts, Cron Jobs, Debugging Shell Scripts, Error Handling,
Regex, CI/CD Automation, Server Administration

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learning roadmap :

Introduction to Shell (https://www.youtube.com/playlist?list=PLT98CRl2KxKHaKA9-4_I38sLzK134p4GJ)
What is Shell → CLI vs GUI → What is Bash → What is Scripting

Popular Shells
bash → zsh → fish → dash → ksh → tcsh → PowerShell → cmd

Shell Setup
Install Bash → Configure Environment → Basic Setup

Shell Fundamentals
Files & Directories
pwd → ls → cd → mkdir → rmdir → cp → mv → rm → cat → touch → find

File Permissions
rwx → chmod → chown → chgrp

Text Processing
less/more → grep → head/tail → cut → paste → sort → uniq → wc → awk → sed

Wildcards
? → * → [] → {}

Redirects & Pipelines
stdin/stdout/stderr → pipes → output redirection → input redirection → error redirection → command substitution → process substitution

Shell Essentials
Tab Completion → Alias → History → Help Commands → Stop Execution

Bash Scripting (https://www.youtube.com/playlist?list=PLIhvC56v63IKioClkSNDjW7iz-6TFvLwS)
Text Editors
nano → vim → vi → emacs

Script Basics
Script Anatomy → Shebang → Running Scripts

Variables
Shell Variables → Environment Variables → Scope → Special Variables

Data Types
Strings → Arrays → Numbers → Associative Arrays

Operators
Arithmetic → Logical → Comparison → String → File Test

Numerics
Arithmetic Expansion → expr → let → bc → awk

Arguments
$1…$n → $@ → $# → shift

Exit Codes
$? → exit

String Manipulation
Length → Substring → Replace → Case Conversion

Input / Output

read → printf → here-doc → here-string

Conditionals
if → case

Loops
for → while → until → break → continue

Functions
Function Scope → Recursive Functions

Advanced Bash (https://www.youtube.com/playlist?list=PLdNWI2lSGGYPte3japrRofMNETags5EcG)
Regular Expressions
Basic Regex → Extended Regex → grep/sed/awk with regex

Debugging
set -x → bash -n → shellcheck

Error Handling
set -e → set -u → trap → logging

System Administration (https://www.youtube.com/playlist?list=PL9ooVrP1hQOH3SvcgkC4Qv2cyCebvs0Ik)
Process Management
jobs → fg/bg → nohup → disown

System Monitoring
ps → top → htop → free → uptime → df → du → vmstat → iostat

Networking
ping → curl → wget → ssh → scp → rsync → ip → netstat → ss

Package Management
apt → yum → dnf → brew

Compression
tar → gzip → zip → bzip2 → xz

Task Scheduling (https://www.youtube.com/playlist?list=PL9ooVrP1hQOH3SvcgkC4Qv2cyCebvs0Ik)
cron → crontab → at → systemd timers

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related roadmaps :

→ Linux Roadmap
→ DevOps Roadmap
→ Backend Roadmap
→ Docker Roadmap
→ Kubernetes Roadmap

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certificates :

**Free

1. **Linux Foundation – Introduction to Linux (edX)**
   https://www.edx.org/course/introduction-to-linux

2. **Bash Scripting Full Course – freeCodeCamp**
   https://www.freecodecamp.org/news/bash-scripting-tutorial-linux-shell-script-and-command-line-for-beginners/

3. **ShellCheck (Official Tool + Learning)**
   https://www.shellcheck.net/

**Paid

1. **Linux Foundation – Linux System Administration (LFCS)**
   https://training.linuxfoundation.org/certification/linux-foundation-certified-sysadmin-lfcs/

2. **Red Hat Certified System Administrator (RHCSA)**

https://www.redhat.com/en/services/certification/rhcsa

3. **Udemy – Complete Bash Shell Scripting**
   https://www.udemy.com/course/bash-shell-scripting-linux/

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role :

SQL Developer

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skills :

SQL Query Writing, Relational Databases, Data Modeling, CRUD Operations, Joins, Subqueries, Indexing, Query Optimization,
Transactions, ACID Properties, Stored Procedures, Views, Window Functions, Performance Tuning, Database Security,
Data Integrity, PostgreSQL / MySQL / SQL Server

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learning roadmap :

Learn the Basics

Relational Databases → RDBMS Concepts → Benefits & Limitations → SQL vs NoSQL → PostgreSQL (reference DB) (https://www.youtube.com/playlist?list=PLL_LQvNX4xKxq57Vs1PYCfH6FI4socDWS)

Basic SQL Syntax

SQL Keywords → Data Types → Operators

Core SQL Statements

SELECT → INSERT → UPDATE → DELETE

DDL (Data Definition Language)

CREATE TABLE → ALTER TABLE → TRUNCATE TABLE → DROP TABLE

DML (Data Manipulation Language)
(https://www.youtube.com/playlist?list=PLL_LQvNX4xKxq57Vs1PYCfH6FI4socDWS)

SELECT
FROM
WHERE
ORDER BY
GROUP BY
HAVING
JOIN
INSERT
UPDATE
DELETE

Aggregate Queries (https://www.youtube.com/playlist?list=PLL_LQvNX4xKxq57Vs1PYCfH6FI4socDWS)

SUM → COUNT → AVG → MIN → MAX → GROUP BY → HAVING

Constraints (https://www.youtube.com/playlist?list=PLL_LQvNX4xKxq57Vs1PYCfH6FI4socDWS)

PRIMARY KEY
FOREIGN KEY
UNIQUE
NOT NULL
CHECK

JOIN Queries (https://www.youtube.com/playlist?list=PLL_LQvNX4xKxq57Vs1PYCfH6FI4socDWS)

INNER JOIN
LEFT JOIN
RIGHT JOIN
FULL OUTER JOIN
SELF JOIN
CROSS JOIN

Subqueries (https://www.youtube.com/playlist?list=PLL_LQvNX4xKxq57Vs1PYCfH6FI4socDWS)

Scalar Subqueries
Column Subqueries
Row Subqueries
Table Subqueries
Nested Subqueries
Correlated Subqueries

Advanced SQL Functions
(https://www.youtube.com/playlist?list=PLL_LQvNX4xKxq57Vs1PYCfH6FI4socDWS)

Numeric Functions
FLOOR → ABS → MOD → ROUND → CEILING

String Functions
(https://www.youtube.com/playlist?list=PLL_LQvNX4xKxq57Vs1PYCfH6FI4socDWS)
CONCAT → LENGTH → SUBSTRING → REPLACE → UPPER → LOWER

Conditional Functions
(https://www.youtube.com/playlist?list=PLL_LQvNX4xKxq57Vs1PYCfH6FI4socDWS)
CASE → NULLIF → COALESCE

Date & Time Functions
(https://www.youtube.com/playlist?list=PLL_LQvNX4xKxq57Vs1PYCfH6FI4socDWS)
DATE → TIME → TIMESTAMP → DATEPART → DATEADD

Views

Create View → Modify View → Drop View

Indexes (https://www.youtube.com/playlist?list=PLL_LQvNX4xKxq57Vs1PYCfH6FI4socDWS)

Index Creation → Index Types → Query Optimization Using Indexes

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Transactions (https://www.youtube.com/playlist?list=PLL_LQvNX4xKxq57Vs1PYCfH6FI4socDWS)

BEGIN → COMMIT → ROLLBACK → SAVEPOINT
ACID Properties
Transaction Isolation Levels

Data Integrity & Security (https://www.youtube.com/playlist?list=PLL_LQvNX4xKxq57Vs1PYCfH6FI4socDWS)

Integrity Constraints
GRANT → REVOKE
Database Security Best Practices

Stored Procedures & Functions

Performance Optimization
(https://www.youtube.com/playlist?list=PLL_LQvNX4xKxq57Vs1PYCfH6FI4socDWS)

Query Optimization Techniques
Index Usage
Optimizing Joins

Reducing Subqueries
Query Execution Plans

Advanced SQL (https://www.youtube.com/playlist?list=PLL_LQvNX4xKxq57Vs1PYCfH6FI4socDWS)

Window Functions
ROW_NUMBER → RANK → DENSE_RANK → LEAD → LAG

Recursive Queries
CTEs (Common Table Expressions)
Pivot / Unpivot
Dynamic SQL

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related roadmaps :

→ Backend Roadmap
→ PostgreSQL Roadmap
→ MySQL Roadmap
→ Data Engineer Roadmap
→ MongoDB Roadmap

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certificates :

Free (Highly Recommended)

1. **SQL for Data Analysis – Mode SQL Tutorial**
   https://mode.com/sql-tutorial/

2. **PostgreSQL Official Docs (Best for Fundamentals)**
   https://www.postgresql.org/docs/

Paid / Industry Recognized

1. **Google Data Analytics (SQL-heavy)**
   https://www.coursera.org/professional-certificates/google-data-analytics

2. **IBM Data Analyst Professional Certificate**
   https://www.coursera.org/professional-certificates/ibm-data-analyst

3. **Udemy – The Complete SQL Bootcamp**
   https://www.udemy.com/course/the-complete-sql-bootcamp/

**role : Swift & SwiftUI Developer

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skills :

Swift Programming, Swift Syntax & Core Types, Optionals & Memory Safety,

Protocols & Generics, Functional Swift (Closures),
OOP (Classes, Structs), Error Handling, Asynchronous Programming (async/await, Tasks, Actors), SwiftUI Framework,
View & Layout Design (Stacks, Navigation), Data Flow & State Management (@State, @Binding, @ObservedObject,
@StateObject, @EnvironmentObject), Animations, User Interaction & Gestures, App Architecture (MVVM, Clean Architecture),
Data Persistence (UserDefaults, Core Data, SwiftData, CloudKit, Firebase), Networking (Alamofire, Moya),
Testing (XCTest, XCUITest), Package Management (SPM), Debugging & Logging, Cross-Platform & Server-Side Swift

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learning roadmap :

Introduction to Swift (https://www.youtube.com/watch?v=8Xg7E9shq0U)
What is Swift? → Why Swift? → Swift vs Objective-C → Where Swift Is Used → Installing Swift

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Swift Basics
Constants & Variables → Type Annotations → Printing → String Interpolation → Comments → Semicolons

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Core Types
Integers → Strings → Floats & Doubles → Tuples → Booleans → Optionals & nil

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Type System
Type Safety → Type Inference → Type Casting → Memory Safety

Operators
Arithmetic → Logical → Comparison → Nil-Coalescing

Control Flow
if/else → switch/case → Loops (for, while, repeat…while, continue, break)

Error Handling
throwing → propagating → catching

Functions & Closures
Function Basics → Parameters → Return Types → Function Types → Nested Functions → Closures → Trailing Closures

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Protocols & Generics
Protocols → Generics

Structures & Classes
Enums → Inheritance → Methods → Initializers → Properties (Stored, Computed, Observers, Wrappers) → Subscripts → Extensions → Optional Chaining

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Advanced Swift
Access Control → Automatic Reference Counting (ARC) → Result Builders → Macros

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Asynchronous Programming (https://www.youtube.com/watch?v=p6q1RmYUsNU)
Async Functions → Async Sequences → Tasks & Task Groups → Unstructured Concurrency → Actors → Strict Concurrency Checking

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SwiftUI Fundamentals (https://www.youtube.com/playlist?list=PL9VJ9OpT-IPSM6dFSwQCIl409gNBsqKTe)
What Is SwiftUI → UIKit vs SwiftUI → SwiftUI Usage → App Lifecycle

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Views & Layouts
Basic Views (Text, Image, Button, List, Form) → Layouts (HStack, VStack, ZStack, Grid, GeometryReader) → Navigation (NavigationStack, NavigationPath, TabView, NavigationLink)

View Modifiers (https://www.youtube.com/playlist?list=PLMRqhzcHGw1Z-lZaaun3A3mV9PbEfHANI)
font → padding → background → cornerRadius

Data Flow & State Management (https://www.youtube.com/playlist?list=PLWHegwAgjOkqVz-pReRmNi-2Jb-YuUR9e)
@State → @Binding → @StateObject → @ObservedObject → @EnvironmentObject → ViewBuilder

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Animations (https://www.youtube.com/playlist?list=PLTz6PJ9dfcu30W4CDtFv7f2PAoFMn0Xmb)
Implicit Animations → Explicit Animations → Transitions → Animatable Protocol

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User Interaction (https://www.youtube.com/playlist?list=PLvUWi5tdh92w0BP5gwJxgSJ12wKmYK9Y8)
UI Controls → Gestures → Drag & Drop

Localization & Accessibility (https://www.youtube.com/playlist?list=PLMRqhzcHGw1Z-lZaaun3A3mV9PbEfHANI)
Localization → Accessibility

Data Persistence (https://www.youtube.com/playlist?

list=PLvUWi5tdh92zaJIMa65f8G2aWTytY3rN3)
UserDefaults & AppStorage → FileManager → Core Data → SwiftData → CloudKit → Firebase → Realm → GRDB → General Databases

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App Architecture (https://www.youtube.com/watch?v=dtBzlIiSxUg)
MVVM → Clean Architecture → Dependency Injection

Advanced Swift & SwiftUI (https://www.youtube.com/playlist?list=PLlc_rDuPW0Y3Vx_qMXrDkCMbxRh36yYRm)
SwiftUI with Async/Await

Ecosystem & Tooling (https://t.me/learnswiftui)
Swift Package Manager → Using & Creating Packages → Plugins → Swift Package Index
Networking: Alamofire → Moya → swift-nio → Pulse
Server Frameworks: Vapor → Hummingbird → MongoKitten
Testing: XCTest → Swift Testing → XCUITest
Debugging & Logging: Xcode Debugging → SwiftUI Inspector → swift-log → CocoaLumberjack
IDEs: Xcode → VSCode → Neovim → Emacs
More Tools: DocC → Swift Charts → Swift Playgrounds

Beyond Apple Platforms (https://www.youtube.com/playlist?list=PLievaKnl8uRQqXGy3xiG9rvu8Wkq5Tndv)
Static Linux SDK → WebAssembly SDKs → Server-Side Swift

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related roadmaps :

Swift & SwiftUI
→ iOS Roadmap
→ Android Roadmap
→ Kotlin Roadmap
→ React Native Roadmap
→ Flutter Roadmap
→ Backend Roadmap
→ System Design Roadmap

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certificates :

Free (Best):

1. **Swift.org Official Guides**
   https://swift.org/documentation/

2. **Developing iOS Apps with Swift – Stanford (Free on YouTube / iTunes)**
   https://cs193p.sites.stanford.edu/

3. **Swift & SwiftUI Tutorials – Apple Developer**
   https://developer.apple.com/tutorials/swiftui

Paid (Industry-Recognized):

1. **iOS App Development with Swift – Coursera (University of Toronto)**
   https://www.coursera.org/specializations/app-development

2. **Udacity iOS Developer Nanodegree**
   https://www.udacity.com/course/ios-developer-nanodegree--nd003

3. **Ray Wenderlich — Swift & iOS Courses + Certificates**
   https://www.raywenderlich.com/

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role :

System Design Engineer

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skills :

System Design Fundamentals, Scalability, High Availability, CAP Theorem, Consistency Models, Load Balancing,
Caching Strategies, Database Design, SQL & NoSQL, Sharding, Replication, Microservices Architecture, Service Discovery,
Message Queues, Event-Driven Architecture, Asynchronous Processing, API Design, REST, gRPC, GraphQL, Networking Basics,
DNS, CDN, Performance Optimization, Reliability Engineering, Fault Tolerance, Monitoring, Observability,
Cloud Design Patterns, Security Patterns

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learning roadmap :

Introduction (https://www.youtube.com/playlist?list=PLMCXHnjXnTnvo6alSjVkgxV-VH6EPyvoX)
→ What is System Design
→ How to approach System Design

Core Concepts & Trade-offs
→ Performance vs Scalability (https://www.youtube.com/playlist?list=PLCRMIe5FDPsd0gVs500xeOewfySTsmEjf)
→ Latency vs Throughput
→ Availability vs Consistency
→ CAP Theorem (https://www.youtube.com/playlist?list=PL6W8uoQQ2c63W58rpNFDwdrBnq5G3EfT7)
→ AP Systems
→ CP Systems

Consistency Patterns
→ Weak Consistency (https://www.youtube.com/playlist?list=PLCRMIe5FDPsd0gVs500xeOewfySTsmEjf)
→ Eventual Consistency
→ Strong Consistency

Availability Patterns
→ Fail-Over (https://www.youtube.com/playlist?

list=PL6W8uoQQ2c63W58rpNFDwdrBnq5G3EfT7)
→ Active-Active
→ Active-Passive
→ Replication (Master-Slave, Master-Master)
(https://www.youtube.com/playlist?list=PLCRMIe5FDPsd0gVs500xeOewfySTsmEjf)
→ Availability Numbers (99.9, 99.99, 99.999)
(https://www.youtube.com/playlist?list=PL6W8uoQQ2c63W58rpNFDwdrBnq5G3EfT7)
→ Parallel vs Sequential Availability

Background Jobs
→ Event-Driven Jobs (https://www.youtube.com/playlist?
list=PLMCXHnjXnTnvo6alSjVkgxV-VH6EPyvoX)
→ Schedule-Driven Jobs
→ Returning Results

DNS & CDNs
→ Domain Name System (https://www.youtube.com/playlist?
list=PLCRMIe5FDPsd0gVs500xeOewfySTsmEjf)
→ Content Delivery Networks
→ Push CDNs
→ Pull CDNs

Load Balancing & Scaling (https://www.youtube.com/playlist?
list=PL6W8uoQQ2c63W58rpNFDwdrBnq5G3EfT7)
→ Load Balancer vs Reverse Proxy
→ Load Balancing Algorithms
→ Layer 4 Load Balancing
→ Layer 7 Load Balancing
→ Horizontal Scaling (https://www.youtube.com/playlist?
list=PLMCXHnjXnTnvo6alSjVkgxV-VH6EPyvoX)

Application Layer
→ Monolithic vs Microservices (https://www.youtube.com/playlist?
list=PLCRMIe5FDPsd0gVs500xeOewfySTsmEjf)
→ Microservices Architecture (https://www.youtube.com/playlist?
list=PL6W8uoQQ2c63W58rpNFDwdrBnq5G3EfT7)
→ Service Discovery

Databases
→ SQL vs NoSQL
→ RDBMS Concepts (https://www.youtube.com/playlist?
list=PLCRMIe5FDPsd0gVs500xeOewfySTsmEjf)
→ Replication
→ Sharding (https://www.youtube.com/playlist?
list=PLMCXHnjXnTnvo6alSjVkgxV-VH6EPyvoX)
→ Federation (https://www.youtube.com/playlist?
list=PLCRMIe5FDPsd0gVs500xeOewfySTsmEjf)
→ Denormalization
→ SQL Tuning
→ NoSQL Types (https://www.youtube.com/playlist?
list=PLMCXHnjXnTnvo6alSjVkgxV-VH6EPyvoX)
→ Key-Value Stores (https://www.youtube.com/playlist?
list=PL6W8uoQQ2c63W58rpNFDwdrBnq5G3EfT7)
→ Document Databases (https://www.youtube.com/playlist?
list=PLMCXHnjXnTnvo6alSjVkgxV-VH6EPyvoX)
→ Wide Column Stores
→ Graph Databases

Caching (https://www.youtube.com/playlist?
list=PL6W8uoQQ2c63W58rpNFDwdrBnq5G3EfT7)
→ Cache-Aside
→ Write-Through
→ Write-Behind
→ Refresh-Ahead

→ Client-side Cache (https://www.youtube.com/playlist?list=PLCRMIe5FDPsd0gVs500xeOewfySTsmEjf)
→ CDN Cache
→ Web Server Cache
→ Application Cache
→ Database Cache

Asynchronism
→ Back Pressure (https://www.youtube.com/playlist?list=PL6W8uoQQ2c63W58rpNFDwdrBnq5G3EfT7)
→ Task Queues (https://www.youtube.com/playlist?list=PLMCXHnjXnTnvo6alSjVkgxV-VH6EPyvoX)
→ Message Queues (https://www.youtube.com/playlist?list=PL6W8uoQQ2c63W58rpNFDwdrBnq5G3EfT7)

Communication
→ HTTP (https://www.youtube.com/playlist?list=PLCRMIe5FDPsd0gVs500xeOewfySTsmEjf)
→ TCP
→ UDP
→ RPC
→ REST
→ gRPC
→ GraphQL
→ Idempotent Operations (https://www.youtube.com/playlist?list=PL6W8uoQQ2c63W58rpNFDwdrBnq5G3EfT7)

Performance Antipatterns
→ Busy Database (https://www.youtube.com/playlist?list=PLCRMIe5FDPsd0gVs500xeOewfySTsmEjf)
→ Chatty I/O
→ No Caching
→ Retry Storm (https://www.youtube.com/playlist?list=PL6W8uoQQ2c63W58rpNFDwdrBnq5G3EfT7)
→ Synchronous I/O (https://www.youtube.com/playlist?list=PLCRMIe5FDPsd0gVs500xeOewfySTsmEjf)

Monitoring & Observability
→ Health Monitoring
→ Availability Monitoring
→ Performance Monitoring
→ Security Monitoring
→ Usage Monitoring
→ Instrumentation
→ Visualization
→ Alerts

Cloud Design Patterns
→ Queue-based Load Leveling (https://www.youtube.com/playlist?list=PL6W8uoQQ2c63W58rpNFDwdrBnq5G3EfT7)
→ Publisher Subscriber (https://www.youtube.com/playlist?list=PLMCXHnjXnTnvo6alSjVkgxV-VH6EPyvoX)
→ Priority Queue (https://www.youtube.com/playlist?list=PL6W8uoQQ2c63W58rpNFDwdrBnq5G3EfT7)
→ Competing Consumers
→ Choreography
→ Claim Check
→ Async Request Reply
→ CQRS
→ Event Sourcing
→ Materialized View
→ Sharding (https://www.youtube.com/playlist?list=PLMCXHnjXnTnvo6alSjVkgxV-VH6EPyvoX)
→ Cache-Aside (https://www.youtube.com/playlist?

list=PL6W8uoQQ2c63W58rpNFDwdrBnq5G3EfT7)
→ Valet Key
→ Strangler Fig
→ Backend for Frontend (https://www.youtube.com/playlist?list=PLCRMIe5FDPsd0gVs500xeOewfySTsmEjf)
→ API Gateway (https://www.youtube.com/playlist?list=PL6W8uoQQ2c63W58rpNFDwdrBnq5G3EfT7)
→ Sidecar
→ Ambassador
→ Anti-Corruption Layer

Reliability Patterns
→ Circuit Breaker
→ Bulkhead
→ Retry
→ Compensating Transaction
→ Deployment Stamps (https://www.youtube.com/playlist?list=PLCRMIe5FDPsd0gVs500xeOewfySTsmEjf)
→ Geodes
→ Health Endpoint Monitoring
→ Federated Identity
→ Gatekeeper
→ Valet Key (https://www.youtube.com/playlist?list=PL6W8uoQQ2c63W58rpNFDwdrBnq5G3EfT7)

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related roadmaps :

Backend Roadmap, DevOps Roadmap, Software Design & Architecture Roadmap, Cloud Engineering Roadmap

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certificates :

**Free certificates

1. **System Design Primer (GitHub – Industry Standard)**
   https://github.com/donnemartin/system-design-primer

2. **Google Cloud Architecture Framework (Free Learning Paths)**

https://cloud.google.com/architecture

3. **AWS Architecture Center (Free)**
   https://aws.amazon.com/architecture/

**Paid certificates

1. **AWS Certified Solutions Architect – Associate (Amazon)**
   https://aws.amazon.com/certification/certified-solutions-architect-associate/

2. **Google Professional Cloud Architect (Google)**

https://cloud.google.com/certification/cloud-architect

3. **Microsoft Azure Solutions Architect Expert (Microsoft)**

https://learn.microsoft.com/en-us/credentials/certifications/azure-solutions-architect/

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role :

Terraform Engineer

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skills :

Terraform, Infrastructure as Code, HashiCorp Configuration Language (HCL), Cloud Infrastructure Automation,
AWS Infrastructure, Azure Infrastructure, GCP Infrastructure, Providers Management, Terraform Modules, State Management,
Remote State, Workspaces, CI/CD Integration, Cloud Security, Secrets Management, Policy as Code, Terraform Cloud,
Terraform Enterprise, GitOps, Linux Basics, Networking Fundamentals

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learning roadmap :

Introduction (https://www.youtube.com/playlist?list=PLGyKZJoyKdOfehoKuFC1PKTGzMtRwnGNd)
→ What is Terraform
→ Usecases and Benefits
→ Infrastructure as Code
→ Configuration as Code vs Infrastructure as Code
→ Installing Terraform

HashiCorp Configuration Language
→ What is HCL
→ Basic Syntax

Getting Started
→ Project Initialization
→ Providers
→ Terraform Registry
→ Provider Configuration
→ Provider Versions
→ Resources
→ Resource Behavior
→ Resource Lifecycle
→ Meta Arguments
→ depends_on
→ count
→ for_each
→ provider
→ lifecycle

Variables
→ Input Variables
→ Type Constraints
→ Environment Variables
→ Variable Definition Files
→ Validation Rules
→ Local Values

Outputs

→ Output Syntax
→ Sensitive Outputs
→ Preconditions

Format & Validate
→ terraform fmt
→ terraform validate
→ TFLint

Deployment
→ terraform plan
→ terraform apply

Clean Up
→ terraform destroy

State Management
→ Terraform State
→ Remote State
→ State Locking
→ Import Existing Resources
→ Splitting State Files
→ State Versioning
→ Handling Sensitive Data
→ State Best Practices

Inspect / Modify State
→ terraform graph
→ terraform state list
→ terraform output
→ terraform show
→ terraform state rm
→ terraform state mv
→ apply -replace
→ state pull
→ state push
→ state replace-provider
→ state force-unlock

Basic & Advanced Commands

Modules
→ Root Modules
→ Child Modules
→ Using Published Modules
→ Creating Local Modules
→ Inputs & Outputs
→ Module Best Practices

Provisioners
→ When to Use Provisioners
→ Creation Time Provisioners
→ Destroy Time Provisioners
→ file
→ local-exec
→ remote-exec
→ Custom Provisioners

Data Sources
→ Template Files

Workspaces

CI / CD Integration

→ GitHub Actions
→ GitLab CI
→ CircleCI
→ Jenkins

Scaling Terraform
→ Splitting Large States
→ Parallelism
→ Deployment Workflows
→ Terragrunt
→ Infracost
→ Terraform Version Management

Testing
→ Unit Testing
→ Contract Testing
→ Integration Testing
→ End-to-End Testing
→ Module Testing

Security
→ Secrets Management
→ Policy as Code
→ Sentinel
→ Terrascan
→ Checkov
→ Trivy
→ KICS

HashiCorp Cloud Platform
→ What is HCP (https://www.youtube.com/playlist?list=PLl4APkPHzsUUHlbhuq9V02n9AMLPySoEQ)
→ When to Use HCP (https://www.youtube.com/playlist?list=PLl4APkPHzsUUHlbhuq9V02n9AMLPySoEQ)
→ Enterprise Features
→ Authentication
→ Workspaces
→ VCS Integration
→ Run Tasks (https://www.youtube.com/playlist?list=PLl4APkPHzsUUHlbhuq9V02n9AMLPySoEQ)

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related roadmaps :

DevOps Roadmap, Linux Roadmap, Docker Roadmap, Kubernetes Roadmap, Cloud Engineer Roadmap

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certificates :

**Free certificates

1. **HashiCorp Terraform Learn (Official Free Learning)**

https://developer.hashicorp.com/terraform/tutorials

2. **Google Cloud Infrastructure as Code (Terraform Labs)**

https://www.cloudskillsboost.google/paths/13

**Paid certificates

1. **HashiCorp Certified: Terraform Associate**
   https://www.hashicorp.com/certification/terraform-associate

2. **AWS Certified DevOps Engineer – Professional**
   https://aws.amazon.com/certification/certified-devops-engineer-professional/

3. **Google Professional Cloud DevOps Engineer**
   https://cloud.google.com/certification/cloud-devops-engineer

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**role : TypeScript Developer

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skills :

TypeScript, JavaScript (ES6+), Static Typing, Type Inference, Interfaces, Generics, Advanced Types,
Object-Oriented Programming, Functional Programming, Type Guards, Utility Types, Module Systems, Node.js,
React with TypeScript, API Development, Code Quality, ESLint, Prettier, Build Tools, tsconfig, Compiler Optimization

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learning roadmap :

JavaScript Prerequisite (https://www.youtube.com/playlist?list=PLZwesj2HKrjosO4_AjF2hTnz18g-JoDSr)
→ JavaScript Fundamentals
→ ES6+ Features (https://www.youtube.com/playlist?list=PL4cUxeGkcC9gKfw25slm4CUDUcM_sXdml)
→ Closures (https://www.youtube.com/playlist?list=PLZwesj2HKrjosO4_AjF2hTnz18g-JoDSr)
→ Promises
→ Async / Await

Introduction to TypeScript
→ TypeScript vs JavaScript (https://www.youtube.com/playlist?list=PL0Zuz27SZ-6NS8GXt5nPrcYpust89zq_b)
→ TypeScript and JavaScript Interoperability
→ Installing TypeScript
→ tsconfig.json
→ Compiler Options

Running TypeScript
→ tsc
→ ts-node
→ TypeScript Playground

TypeScript Types
→ Primitive Types
→ Object Types
→ Interfaces

→ Classes
→ Enums
→ Arrays
→ Tuples
→ Object
→ Top Types (any, unknown)
→ Bottom Type (never)

Assertions
→ as type
→ as any
→ as const
→ Non-null Assertion
→ satisfies

Type Inference

Type Compatibility

Combining Types
→ Union Types
→ Intersection Types
→ Type Aliases
→ keyof

Type Guards and Narrowing
→ typeof
→ instanceof
→ Equality Narrowing
→ Truthiness
→ Type Predicates

Functions
→ Typing Functions
→ Optional and Default Params
→ Function Overloading

Interfaces
→ Types vs Interfaces
→ Extending Interfaces
→ Interface Declaration
→ Hybrid Types

Generics
→ Generic Functions
→ Generic Interfaces
→ Generic Classes
→ Generic Constraints

Classes
→ Constructors
→ Parameter Properties
→ Access Modifiers
→ Abstract Classes
→ Inheritance
→ Polymorphism
→ Method Overriding
→ Constructor Overloading

Decorators

Utility Types (https://www.youtube.com/playlist?list=PL0Zuz27SZ-6NS8GXt5nPrcYpust89zq_b)
→ Partial

→ Pick
→ Omit
→ Readonly
→ Record
→ Exclude
→ Extract
→ NonNullable
→ Parameters
→ ReturnType
→ InstanceType
→ Awaited

Advanced Types (https://www.youtube.com/playlist?list=PL0Zuz27SZ-6NS8GXt5nPrcYpust89zq_b)
→ Mapped Types
→ Conditional Types
→ Literal Types
→ Template Literal Types
→ Recursive Types

Modules (https://www.youtube.com/playlist?list=PL0Zuz27SZ-6NS8GXt5nPrcYpust89zq_b)
→ Namespaces
→ Ambient Modules
→ External Modules
→ Namespace Augmentation
→ Global Augmentation

Ecosystem (https://www.youtube.com/playlist?list=PL0Zuz27SZ-6NS8GXt5nPrcYpust89zq_b)
→ Formatting
→ Linting
→ Build Tools
→ Useful Packages

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related roadmaps :

JavaScript Roadmap, Frontend Roadmap, Backend Roadmap, Node.js Roadmap, React Roadmap

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certificates :

Free certificates (best):

1. **TypeScript Official Handbook & Playground (Microsoft)**
   https://www.typescriptlang.org/docs/

2. **Google TypeScript Course (Cloud Skills Boost)**
   https://www.cloudskillsboost.google/course_templates/536

3. **freeCodeCamp – TypeScript Course**
   https://www.freecodecamp.org/news/learn-typescript-beginners-guide/

Paid certificates (best – MNC level):

1. **Microsoft Certified: JavaScript / TypeScript Developer (via Applied

Skills)**
   https://learn.microsoft.com/credentials/

2. **Meta Front-End Developer Professional Certificate (TypeScript included)**
   https://www.coursera.org/professional-certificates/meta-front-end-developer

3. **Udemy – TypeScript: The Complete Developer’s Guide (Industry-recognized)**
   https://www.udemy.com/course/typescript-the-complete-developers-guide/

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role :

WordPress Developer

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skills :

WordPress Core, PHP, HTML, CSS, JavaScript, MySQL, Theme Development, Plugin Development, Gutenberg Blocks,
Full Site Editing (FSE), Custom Post Types, Taxonomies, Hooks (Actions & Filters), REST API, WP-CLI,
Performance Optimization, Security Hardening, SEO Basics, Headless WordPress, WooCommerce, Deployment & Hosting

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learning roadmap :

Frontend Fundamentals
→ HTML (https://www.youtube.com/playlist?list=PLpcSpRrAaOaqMA4RdhSnnNcaqOVpX7qi5)
→ CSS
→ JavaScript

Backend Fundamentals
→ PHP (https://www.youtube.com/playlist?list=PLz-tsD7hAvutwJ4Km5O7tGkg0iVWunHij)
→ SQL (https://www.youtube.com/playlist?list=PLjVLYmrlmjGcXDYZC67Us13PBD-3us9jB)

Introduction
→ How the Web Works
→ HTTP Basics
→ CMS Concepts
→ wordpress.com vs wordpress.org

Installation & Setup
→ Local Development
→ LocalWP
→ XAMPP
→ VVV
→ Docker

WordPress Core Concepts
→ File Structure
→ The Loop (https://www.youtube.com/playlist?list=PLpcSpRrAaOaqMA4RdhSnnNcaqOVpX7qi5)
→ Hooks (Actions, Filters) (https://www.youtube.com/playlist?

list=PLjVLYmrlmjGcXDYZC67Us13PBD-3us9jB)
→ Shortcodes
→ Enqueue Scripts & Styles (https://www.youtube.com/playlist?list=PLpcSpRrAaOaqMA4RdhSnnNcaqOVpX7qi5)
→ Custom Post Types (https://www.youtube.com/playlist?list=PLjVLYmrlmjGcXDYZC67Us13PBD-3us9jB)
→ Taxonomies
→ Coding Standards
→ WP REST API

Page Editors
→ Gutenberg
→ Block Editor
→ Classic Editor
→ Hybrid Editors
→ Universal Editors
→ Third-Party Page Builders

Themes & Plugins
→ Theme Types (https://www.youtube.com/playlist?list=PLpcSpRrAaOaqMA4RdhSnnNcaqOVpX7qi5)
→ Plugin Concepts (https://www.youtube.com/playlist?list=PL6Kd_lvAfBuYNB4CH0EFpgkd0bQ5NUTQ)

Theme Development
→ Classic Themes (https://www.youtube.com/playlist?list=PLpcSpRrAaOaqMA4RdhSnnNcaqOVpX7qi5)
→ Theme Structure
→ Template Files
→ Template Hierarchy
→ Child Themes
→ Widgets & Menus
→ Customizer
→ Custom Fields (ACF, Metabox) (https://www.youtube.com/playlist?list=PLjVLYmrlmjGcXDYZC67Us13PBD-3us9jB)

Block Themes
→ Block Theme Structure (https://www.youtube.com/playlist?list=PLD8nQCAhR3tT3ehpyOpoYeUj3KHDEVK9h)
→ Full Site Editing (FSE)
→ Block Patterns
→ Template Parts
→ Global Styles
→ Theme Overrides

Plugin Development
→ Plugin File Structure (https://www.youtube.com/playlist?list=PL6Kd_lvAfBuYNB4CH0EFpgkd0bQ5NUTQ)
→ Transients API
→ Cron Jobs
→ AJAX in WordPress
→ REST API
→ Admin Menus
→ Settings API
→ Options API
→ Plugin Licensing & Updates

Database & Integrations
→ Custom Tables (https://www.youtube.com/playlist?list=PLjVLYmrlmjGcXDYZC67Us13PBD-3us9jB)
→ External APIs

Advanced WordPress
→ Headless WordPress

→ Multisite
→ Localization
→ Accessibility

Advanced PHP & Architecture
→ OOP in PHP (https://www.youtube.com/playlist?list=PLz-tsD7hAvutwJ4Km5O7tGkg0iVWunHij)
→ Composer (https://www.youtube.com/playlist?list=PLjVLYmrlmjGcXDYZC67Us13PBD-3us9jB)
→ Autoloading
→ Design Patterns

Workflows & Automation (https://www.youtube.com/playlist?list=PLjVLYmrlmjGcXDYZC67Us13PBD-3us9jB)
→ WP-CLI
→ Bedrock
→ Sage
→ Composer

Security & Hardening (https://www.youtube.com/playlist?list=PLjVLYmrlmjGcXDYZC67Us13PBD-3us9jB)
→ Roles & Capabilities
→ Login Security
→ Firewalls
→ Malware Scanning
→ Activity Logs
→ Vulnerability Monitoring

Performance Optimization (https://www.youtube.com/playlist?list=PLjVLYmrlmjGcXDYZC67Us13PBD-3us9jB)
→ Asset Optimization
→ Image Optimization
→ Database Optimization
→ Caching
→ CDN
→ Core Web Vitals

Testing & Debugging (https://www.youtube.com/playlist?list=PLjVLYmrlmjGcXDYZC67Us13PBD-3us9jB)
→ WP_DEBUG
→ PHPUnit
→ Logs
→ Query Monitor

Monitoring (https://www.youtube.com/playlist?list=PLjVLYmrlmjGcXDYZC67Us13PBD-3us9jB)
→ Performance Monitoring

Deployment (https://www.youtube.com/playlist?list=PLjVLYmrlmjGcXDYZC67Us13PBD-3us9jB)
→ Shared Hosting
→ VPS
→ Managed WordPress Hosting
→ Migrations
→ Automated Deployments
→ Backups & Recovery

SEO Basics (https://www.youtube.com/playlist?list=PLjVLYmrlmjGcXDYZC67Us13PBD-3us9jB)

Career Growth
→ WordPress Contributions
→ Core, Themes, Plugins
→ WordPress Credit Program

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related roadmaps :

Frontend Roadmap, HTML Roadmap, CSS Roadmap, JavaScript Roadmap, PHP Roadmap, Backend Roadmap

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certificates :

Free certificates (best):

1. **WordPress Learn – Official (WordPress.org)**
   https://learn.wordpress.org/

3. **freeCodeCamp – WordPress Development Course**

https://www.freecodecamp.org/news/tag/wordpress/

Paid certificates (best – MNC / industry recognized):

1. **Coursera – WordPress Website Development (University-backed)**
   https://www.coursera.org/courses?query=wordpress

2. **Udemy – Become a WordPress Developer (Industry Standard)**
   https://www.udemy.com/course/become-a-wordpress-developer-php-javascript/

3. **LinkedIn Learning – WordPress Developer Path**

https://www.linkedin.com/learning/topics/wordpress